Canonical Degree-9 Asymptotic Self-Maps, Biquadratic Conic-Square Discriminants, and Closed-Form Point Duplication on $na^4 + b^4 + c^4 = d^4$

October 30, 2026

Abstract

We investigate primitive polynomial parametrizations $(a(p,q), b(p,q), c(p,q), d(p,q)) \in \mathbb{Z}[p,q]^4$, genus-1 conic-bundle fibrations, and closed-form point-duplication maps on the diagonal quartic K3 surface $S_n : na^4 + b^4 + c^4 = d^4$ ($\gcd(a,b,c,d) = 1$). First, for $n = -s^2$ ($s \in \mathbb{Q}^\times$), we prove the *Canonical Degree-9 Asymptotic Reduction Theorem*: the degree-16 and degree-12 invariants $K_4(P_1)$ and $K_3^{\pm}(P_1)$ of the degree-17 principal asymptotic tangent self-map $\mathcal{A}_\pm : S_{-s^2} \dashrightarrow S_{-s^2}$ share an exact degree-8 common polynomial factor $M_\pm(P_1) = (sa_1^2b_1^2 \mp c_1^2d_1^2)(sa_1^2d_1^2 \mp b_1^2c_1^2)$ on the ambient projective space $\mathbb{P}^3$. Canceling $M_\pm(P_1)$ reduces $\mathcal{A}_\pm$ to a symmetric degree-9 polynomial self-map $\mathcal{A}_9^{\pm}(P_1) = \big(a_1(S^\pm - 4W),\, b_1(S^\pm - 4Z^\pm),\, c_1(S^\pm - 4U),\, d_1(S^\pm - 4V^\pm)\big)$ in the four degree-8 monomials $(U, V^\pm, W, Z^\pm) = (s^2a_1^4b_1^2d_1^2,\, \pm sa_1^2b_1^4c_1^2,\, b_1^2c_1^4d_1^2,\, \pm sa_1^2c_1^2d_1^4)$. We establish an explicit ambient quartic divisibility identity $F_{-s^2}(\mathcal{A}_9^{\pm}(P_1)) = F_{-s^2}(P_1)\,\Phi_8^{\pm,+}(P_1)\,\Phi_8^{\pm,-}(P_1)\,\Psi_{16}^{\pm}(P_1)$ via a difference-of-squares decomposition of the degree-32 Taylor cofactor, determine the exact scheme-theoretic and set-theoretic base locus of $\mathcal{A}_9^{\pm}$ on S_{-s^2}—proving via ideal saturation and the Jacobian criterion that its 1-dimensional saturated component is the radical ideal of eight smooth non-coordinate projective lines where $c_1^2 = \mp\epsilon sa_1^2$ and $d_1^2 = \epsilon b_1^2$ ($\epsilon \in \{\pm 1\}$), complemented by the codimension-2 coordinate-axis locus where at least two coordinates vanish—and establish how the saturated pullback ideal $(C^*(I_{\mathrm{Bs}}^{\pm}) : \langle p,q\rangle^\infty) = \langle G(p,q)\rangle$ on $\mathbb{P}^1_{p,q}$ governs polynomial degree drops. Using $\mathcal{A}_9^{\pm}$, we prove that S_{-16} ($s = 4$) is birationally isomorphic over $\mathbb{Q}$ to Euler's surface S_{-1} via $2a = A_{\mathrm{eul}}$ and admits explicit primitive polynomial families at degrees $7, 13, 19, 25, 31, 37, 43, 45, 49$, whereas for S_{-9} ($s = 3$) and general non-square s, we derive the universal Four-Eckardt-Plane conic-bundle reduction $sa^2 = G(k,t)$ whose completed-square discriminant in $K = k^4$ collapses to $\Delta_K(-1+u) = 128u^3(u-1)^2$ and becomes a rational square along $t = 2u^2 - 1$, splitting $G(k, 2u^2-1)$ into two conjugate rational factors. Second, we classify the discriminant-zero locus $\Delta_Q = A_2^2 - 4A_4A_0 = 0$ of the biquadratic conic-square pencil across all algebraic regimes (separating the degenerate quadratic locus at $s = \pm 1$ from the irreducible cubic locus at $s \neq 0, \pm 1$), proving a joint parity theorem explaining the role of $(t^2+1)^2$ divisibility and showing that the common biquadratic factor $M_4(t) = (t^2+1)^2 + s\alpha^2t^2$ between $d(t)^2 + b(t)^2$ and $c(t)^2 + sa(t)^2$ is governed by the rational conic $\beta^2 - s\alpha^2 = 4$. Third, we prove the *Fermat–Eckardt Divisibility Theorem*, the *UFD Binomial-Term Cancellation Obstruction*, the *Vepir–Elkies Master T-Polynomial Classification Theorem*, the *Degree-49 Alternating Iterate Reduction Theorem* for $\mathcal{A}_9^{\mp} \circ \mathcal{A}_9^{\pm}$ (with explicit closed-form degree-40 factors $\Omega_a, \Omega_b, \Omega_c, \Omega_d$ and a bihomogeneous specialization irreducibility criterion), a 2-adic valuation and quadratic-reciprocity law on S_{-s^2} and S_{-4}, and closed-form second-solution parametrizations $P_0 \mapsto P_2(P_0)$ from a single known solution $P_0 = (a_0, b_0, c_0, d_0)$ over $\mathbb{Q}(\sqrt{-n})$ for general n and over $\mathbb{Q}$ across all rational geometric regimes of n. We also situate these results within a comprehensive review of the classical and modern literature on diagonal quartic K3 surfaces.

Contents

1 Canonical Reduction of the Degree-17 Self-Map $\mathcal{A}_\pm$ to Degree 9 on S_{-s^2} **3**

1.1 Exact Base Locus, Curve Degree Formula, and Explicit Families for $s = 4$ ($n = -16$) and Variable $s(u,v)$ 6

2 Universal Four-Eckardt-Plane Conic-Bundle Reduction and Fermat Chord–Tangent Theory on S_{-s^2} 10

3 Discriminant-Zero Locus of the Biquadratic Conic-Square Pencil 12

3.1 Joint Parity Analysis of the Inhomogeneous Quadratic Ansatz 12

3.2 The Exact Reciprocal (t^2+1, t)-Pencil and Its Factored Discriminant Locus 13

3.3 The Two-Factor Conic-Square Biquadratic Pencil and the Universal Conic $\beta^2 - s\alpha^2 = 4$ 14

4 Higher-Degree Undetermined-Coefficient Elimination and Master T-Polynomial Classification 15

4.1 The Fermat–Eckardt Divisibility Theorem and UFD Cancellation Obstruction 15

4.2 The Vepir–Elkies Master T-Polynomial Classification Theorem 16

4.3 Universal Exact Binomial and Trinomial $n(p, q)$ Engines 19

5 Iterated Asymptotic Dynamics and the Infinite Master-Polynomial Tower on S_{-1} 21

5.1 Segre Factorization and Exact Degree-49 Reduction of the Alternating Iterate $\mathcal{A}_9^{\mp} \circ \mathcal{A}_9^{\pm}$ 22

5.2 Action of $\mathcal{A}_9^{\pm}$ on Euler's Curve and the Universal V_4-Symmetric Tower on S_{-1} . . . 25

6 Dual Conic-Bundle Fibrations, 2-Adic Reciprocity Laws, and New Primitive Solutions on $S_{-9}, S_{-25}, S_{-100}$ 27

7 Closed-Form Parametrizations of a Second Solution $P_2(P_0)$ from One Known Solution $P_0 = (a_0, b_0, c_0, d_0)$ 29

7.1 Universal Tangent-Plane Taylor Expansion and Degree-9 Asymptotic Formula for General n 30

7.2 Universal Diagonal-Quadric Tangent-Duplication Theorem and the (b_0, c_0)-Symmetric Map $\mathcal{E}_9^{\pm}$ 31

7.3 Complete Recipes for a Second Solution $P_2(P_0)$ Across All Regimes of n 34

8 Literature Review and Diophantine Context of $na^4 + b^4 + c^4 = d^4$ 36

8.1 Euler's Quartic Conjecture, Classical Parametrizations, and Inflection Tangents . . . 36

8.2 Elliptic Fibrations, Disproof of Euler's Conjecture, and Zariski Density 37

8.3 Local-to-Global Principles and Brauer–Manin Obstructions 38

1 Canonical Reduction of the Degree-17 Self-Map $\mathcal{A}_\pm$ to Degree 9 on S_{-s^2}

On the diagonal quartic K3 surface

$$S_{-s^2} : -s^2a^4 + b^4 + c^4 = d^4, \qquad s \in \mathbb{Q}^\times, \tag{1}$$

the product of the four surface coefficients $ABCD = (-s^2)(1)(1)(-1) = s^2$ is a non-zero rational square. By Richmond's theorem [12, 10], the two principal asymptotic (inflection-tangent) directions at any generic rational point $P_1 = (a_1 : b_1 : c_1 : d_1) \in S_{-s^2}(\mathbb{Q})$ are individually defined over $\mathbb{Q}$ and given by the 4-monomial degree-5 vector

$$V_\pm(P_1) = \big(b_1c_1^3d_1,\ \pm sa_1c_1d_1^3,\ s^2a_1^3b_1d_1,\ \pm sa_1b_1^3c_1\big). \tag{2}$$

Along the asymptotic tangent line $P(t) = P_1 + tV_\pm(P_1)$, the t^0, t^1, and t^2 coefficients of $F_{-s^2}(P_1 + tV_\pm(P_1))$ vanish modulo the surface equation $F_{-s^2}(P_1) = -s^2a_1^4 + b_1^4 + c_1^4 - d_1^4 = 0$, and the fourth intersection point with S_{-s^2} is given in unreduced form by the degree-17 rational map

$$\mathcal{A}_\pm(P_1) = K_4(P_1)\,P_1 - K_3^\pm(P_1)\,V_\pm(P_1), \tag{3}$$

where the homogeneous degree-16 and degree-12 polynomials $K_4, K_3^\pm \in \mathbb{Z}[s][a_1, b_1, c_1, d_1]$ are

$$K_4(P_1) = \big(s^2a_1^4b_1^4 - c_1^4d_1^4\big)\big(s^2a_1^4d_1^4 - b_1^4c_1^4\big), \tag{4}$$

$$K_3^\pm(P_1) = 4a_1b_1c_1d_1\Big[b_1^2d_1^2\big(s^2a_1^4 + c_1^4\big) \mp sa_1^2c_1^2\big(b_1^4 + d_1^4\big)\Big]. \tag{5}$$

Our first theorem establishes that the bracket inside $K_3^\pm(P_1)$ factors over $\mathbb{Z}[s, a_1, b_1, c_1, d_1]$ into two quadratic-monomial binomials that divide $K_4(P_1)$ identically on $\mathbb{P}^3$. Canceling this degree-8 factor drops the degree of the self-map from 17 to 9, exposes a completely symmetric 4-monomial structure, and yields an explicit ambient quartic divisibility identity on $\mathbb{P}^3$.

Theorem 1.1 (The Canonical Degree-9 Asymptotic Self-Map $\mathcal{A}_9^\pm$ on S_{-s^2})**.** *Define the conjugate degree-8 quadratic-monomial products*

$$M_\pm(P_1) = \big(sa_1^2b_1^2 \mp c_1^2d_1^2\big)\big(sa_1^2d_1^2 \mp b_1^2c_1^2\big). \tag{6}$$

Then:

1. ***Exact Factorization of*** K_4 ***and*** $K_3^\pm$ ***in*** $\mathbb{Z}[s, a_1, b_1, c_1, d_1]$***:*** *In the polynomial ring $\mathbb{Z}[s, a_1, b_1, c_1, d_1]$, we have the exact identities*

$$K_4(P_1) = M_+(P_1)\,M_-(P_1), \qquad K_3^\pm(P_1) = 4a_1b_1c_1d_1\,M_\pm(P_1). \tag{7}$$

2. ***Reduction to the Symmetric Degree-9 Self-Map*** $\mathcal{A}_9^\pm$***:*** *Dividing $\mathcal{A}_\pm(P_1)$ in (3) by the common degree-8 factor $M_\pm(P_1)$ yields the reduced degree-9 rational self-map $\mathcal{A}_9^\pm : S_{-s^2} \dashrightarrow S_{-s^2}$,*

$$\mathcal{A}_9^\pm(P_1) = M_\mp(P_1)\,P_1 - 4a_1b_1c_1d_1\,V_\pm(P_1). \tag{8}$$

Introduce the four degree-8 monomials (which satisfy the Segre syzygy $UW = V^\pm Z^\pm = P := s^2a_1^4b_1^4c_1^4d_1^4$):

$$U = s^2a_1^4b_1^2d_1^2, \qquad V^\pm = \pm sa_1^2b_1^4c_1^2, \qquad W = b_1^2c_1^4d_1^2, \qquad Z^\pm = \pm sa_1^2c_1^2d_1^4, \tag{9}$$

and their sum $S^\pm = U + V^\pm + W + Z^\pm = M_\mp(P_1)$. Then every coordinate of $\mathcal{A}_9^\pm(P_1) = (a_2, b_2, c_2, d_2)$ is divisible by its corresponding base coordinate a_1, b_1, c_1, d_1 and takes the symmetric form

$$\begin{aligned} a_2 &= a_1\big(S^\pm - 4W\big) &&= a_1\big(U + V^\pm - 3W + Z^\pm\big),\\ b_2 &= b_1\big(S^\pm - 4Z^\pm\big) &&= b_1\big(U + V^\pm + W - 3Z^\pm\big),\\ c_2 &= c_1\big(S^\pm - 4U\big) &&= c_1\big(-3U + V^\pm + W + Z^\pm\big),\\ d_2 &= d_1\big(S^\pm - 4V^\pm\big) &&= d_1\big(U - 3V^\pm + W + Z^\pm\big). \end{aligned} \tag{10}$$

3. ***Ambient Quartic Divisibility Identity and Cofactor Decomposition:*** *On $\mathbb{P}^3$, the quartic form $F_{-s^2}(P) = -s^2a^4+b^4+c^4-d^4$ evaluated at $\mathcal{A}_9^{\pm}(P_1)$ is identically divisible in $\mathbb{Z}[s,a_1,b_1,c_1,d_1]$ by $F_{-s^2}(P_1)$:*

$$F_{-s^2}\big(\mathcal{A}_9^{\pm}(P_1)\big) = F_{-s^2}(P_1)\,Q_{32}^{\pm}(P_1) = F_{-s^2}(P_1)\,\Phi_8^{\pm,+}(P_1)\,\Phi_8^{\pm,-}(P_1)\,\Psi_{16}^{\pm}(P_1), \tag{11}$$

where the degree-32 cofactor $Q_{32}^{\pm}(P_1)$ satisfies the exact difference-of-squares identity

$$Q_{32}^{\pm}(P_1) = \big((S^{\pm})^2 - 16P\big)^2 - 64P\,M_{\pm}(P_1)^2, \tag{12}$$

and its factors for $\epsilon\in\{+1,-1\}$ are given explicitly by

$$\Phi_8^{\pm,\epsilon}(P_1) = S^{\pm} \pm 4sa_1^2b_1c_1^2d_1\big(b_1d_1+\epsilon(b_1^2+d_1^2)\big), \tag{13}$$

$$\begin{aligned}\Psi_{16}^{\pm}(P_1) &= (S^{\pm})^2 - 16P \mp 8sa_1^2b_1^2c_1^2d_1^2\,M_{\pm}(P_1)\\ &= (S^{\pm})^2 \mp 8sa_1^2b_1^2c_1^2d_1^2\Big[(s^2a_1^4+c_1^4)b_1^2d_1^2 \mp sa_1^2c_1^2(b_1^2-d_1^2)^2\Big].\end{aligned} \tag{14}$$

Proof. We first establish the ambient Taylor expansion of $F_{-s^2}(P_1+tV_{\pm}(P_1))$ on $\mathbb{P}^3$ for $V_{\pm}(P_1) = (v_a,v_b,v_c,v_d)$ in (2). By Taylor's theorem in $\mathbb{Z}[s,a_1,b_1,c_1,d_1][t]$,

$$F_{-s^2}\big(P_1+tV_{\pm}(P_1)\big) = F_{-s^2}(P_1) + 4C_1t + 6C_2t^2 + 4C_3t^3 + C_4t^4,$$

where direct substitution of (2) yields:

$$\begin{aligned}C_1 &= -s^2a_1^3v_a + b_1^3v_b + c_1^3v_c - d_1^3v_d = \big(-s^2a_1^3b_1c_1^3d_1 + s^2a_1^3b_1c_1^3d_1\big) \pm sa_1b_1^3c_1d_1^3(1-1) = 0,\\ C_2 &= -s^2a_1^2v_a^2 + b_1^2v_b^2 + c_1^2v_c^2 - d_1^2v_d^2 = s^2a_1^2b_1^2c_1^2d_1^2\big(-c_1^4+d_1^4+s^2a_1^4-b_1^4\big) = -s^2a_1^2b_1^2c_1^2d_1^2\,F_{-s^2}(P_1).\end{aligned}$$

Thus $C_1=0$ identically on $\mathbb{P}^3$ and C_2 is an exact multiple of $F_{-s^2}(P_1)$ on $\mathbb{P}^3$. For $4C_3$ and C_4, substituting (2) gives

$$\begin{aligned}4C_3 &= 4\big(-s^2a_1v_a^3 + b_1v_b^3 + c_1v_c^3 - d_1v_d^3\big)\\ &= 4s^2a_1b_1c_1d_1\Big[b_1^2d_1^2\big(s^2a_1^4-c_1^4\big)\big(s^2a_1^4+c_1^4\big) \pm sa_1^2c_1^2\big(d_1^4-b_1^4\big)\big(b_1^4+d_1^4\big)\Big],\\ C_4 &= -s^2v_a^4 + v_b^4 + v_c^4 - v_d^4 = s^2b_1^4c_1^4d_1^4\big(s^6a_1^{12}-c_1^{12}\big) + s^4a_1^4b_1^4c_1^4d_1^4\big(d_1^8-b_1^8\big).\end{aligned}$$

In the ambient polynomial ring $\mathbb{Z}[s,a_1,b_1,c_1,d_1]$, we have the exact identity $d_1^4-b_1^4 = -(s^2a_1^4-c_1^4) - F_{-s^2}(P_1)$. Substituting this relation for $d_1^4-b_1^4$ into $4C_3$ immediately gives the ambient decomposition

$$4C_3 = s^2(s^2a_1^4-c_1^4)\,K_3^{\pm}(P_1) \mp 4s^3a_1^3b_1c_1^3d_1(b_1^4+d_1^4)\,F_{-s^2}(P_1), \tag{15}$$

with $K_3^{\pm}(P_1)$ as in (5). Similarly, for C_4, we have the exact ambient identity in $\mathbb{Z}[s,a_1,b_1,c_1,d_1]$:

$$C_4 = s^2(s^2a_1^4-c_1^4)\,K_4(P_1) - s^4a_1^4c_1^4\big(b_1^8+b_1^4d_1^4+d_1^8\big)\,F_{-s^2}(P_1), \tag{16}$$

which is verified directly by expanding $C_4 - s^2(s^2a_1^4-c_1^4)K_4(P_1)$:

$$\begin{aligned}C_4 - s^2(s^2a_1^4-c_1^4)K_4(P_1) &= s^2b_1^4c_1^4d_1^4(s^6a_1^{12}-c_1^{12}) + s^4a_1^4b_1^4c_1^4d_1^4(d_1^8-b_1^8)\\ &\quad - s^2(s^2a_1^4-c_1^4)\Big[b_1^4d_1^4\big(s^4a_1^8+c_1^8\big) - s^2a_1^4c_1^4\big(b_1^8+d_1^8\big)\Big]\\ &= s^4a_1^4c_1^4\Big[b_1^4d_1^4(s^2a_1^4-c_1^4) + b_1^4d_1^4(d_1^8-b_1^8) + (s^2a_1^4-c_1^4)(b_1^8+d_1^8)\Big]\\ &= -s^4a_1^4c_1^4\big(b_1^8+b_1^4d_1^4+d_1^8\big)\big(-s^2a_1^4+b_1^4+c_1^4-d_1^4\big).\end{aligned}$$

On the surface S_{-s^2} where $F_{-s^2}(P_1)=0$, equations (15) and (16) reduce to $4C_3 = s^2(s^2a_1^4-c_1^4)K_3^{\pm}(P_1)$ and $C_4 = s^2(s^2a_1^4-c_1^4)K_4(P_1)$, so the residual intersection root is $t = -K_3^{\pm}(P_1)/K_4(P_1)$, proving (3).

(1) To prove (7), expand the product in (6):

$$M_\pm(P_1) = s^2a_1^4b_1^2d_1^2 \mp sa_1^2b_1^4c_1^2 \mp sa_1^2c_1^2d_1^4 + b_1^2c_1^4d_1^2 = b_1^2d_1^2\big(s^2a_1^4+c_1^4\big) \mp sa_1^2c_1^2\big(b_1^4+d_1^4\big),$$

which matches the bracket in (5) term-for-term, establishing $K_3^\pm(P_1) = 4a_1b_1c_1d_1M_\pm(P_1)$. Similarly, multiplying $M_+(P_1)$ and $M_-(P_1)$ by the difference-of-squares identity $(A-B)(A+B)=A^2-B^2$ on each factor gives

$$\begin{aligned} M_+(P_1)\,M_-(P_1) &= \Big[(sa_1^2b_1^2)^2 - (c_1^2d_1^2)^2\Big]\Big[(sa_1^2d_1^2)^2 - (b_1^2c_1^2)^2\Big] \\ &= \big(s^2a_1^4b_1^4 - c_1^4d_1^4\big)\big(s^2a_1^4d_1^4 - b_1^4c_1^4\big) = K_4(P_1). \end{aligned}$$

(2) Substituting (7) into (3) gives $\mathcal{A}_\pm(P_1) = M_\pm(P_1)\big(M_\mp(P_1)P_1 - 4a_1b_1c_1d_1V_\pm(P_1)\big)$. Canceling the scalar polynomial factor $M_\pm(P_1)$ in projective coordinates yields (8). With $(U, V^\pm, W, Z^\pm)$ defined as in (9), we have $M_\mp(P_1) = U + V^\pm + W + Z^\pm = S^\pm$, while multiplying $V_\pm(P_1)$ in (2) by $4a_1b_1c_1d_1$ gives

$$\begin{aligned} 4a_1b_1c_1d_1\,V_\pm(P_1) &= \big(4a_1b_1^2c_1^4d_1^2,\ \pm 4sa_1^2b_1c_1^2d_1^4,\ 4s^2a_1^4b_1^2c_1d_1^2,\ \pm 4sa_1^2b_1^4c_1^2d_1\big) \\ &= \big(4a_1W,\ 4b_1Z^\pm,\ 4c_1U,\ 4d_1V^\pm\big). \end{aligned}$$

Subtracting this vector from $S^\pm P_1 = (a_1S^\pm, b_1S^\pm, c_1S^\pm, d_1S^\pm)$ immediately yields (10).

(3) To prove the ambient divisibility identity (11)–(14) explicitly in $\mathbb{Z}[s,a_1,b_1,c_1,d_1]$, write $\mathcal{A}_9^\pm(P_1) = S^\pm P_1 + T_0V_\pm(P_1)$ where $S^\pm = M_\mp(P_1)$ and $T_0 = -4a_1b_1c_1d_1$. By homogeneity of F_{-s^2} of degree 4 on $\mathbb{P}^3$, the Taylor expansion on $\mathbb{P}^3$ gives

$$F_{-s^2}\big(\mathcal{A}_9^\pm(P_1)\big) = (S^\pm)^4F_{-s^2}(P_1) + 4C_1(S^\pm)^3T_0 + 6C_2(S^\pm)^2T_0^2 + 4C_3S^\pm T_0^3 + C_4T_0^4.$$

Substituting $C_1=0$, $6C_2 = -6s^2a_1^2b_1^2c_1^2d_1^2F_{-s^2}(P_1)$, and the ambient formulas (15)–(16) into this expansion, observe that since $K_3^\pm(P_1) = -T_0M_\pm(P_1)$ and $K_4(P_1) = M_+(P_1)M_-(P_1) = M_\pm(P_1)S^\pm$, the non-$F_{-s^2}(P_1)$ parts of $4C_3S^\pm T_0^3$ and $C_4T_0^4$ cancel identically on $\mathbb{P}^3$:

$$s^2(s^2a_1^4 - c_1^4)\Big[(-T_0M_\pm(P_1))S^\pm T_0^3 + M_\pm(P_1)S^\pm T_0^4\Big] = 0.$$

Consequently, $F_{-s^2}(P_1)$ factors out of every remaining term, giving $F_{-s^2}(\mathcal{A}_9^\pm(P_1)) = F_{-s^2}(P_1)\,Q_{32}^\pm(P_1)$ where, with $P = s^2a_1^4b_1^4c_1^4d_1^4$ and $T_0^2 = 16a_1^2b_1^2c_1^2d_1^2$,

$$Q_{32}^\pm(P_1) = (S^\pm)^4 - 96P(S^\pm)^2 \pm 256s^3a_1^6b_1^4c_1^6d_1^4(b_1^4+d_1^4)S^\pm - 256s^4a_1^8b_1^4c_1^8d_1^4(b_1^8+b_1^4d_1^4+d_1^8).$$

Now introduce $A_8 = b_1^2d_1^2(s^2a_1^4+c_1^4)$ and $B_8^\pm = \pm sa_1^2c_1^2(b_1^4+d_1^4)$, so that $S^\pm = A_8 + B_8^\pm$ and $M_\pm(P_1) = A_8 - B_8^\pm$. Substituting $\pm sa_1^2c_1^2(b_1^4+d_1^4)S^\pm = B_8^\pm(A_8+B_8^\pm) = A_8B_8^\pm + (B_8^\pm)^2$ and using $(b_1^4+d_1^4)^2 - (b_1^8+b_1^4d_1^4+d_1^8) = b_1^4d_1^4$ combines the last two terms of $Q_{32}^\pm(P_1)$ into

$$256s^2a_1^4b_1^4c_1^4d_1^4\big(A_8B_8^\pm + s^2a_1^4b_1^4c_1^4d_1^4\big) = 64P\big(4A_8B_8^\pm\big) + 256P^2 = 64P\big((S^\pm)^2 - M_\pm(P_1)^2\big) + 256P^2.$$

Hence $Q_{32}^\pm(P_1)$ simplifies immediately to the difference of two squares (12):

$$\begin{aligned} Q_{32}^\pm(P_1) &= (S^\pm)^4 - 96P(S^\pm)^2 + 64P\big((S^\pm)^2 - M_\pm(P_1)^2\big) + 256P^2 \\ &= (S^\pm)^4 - 32P(S^\pm)^2 + 256P^2 - 64PM_\pm(P_1)^2 = \big((S^\pm)^2 - 16P\big)^2 - \big(8sa_1^2b_1^2c_1^2d_1^2M_\pm(P_1)\big)^2. \end{aligned}$$

Factoring this difference of squares as $\big[(S^\pm)^2 - 16P \pm 8sa_1^2b_1^2c_1^2d_1^2M_\pm(P_1)\big]\big[(S^\pm)^2 - 16P \mp 8sa_1^2b_1^2c_1^2d_1^2M_\pm(P_1)\big]$, the second factor is $\Psi_{16}^\pm(P_1)$ in (14) (since $M_\pm(P_1) - 2sa_1^2b_1^2c_1^2d_1^2 = (s^2a_1^4+c_1^4)b_1^2d_1^2 \mp sa_1^2c_1^2(b_1^2-d_1^2)^2$). Finally, using $M_\pm(P_1) = S^\pm - 2B_8^\pm$, the first factor is a difference of squares in $S^\pm \pm 4sa_1^2b_1^2c_1^2d_1^2$:

$$\begin{aligned} (S^\pm)^2 - 16P \pm 8sa_1^2b_1^2c_1^2d_1^2\big(S^\pm - 2B_8^\pm\big) &= \big(S^\pm \pm 4sa_1^2b_1^2c_1^2d_1^2\big)^2 - 16s^2a_1^4b_1^2c_1^4d_1^2\big(b_1^4+2b_1^2d_1^2+d_1^4\big) \\ &= \big(S^\pm \pm 4sa_1^2b_1^2c_1^2d_1^2\big)^2 - \big(4sa_1^2b_1c_1^2d_1(b_1^2+d_1^2)\big)^2 \\ &= \Phi_8^{\pm,+}(P_1)\,\Phi_8^{\pm,-}(P_1), \end{aligned}$$

completing the proof of (11)–(14). □

Algebraic Significance of the Degree-9 Reduction: Applying the unreduced degree-17 formula $\mathcal{A}_+$ to Euler's smallest point $P_1 = (59, 133, 134, 158)$ on S_{-1} ($s = 1$) requires dividing out the 22-digit greatest common divisor gcd = 11,830,033,013,977,620,000,000. By Theorem 1.1, the exact factor $M_+(59, 133, 134, 158) = 89{,}215{,}935{,}248{,}700{,}000$ accounts for almost the entire GCD: evaluating the degree-9 formula $\mathcal{A}_9^+(59, 133, 134, 158)$ directly leaves only the small residual gcd = 132,600 and produces the identical primitive solution $(-161634227550911, 50577474852863, 186785699397086, 152526608743862)$.

1.1 Exact Base Locus, Curve Degree Formula, and Explicit Families for $s = 4$ ($n = -16$) and Variable $s(u, v)$

Theorem 1.1 allows us to determine both the set-theoretic and scheme-theoretic base locus of $\mathcal{A}_9^\pm$ on S_{-s^2} and its exact degree-drop law on algebraic curves on S_{-s^2}.

Corollary 1.2 (Base Scheme Saturation and Curve Degree Law under $\mathcal{A}_9^\pm$)**.** *On S_{-s^2}, let $I_{\mathrm{Bs}}^\pm = \langle a_1 L_a^\pm, b_1 L_b^\pm, c_1 L_c^\pm, d_1 L_d^\pm, F_{-s^2}\rangle \subset \overline{\mathbb{Q}}[a_1, b_1, c_1, d_1]$ denote the homogeneous base ideal of $\mathcal{A}_9^\pm$, where the four multipliers $L_a^\pm = S^\pm - 4W$, $L_b^\pm = S^\pm - 4Z^\pm$, $L_c^\pm = S^\pm - 4U$, and $L_d^\pm = S^\pm - 4V^\pm$ in (10) have pairwise differences $4(U - W)$, $4(V^\pm - Z^\pm)$, $4(U - V^\pm)$, and $4(W - Z^\pm)$. Then the indeterminacy locus $\mathrm{Bs}(\mathcal{A}_9^\pm) = V(I_{\mathrm{Bs}}^\pm) \subset S_{-s^2}(\overline{\mathbb{Q}})$ decomposes into two geometric strata with the following ideal and scheme structure:*

1. ***Eight Reduced Projective Lines Defined by $U = V^\pm = W = Z^\pm$:*** *Saturating $I_{\mathrm{Bs}}^\pm$ with respect to the coordinate-hyperplane product $a_1b_1c_1d_1$ isolates the pure 1-dimensional component*

$$J_{\mathrm{lines}}^\pm = \left(I_{\mathrm{Bs}}^\pm : (a_1b_1c_1d_1)^\infty\right) = \left\langle sa_1^2b_1^2 \pm c_1^2d_1^2,\ sa_1^2d_1^2 \pm b_1^2c_1^2,\ c_1^4 - s^2a_1^4,\ d_1^4 - b_1^4\right\rangle.$$

Over $\overline{\mathbb{Q}}$, $J_{\mathrm{lines}}^\pm$ is a radical ideal (reduced of multiplicity 1) that splits into the intersection of the eight prime ideals of eight smooth projective lines:

$$sa_1^2b_1^2 = \mp c_1^2d_1^2, \quad sa_1^2d_1^2 = \mp b_1^2c_1^2 \iff \begin{cases} d_1^2 = b_1^2, & c_1^2 = \mp sa_1^2 \quad (4 \text{ lines } \mathcal{L}_{\epsilon_1,\epsilon_2}^{\pm,\mathrm{I}}), \\ d_1^2 = -b_1^2, & c_1^2 = \pm sa_1^2 \quad (4 \text{ lines } \mathcal{L}_{\epsilon_1,\epsilon_2}^{\pm,\mathrm{II}}), \end{cases} \tag{17}$$

all eight of which lie entirely on S_{-s^2} since $c_1^4 = s^2a_1^4$ and $d_1^4 = b_1^4$ imply $-s^2a_1^4 + b_1^4 + c_1^4 - d_1^4 = 0$ identically. (In particular, for $s = 1$ on S_{-1}, the four lines $d_1 = \pm b_1, c_1 = \pm a_1$ are rational base lines of $\mathcal{A}_9^-$, whereas on the open coordinate-hyperplane sections where exactly one coordinate of P_1 vanishes, $\mathcal{A}_9^\pm$ is well-defined and restricts to the identity map.)

2. ***Codimension-2 Coordinate-Axis Locus (≥ 2 Coordinates Vanishing):*** *On the coordinate hyperplanes $\{a_1b_1c_1d_1 = 0\} \cap S_{-s^2}$, a point belongs to $\mathrm{Bs}(\mathcal{A}_9^\pm)$ if and only if at least two of $\{a_1, b_1, c_1, d_1\}$ vanish (where $U = V^\pm = W = Z^\pm = 0$). Besides the coordinate-axis points lying on the closures of the eight lines (17) (namely $a_1 = c_1 = 0, b_1^4 = d_1^4$ and $b_1 = d_1 = 0, c_1^4 = s^2a_1^4$), this stratum contributes the isolated 0-dimensional Eckardt points $a_1 = b_1 = 0, c_1^4 = d_1^4$ (including the rational points $(0 : 0 : 1 : \pm 1)$), $a_1 = d_1 = 0, b_1^4 + c_1^4 = 0$, $b_1 = c_1 = 0, d_1^4 = -s^2a_1^4$, and $c_1 = d_1 = 0, b_1^4 = s^2a_1^4$.*

In particular, if $C(p, q) = (a_1(p, q), b_1(p, q), c_1(p, q), d_1(p, q)) \in \mathbb{Z}[p, q]^4$ is a primitive polynomial curve of degree m on S_{-s^2} whose image is not contained in any of the eight base lines (17), then the four pulled-back degree-$9m$ coordinate forms $A_2^\pm(p, q) = a_1(p, q)L_a^\pm(p, q)$, $B_2^\pm(p, q) = b_1(p, q)L_b^\pm(p, q)$, $C_2^\pm(p, q) = c_1(p, q)L_c^\pm(p, q)$, $D_2^\pm(p, q) = d_1(p, q)L_d^\pm(p, q)$ are not all zero and have a well-defined greatest common divisor

$$G(p, q) = \gcd\left(A_2^\pm(p, q),\ B_2^\pm(p, q),\ C_2^\pm(p, q),\ D_2^\pm(p, q)\right) \in \mathbb{Q}[p, q],$$

whose saturation with respect to the irrelevant ideal $\mathfrak{m} = \langle p, q\rangle$ satisfies $\big(C^*(I^{\pm}_{\mathrm{Bs}}) : \langle p, q\rangle^{\infty}\big) = \langle G(p, q)\rangle$ and defines the effective base pullback divisor $D^{\pm}_{\mathrm{Bs}} = \mathrm{div}(G)$ supported at $C(p, q) \cap \mathrm{Bs}(\mathcal{A}^{\pm}_9)$ on $\mathbb{P}^1_{p,q}$. Dividing the four coordinates by $G(p, q)$ yields a primitive polynomial curve $\mathcal{A}^{\pm}_9(C(p, q))$ on S_{-s^2} of exact degree $9m - \deg(G)$.

Proof. A point $P_1 = (a_1 : b_1 : c_1 : d_1) \in S_{-s^2}(\overline{\mathbb{Q}})$ lies in $V(I^{\pm}_{\mathrm{Bs}})$ if and only if $a_1L^{\pm}_a = b_1L^{\pm}_b = c_1L^{\pm}_c = d_1L^{\pm}_d = 0$.

Case 1: $a_1b_1c_1d_1 = 0$. Every monomial $U, V^{\pm}, W, Z^{\pm}$ in (9) contains three of the four variables $\{a_1, b_1, c_1, d_1\}$ to positive even powers: U omits c_1, $V^{\pm}$ omits d_1, W omits a_1, and $Z^{\pm}$ omits b_1. Thus, if at least two coordinates of P_1 vanish (for instance at the rational Eckardt points $a_1 = b_1 = 0, c_1 = \pm d_1$ or $a_1 = c_1 = 0, b_1 = \pm d_1$), then every monomial $U, V^{\pm}, W, Z^{\pm}$ contains at least one vanishing coordinate, so $U = V^{\pm} = W = Z^{\pm} = 0$ and $L^{\pm}_a = L^{\pm}_b = L^{\pm}_c = L^{\pm}_d = 0$; hence all such points lie in $\mathrm{Bs}(\mathcal{A}^{\pm}_9)$. Conversely, if exactly one coordinate of P_1 vanishes—say $a_1 = 0$ and $b_1c_1d_1 \neq 0$ (a generic point of the Fermat curve $b_1^4 + c_1^4 = d_1^4$)—then $U = V^{\pm} = Z^{\pm} = 0$ while $W = b_1^2c_1^4d_1^2 \neq 0$, so $S^{\pm} = W \neq 0$ and $\mathcal{A}^{\pm}_9(0 : b_1 : c_1 : d_1) = (0 : b_1W : c_1W : d_1W) = (0 : b_1 : c_1 : d_1)$ is well-defined and non-zero. By the same calculation, if $b_1 = 0$ and $a_1c_1d_1 \neq 0$ (only $Z^{\pm} \neq 0$), or $c_1 = 0$ and $a_1b_1d_1 \neq 0$ (only $U \neq 0$), or $d_1 = 0$ and $a_1b_1c_1 \neq 0$ (only $V^{\pm} \neq 0$), $\mathcal{A}^{\pm}_9(P_1) = P_1$ is well-defined. Hence $\mathrm{Bs}(\mathcal{A}^{\pm}_9) \cap \{a_1b_1c_1d_1 = 0\}$ consists precisely of the points where at least two coordinates vanish.

Case 2: $a_1b_1c_1d_1 \neq 0$ and Ideal Saturation. Inverting $a_1b_1c_1d_1$ replaces the generators $a_1L^{\pm}_a, b_1L^{\pm}_b, c_1L^{\pm}_c, d_1L^{\pm}_d$ by $L^{\pm}_a, L^{\pm}_b, L^{\pm}_c, L^{\pm}_d$. Since $S^{\pm} = U + V^{\pm} + W + Z^{\pm}$, the linear span of $\{L^{\pm}_a, L^{\pm}_b, L^{\pm}_c, L^{\pm}_d\}$ equals the linear span of $\{U - W, V^{\pm} - Z^{\pm}, U - V^{\pm}, W - Z^{\pm}\}$, where

$$\begin{aligned} U - W &= b_1^2d_1^2(s^2a_1^4 - c_1^4), & V^{\pm} - Z^{\pm} &= \pm sa_1^2c_1^2(b_1^4 - d_1^4),\\ U - V^{\pm} &= sa_1^2b_1^2(sa_1^2d_1^2 \mp b_1^2c_1^2), & W - Z^{\pm} &= c_1^2d_1^2(b_1^2c_1^2 \mp sa_1^2d_1^2),\\ U - Z^{\pm} &= sa_1^2d_1^2(sa_1^2b_1^2 \mp c_1^2d_1^2). \end{aligned}$$

Canceling the invertible monomials $b_1^2d_1^2$, $sa_1^2c_1^2$, $sa_1^2b_1^2$, and $sa_1^2d_1^2$ shows that the saturated ideal $J^{\pm}_{\mathrm{lines}} = (I^{\pm}_{\mathrm{Bs}} : (a_1b_1c_1d_1)^{\infty})$ is generated by $f_1 = sa_1^2b_1^2 \pm c_1^2d_1^2$, $f_2 = sa_1^2d_1^2 \pm b_1^2c_1^2$, $f_3 = c_1^4 - s^2a_1^4$, and $f_4 = d_1^4 - b_1^4$ (note that $F_{-s^2} = f_3 - f_4 \in J^{\pm}_{\mathrm{lines}}$). Over $\overline{\mathbb{Q}}$, $d_1^4 - b_1^4 = (d_1^2 - b_1^2)(d_1^2 + b_1^2)$ and $c_1^4 - s^2a_1^4 = (c_1^2 - sa_1^2)(c_1^2 + sa_1^2)$ split into coprime quadratic factors on $\{a_1b_1c_1d_1 \neq 0\}$, and substituting $d_1^2 = \epsilon b_1^2$ ($\epsilon \in \{\pm 1\}$) into $f_1 = b_1^2(sa_1^2 \pm \epsilon c_1^2) = 0$ forces $c_1^2 = \mp\epsilon sa_1^2$. Since the Jacobian matrix of $(d_1^2 - \epsilon b_1^2, c_1^2 \pm \epsilon sa_1^2)$ with respect to (d_1, c_1) is $\mathrm{diag}(2d_1, 2c_1)$, which is invertible everywhere on $\{a_1b_1c_1d_1 \neq 0\}$, $J^{\pm}_{\mathrm{lines}}$ is smooth of codimension 2 at every point of $\{a_1b_1c_1d_1 \neq 0\}$ and is therefore radical, equal to the intersection of the eight linear prime ideals in (17).

Finally, let $C(p, q) \subset S_{-s^2}$ be a primitive polynomial curve of degree m not contained in any of the eight lines (17). Because $C(p, q)$ is not contained in $\mathrm{Bs}(\mathcal{A}^{\pm}_9)$, the four pulled-back homogeneous forms $A^{\pm}_2, B^{\pm}_2, C^{\pm}_2, D^{\pm}_2 \in \mathbb{Q}[p, q]$ of degree $9m$ do not all vanish identically. Let $G(p, q) = \gcd(A^{\pm}_2, B^{\pm}_2, C^{\pm}_2, D^{\pm}_2)$ in the UFD $\mathbb{Q}[p, q]$, and write $(A^{\pm}_2, B^{\pm}_2, C^{\pm}_2, D^{\pm}_2) = G(p, q) \cdot (\tilde{a}_2, \tilde{b}_2, \tilde{c}_2, \tilde{d}_2)$. Then $\gcd(\tilde{a}_2, \tilde{b}_2, \tilde{c}_2, \tilde{d}_2) = 1$ in $\mathbb{Q}[p, q]$, so the four homogeneous polynomials $\tilde{a}_2, \tilde{b}_2, \tilde{c}_2, \tilde{d}_2$ have no common zero in $\mathbb{P}^1(\overline{\mathbb{Q}})$. By the projective Nullstellensatz in $\mathbb{Q}[p, q]$, the ideal $\mathfrak{a} = \langle \tilde{a}_2, \tilde{b}_2, \tilde{c}_2, \tilde{d}_2\rangle$ satisfies $\sqrt{\mathfrak{a}} = \langle p, q\rangle$, so $\langle p, q\rangle^N \subset \mathfrak{a}$ for $N \gg 1$ and $(\mathfrak{a} : \langle p, q\rangle^{\infty}) = \mathbb{Q}[p, q]$. Therefore the pullback ideal $C^*(I^{\pm}_{\mathrm{Bs}}) = \langle A^{\pm}_2, B^{\pm}_2, C^{\pm}_2, D^{\pm}_2\rangle = G(p, q) \cdot \mathfrak{a}$ has saturation $(C^*(I^{\pm}_{\mathrm{Bs}}) : \langle p, q\rangle^{\infty}) = \langle G(p, q)\rangle$, defining the principal base divisor $\mathrm{div}(G)$ on $\mathbb{P}^1_{p,q}$, and $(\tilde{a}_2, \tilde{b}_2, \tilde{c}_2, \tilde{d}_2)$ is a primitive polynomial curve on S_{-s^2} of exact degree $9m - \deg(G)$. □

We now apply Theorem 1.1 and Corollary 1.2 to establish explicit polynomial families for $s = 4$ ($n = -16$) and 2-parameter variable-s surfaces.

Proposition 1.3 (Complete Resolution of $s = 4$ ($n = -16$) and Variable-s Curve Extraction)**.**

1. ***Exact Rational Curves of Degrees*** $7, 13, 19, 25, 31, 37, 43, 45, 49$ ***on*** S_{-16} ***($s = 4$):*** *Because $s = 4 = 2^2$ is a rational square, $n = -16 = -2^4 \equiv -1 \pmod{\mathbb{Q}^{\times 4}}$. Thus*

$S_{-16} : -16a^4+b^4+c^4=d^4$ is birationally isomorphic over $\mathbb{Q}$ to Euler's surface $S_{-1}: -A^4+B^4+C^4=D^4$ via $(A,B,C,D)=(2a,b,c,d)$. Substituting Euler's degree-7 family $(A_{\mathrm{eul}}(p,q), B_{\mathrm{eul}}(p,q), C_{\mathrm{eul}}(p,q), D_{\mathrm{eul}}(p,q))$ and the degree-$13,19,25,31,37,43,45,49$ families on S_{-1} from Theorem 5.3 via

$$\big(a(p,q),\, b(p,q),\, c(p,q),\, d(p,q)\big) = \frac{1}{\gcd}\Big(A_m(p,q),\, 2B_m(p,q),\, 2C_m(p,q),\, 2D_m(p,q)\Big) \tag{18}$$

yields primitive polynomial parametrizations of S_{-16} at every degree $m \in \{7,13,19,25,31,37,43,45,49\}$.

2. ***Universal Degree Drop from 18 to 10 on the (k,t)-Conic-Bundle Section:*** *On the 2-parameter family $a_1(k,t)=1$, $b_1(k,t)=-k(k^4-1)t^2+4kt+2k$, $c_1(k,t)=-(k^4-1)t^2-2(k^4-1)t+2$, $d_1(k,t)=k(k^4-1)t^2+2k$ of Theorem 2.1, which satisfies $-s(k,t)^2a_1^4+b_1^4+c_1^4=d_1^4$ identically with $s(k,t)=G(k,t)$, applying $\mathcal{A}_9^+$ divides out the universal degree-8 base-locus factor in t:*

$$g(k,t)=32k^4t(t+1)\big((k^4-1)t-2\big)\Big((k^2-1)^2t^2-2(k^2-1)t+2\Big)\Big((k^2+1)^2t^2+2(k^2+1)t+2\Big),$$

reducing the degree in t from 18 to 10 in (b_2,c_2,d_2) and from 16 to 8 in a_2. In particular, at $k=2$ and $t=u/v$—homogenizing (23) by v^2 so that $a_1(u,v)=1$, $b_1(u,v)=-30u^2+8uv+4v^2=-2(15u^2-4uv-2v^2)$, $c_1(u,v)=-15u^2-30uv+2v^2=-(15u^2+30uv-2v^2)$, $d_1(u,v)=30u^2+4v^2=2(15u^2+2v^2)$, and $s(u,v)=s_2(u,v)=v^4G(2,u/v)$ with

$$s_2(u,v)=v^4G(2,u/v)=225u^4-1020u^3v-824u^2v^2+136uv^3+4v^4$$

(which satisfies $-s_2(u,v)^2a_1(u,v)^4+b_1(u,v)^4+c_1(u,v)^4=d_1(u,v)^4$ identically)—dividing $\mathcal{A}_9^+(a_1,b_1,c_1,d_1)$ by $2v^8g(2,u/v)$ yields the exact homogeneous polynomial solution $(a_2(u,v),b_2(u,v),c_2(u,v),d_2(u,v))$ on the same surface $-s_2(u,v)^2a_2(u,v)^4+b_2(u,v)^4+c_2(u,v)^4=d_2(u,v)^4$ (with $\deg(a_2)=8$ and $\deg(b_2)=\deg(c_2)=\deg(d_2)=10$, of homogeneous degree 40 in (u,v)):

$$\begin{aligned}
a_2(u,v) &= -\Big(405000u^8+644625u^7v+545175u^6v^2-311850u^5v^3+81000u^4v^4\\
&\quad +41580u^3v^5+9692u^2v^6-1528uv^7+128v^8\Big),\\
b_2(u,v) &= (15u^2-4uv-2v^2)\Big(50625u^8-1748250u^7v-1947150u^6v^2-1028700u^5v^3\\
&\quad -621000u^4v^4+137160u^3v^5-34616u^2v^6+4144uv^7+16v^8\Big),\\
c_2(u,v) &= -(15u^2+30uv-2v^2)\Big(405000u^8-1076625u^7v-487575u^6v^2+254250u^5v^3\\
&\quad -94440u^4v^4-33900u^3v^5-8668u^2v^6+2552uv^7+128v^8\Big),\\
d_2(u,v) &= (15u^2+2v^2)\Big(50625u^8+1694250u^7v+118350u^6v^2-2160900u^5v^3\\
&\quad -270120u^4v^4+288120u^3v^5+2104u^2v^6-4016uv^7+16v^8\Big).
\end{aligned} \tag{19}$$

(Equivalently, if one rescales the surface parameter by $s_{\mathrm{red}}(u,v)=\frac{1}{4}s_2(u,v)$, then replacing $a_1\mapsto 2$ and $a_2(u,v)\mapsto 2a_2(u,v)$ satisfies $-s_{\mathrm{red}}(u,v)^2(2a_i)^4+b_i^4+c_i^4=d_i^4$ for both $i=1,2$.)

3. ***Birationality and Genus-1 Preservation on Constant Non-Square s_0 Fibers (e.g., $s_0=3, n=-9$):*** *For any fixed non-square rational $s_0\in\mathbb{Q}^\times\setminus\mathbb{Q}^{\times 2}$ and fixed $k\in\mathbb{Q}\setminus\{0,\pm1\}$, specializing $s_0a^2=G(k,t)$ defines a smooth genus-1 quartic curve $E_{k,s_0}\subset\mathbb{P}(1,1,2)$ with function field $\mathbb{Q}(E_{k,s_0})=\mathbb{Q}(t,a)$ where $a=\sqrt{G(k,t)/s_0}$. Restricting $\mathcal{A}_9^+$ to E_{k,s_0} and canceling $g(k,t)$ gives the image point $\big[a\cdot\widetilde{L}_a(k,t):b_2(k,t):c_2(k,t):d_2(k,t)\big]\in\mathbb{P}^3$, which induces an isomorphism of function fields $\mathbb{Q}(\mathcal{A}_9^+(E_{k,s_0}))\cong\mathbb{Q}(t,a)=\mathbb{Q}(E_{k,s_0})$. Consequently, $\mathcal{A}_9^+(E_{k,s_0})$ is birational to E_{k,s_0} and has geometric genus 1 (it is neither contracted to a point nor mapped to a genus-0 curve).*

Proof. (1) If $-A_m(p,q)^4+B_m(p,q)^4+C_m(p,q)^4=D_m(p,q)^4$, multiplying both sides by $16=2^4$ gives

$$-16A_m(p,q)^4+\big(2B_m(p,q)\big)^4+\big(2C_m(p,q)\big)^4=\big(2D_m(p,q)\big)^4,$$

so $(a,b,c,d)=(A_m,2B_m,2C_m,2D_m)/\gcd$ lies on $S_{-16}:-16a^4+b^4+c^4=d^4$.

(2) On the universal (k,t)-section of Theorem 2.1, each of b_1,c_1,d_1 has degree 2 in t while $s(k,t)=G(k,t)$ has degree 4 in t and $a_1=1$ has degree 0 in t. Thus the monomials $U=G^2b_1^2d_1^2$, $V^+=Gb_1^4c_1^2$, $W=b_1^2c_1^4d_1^2$, and $Z^+=Gc_1^2d_1^4$ have degrees $12,16,12,16$ in t, so the raw coordinates of $\mathcal{A}_9^+(P_1)$ have degree 16 in a_2 and degree 18 in (b_2,c_2,d_2). By Corollary 1.2(1), the common factors in t of the four multipliers $S^+-4W, S^+-4Z^+, S^+-4U, S^+-4V^+$ occur precisely at values of t where $P_1(k,t)$ meets the eight base lines (17), i.e., where $U=V^+=W=Z^+$. Checking the factorizations of $c_1(k,t)^2-G(k,t)$ and $d_1(k,t)^2-b_1(k,t)^2$:

$$d_1(k,t)^2-b_1(k,t)^2=8k^2t\big((k^4-1)t^2-2t-2\big),\qquad c_1(k,t)^2-G(k,t)=4(k^4-1)t(t+1)\big((k^4-1)t-2\big).$$

At $t=0$, $t=-1$, and $t=2/(k^4-1)$, we have $c_1^2=G$ and $b_1^2=d_1^2$ with signs making $U=V^+=W=Z^+$ (meeting the lines $\mathcal{L}^{+,\mathrm{II}}_{\epsilon_1,\epsilon_2}$), accounting for the three linear factors $t(t+1)((k^4-1)t-2)$. Similarly, the two quadratic factors $(k^2\mp1)^2t^2\mp2(k^2\mp1)t+2$ are the roots of $d_1^2+b_1^2=0$ and $c_1^2+G=0$ where $P_1(k,t)$ meets the lines $\mathcal{L}^{+,\mathrm{I}}_{\epsilon_1,\epsilon_2}$, so they divide $M_-(P_1)=(Gb_1^2+c_1^2d_1^2)(Gd_1^2+b_1^2c_1^2)$ and all four multipliers $S^+-4W, S^+-4Z^+, S^+-4U, S^+-4V^+$. Dividing out this degree-8 polynomial $g(k,t)$ leaves degree $18-8=10$ in (b_2,c_2,d_2) and degree $16-8=8$ in a_2. Specializing $k=2$ and $t=u/v$ with $a_1(u,v)=1$, $b_1(u,v)=-30u^2+8uv+4v^2$, $c_1(u,v)=-15u^2-30uv+2v^2$, $d_1(u,v)=30u^2+4v^2$, and $s_2(u,v)=v^4G(2,u/v)=225u^4-1020u^3v-824u^2v^2+136uv^3+4v^4$, and dividing out $2v^8g(2,u/v)=1024uv(u+v)(15u-2v)(9u^2-6uv+2v^2)(25u^2+10uv+2v^2)$ yields (19), and Theorem 1.1(3) guarantees $-s_2(u,v)^2a_2(u,v)^4+b_2(u,v)^4+c_2(u,v)^4-d_2(u,v)^4=0$ identically because $F_{-s_2^2}(P_1)=0$. (As an explicit specialization check at $(u,v)=(1,1)$, where $s_2(1,1)=-1479$, (19) evaluates to $a_2(1,1)=-1407422$, $b_2(1,1)=-9(-5074192)=45667728$, $c_2(1,1)=-43(-1034250)=44472750$, $d_2(1,1)=17(-325704)=-5536968$, which satisfies $-(-1479)^2(-1407422)^4+45667728^4+44472750^4=(-5536968)^4$ identically.)

(3) For fixed $k\in\mathbb{Q}\setminus\{0,\pm1\}$ and $s_0\in\mathbb{Q}^\times\setminus\mathbb{Q}^{\times2}$, the polynomial $G(k,t)$ in (24) has discriminant $\Delta_t(G)=2^{24}k^{12}(k^4-1)^6(k^4+1)^2\neq0$, so $G(k,t)$ has four distinct roots in $\overline{\mathbb{Q}}$ and $E_{k,s_0}:s_0a^2=G(k,t)$ is a smooth curve of genus 1 with function field $\mathbb{Q}(E_{k,s_0})=\mathbb{Q}(t,a)$. On E_{k,s_0}, substituting $s_0a^2=G(k,t)$ into U,V^+,W,Z^+ replaces s_0a^2 by $G(k,t)$, so the multipliers L_a^+,L_b^+,L_c^+,L_d^+ are identical to those in Part (2). After dividing all four coordinates by $g(k,t)$, the image of $(a,b_1(k,t),c_1(k,t),d_1(k,t))\in E_{k,s_0}$ under $\mathcal{A}_9^+$ is

$$\mathcal{A}_9^+(P_1)=\big[a\,\widetilde{L}_a(k,t):b_2(k,t):c_2(k,t):d_2(k,t)\big]\in\mathbb{P}^3,$$

where $\widetilde{L}_a(k,t)=(S^+-4W)/g(k,t)\in\mathbb{Q}[k,t]$ has degree 8 in t and $b_2(k,t),c_2(k,t),d_2(k,t)\in\mathbb{Q}[k,t]$ have degree 10 in t. Let $F'=\mathbb{Q}\big(b_2(k,t)/d_2(k,t),\,c_2(k,t)/d_2(k,t),\,a\widetilde{L}_a(k,t)/d_2(k,t)\big)\subset\mathbb{Q}(t,a)$ be the function field of the image curve $\mathcal{A}_9^+(E_{k,s_0})$. Because $b_2(k,t)/d_2(k,t)$ and $c_2(k,t)/d_2(k,t)$ separate points at $t=0$—specifically, the greatest common divisor $\gcd_t\big(b_2(k,t)d_2(k,0)-d_2(k,t)b_2(k,0),\,c_2(k,t)d_2(k,0)-d_2(k,t)c_2(k,0)\big)=t$ is linear in t—the subfield $\mathbb{Q}(b_2/d_2,c_2/d_2)$ equals the entire rational function field $\mathbb{Q}(t)$. Since $\widetilde{L}_a(k,0)=16k^8(k^4-1)^4\neq0$ and $d_2(k,0)=32k^9(k^4-1)^4\neq0$ for $k\notin\{0,\pm1\}$, both $\widetilde{L}_a(k,t)$ and $d_2(k,t)$ are non-zero in $\mathbb{Q}(t)$, so adjoining the first affine coordinate $a\widetilde{L}_a(k,t)/d_2(k,t)$ to $\mathbb{Q}(t)$ adjoins $a=\sqrt{G(k,t)/s_0}$, proving $F'=\mathbb{Q}(t,a)=\mathbb{Q}(E_{k,s_0})$. Thus $\mathcal{A}_9^+|_{E_{k,s_0}}$ is birational onto its image, which therefore has geometric genus 1. □

Table 1: Explicit Primitive Integer Solutions Generated by the Canonical Degree-9 Self-Map $\mathcal{A}_9^{\pm}$ on $S_{-s^2}: -s^2a^4+b^4+c^4=d^4$ (normalized projectively with $d_2>0$; all seed and image points satisfy $F_{-s^2}(a_i,b_i,c_i,d_i)=0$ identically).

Surface S_{-s^2}	**Seed Point** $P_1=(a_1,b_1,c_1,d_1)$	**Branch**	**Residual** gcd **of** (10) **& Primitive Image** $\mathcal{A}_9^{\pm}(P_1)=(a_2,b_2,c_2,d_2)$ $(d_2>0)$
$s=1$ $(n=-1)$	$(59,133,134,158)$	$\mathcal{A}_9^{+}$	gcd = 132,600: $(-161634227550911, 50577474852863, 186785699397086, 152526608743862)$
$s=4$ $(n=-16)$	$(67,59,158,133)$	$\mathcal{A}_9^{+}$	gcd = 222,720: $(151682297322 47, -94602936319406, 88325451977522, 108797492863822)$
$s=4$ $(n=-16)$	$(67,59,158,133)$	$\mathcal{A}_9^{-}$	gcd = 172,864: $(93392849698543, -161634227550911, 152526608743862, 50577474852863)$
$s=6$ $(n=-36)$	$(80,108,295,281)$	$\mathcal{A}_9^{+}$	gcd = 338,430,000: $(407259073840, -17142597347004, 21086472560885, 23086242547147)$
$s=6$ $(n=-36)$	$(80,108,295,281)$	$\mathcal{A}_9^{-}$	gcd = 129,750,000: $(25191276484496, -58585946249268, 41809582553311, 24122195114849)$
$s=15$ $(n=-225)$	$(86,45,334,109)$	$\mathcal{A}_9^{+}$	gcd = 513,965,760: $(200057337374, -40228312 4865, 78553829 7046, 464499774271)$
$s=15$ $(n=-225)$	$(86,45,334,109)$	$\mathcal{A}_9^{-}$	gcd = 15,360: $(13436605124643299, -16950291699510315, 51897213301552741, 7090741712018341)$

2 Universal Four-Eckardt-Plane Conic-Bundle Reduction and Fermat Chord–Tangent Theory on S_{-s^2}

To study rational points and curves on S_{-s^2} for arbitrary rational $s\in\mathbb{Q}^\times$ (including $s=2,3,5,6,15$), we combine the Four-Eckardt-Plane Identity with the rational square $n=-s^2$.

Theorem 2.1 (The Universal (k,t)-Conic-Bundle Fibration on S_{-s^2})**.** *On $S_{-s^2}: -s^2a^4+b^4+c^4=d^4$, the Four-Eckardt-Plane Identity $(b^2+c^2-d^2)^2+na^4=2(d^2-b^2)(d^2-c^2)$ splits over $\mathbb{Q}$ into a difference of two squares:*

$$\big(b^2+c^2-d^2-sa^2\big)\big(b^2+c^2-d^2+sa^2\big)=2\big(d^2-b^2\big)\big(d^2-c^2\big). \tag{20}$$

Introducing the projective pencil parameter $\lambda\in\mathbb{P}^1(\mathbb{Q})\setminus\{0,-1,-1/2,\infty\}$ via

$$b^2+c^2-d^2-sa^2=2\lambda\big(d^2-b^2\big),\qquad b^2+c^2-d^2+sa^2=\frac{1}{\lambda}\big(d^2-c^2\big), \tag{21}$$

and eliminating sa^2 yields the diagonal ternary quadratic form in (b,c,d):

$$2\lambda(\lambda+1)\,b^2+(2\lambda+1)\,c^2=\big(2\lambda^2+2\lambda+1\big)\,d^2. \tag{22}$$

1. ***Rational Base-Point Parametrization ($2\lambda+1=k^2$):*** *At $b=d$, equation (22) reduces to $(2\lambda+1)c^2=d^2$. Setting $\lambda=(k^2-1)/2$ ($k\in\mathbb{Q}\setminus\{0,\pm1\}$) gives the smooth conic $(k^4-1)b^2+2k^2c^2=(k^4+1)d^2$ with rational base point $(b:c:d)=(k:1:k)$, which is rationally parametrized over $\mathbb{Q}(k,t)$ by*

$$\begin{aligned} b(k,t)&=-k(k^4-1)t^2+4kt+2k,\\ c(k,t)&=-(k^4-1)t^2-2(k^4-1)t+2,\\ d(k,t)&=k(k^4-1)t^2+2k. \end{aligned} \tag{23}$$

2. ***Exact Quartic Reduction to $sa^2=G(k,t)$:*** *Substituting (23) into the difference of (21) satisfies $b(k,t)^4+c(k,t)^4-d(k,t)^4=G(k,t)^2$ identically on $\mathbb{Q}[k,t]$, reducing S_{-s^2} to the double cover $sa^2=G(k,t)$, where*

$$\begin{aligned} G(k,t)&=(k^4-1)^2t^4-4(k^4-1)(k^4+1)t^3-4(k^8-3k^4-2)t^2+8(k^4+1)t+4\\ &=t^2(t^2-4t-4)k^8-2t(t+2)(t^2-2t-2)k^4+(t^2+2t+2)^2. \end{aligned} \tag{24}$$

3. ***Split Discriminant and Difference-of-Two-Squares Decomposition along $t=2u^2-1$:*** *Viewing $G(k,t)$ as a quadratic in $K=k^4$, its discriminant is $\Delta_K(t)=128t^2(t+1)^3=2(t+1)\big[8t(t+1)\big]^2$, which is a square in $\mathbb{Q}(t)$ if and only if $2(t+1)=(2u)^2$, i.e., $t=2u^2-1$. Setting $t=2u^2-1$ makes $\Delta_K(2u^2-1)=\big(32u^3(2u^2-1)\big)^2$ a perfect square in $\mathbb{Q}[u]$ and factors $G(k,2u^2-1)=L_+(k,u)\,L_-(k,u)$ over $\mathbb{Q}[k,u]$ into two conjugate factors:*

$$L_\pm(k,u)=(2u^2-1)(2u^2\pm4u+1)k^4-(2u^2\pm2u+1)^2. \tag{25}$$

Consequently, $sa^2=X(k,u)^2-Y(k,u)^2$ is a difference of two polynomial squares in $\mathbb{Z}[k,u]$ with $X(k,u)=(4u^4-1)k^4-(4u^4+8u^2+1)$ and $Y(k,u)=4u\big((2u^2-1)k^4-(2u^2+1)\big)$.

4. ***Shifted Coordinate*** $t=-1+v$ ***and Completed-Square Identity:*** *Under the linear shift* $t=-1+v$ *centered at the rational root* $t=-1$ *of* $\Delta_K(t)$, $G(k,-1+v)$ *takes the symmetric form*

$$\begin{aligned} G(k,-1+v) &= \Big[(k^4+1)-4k^4v-(k^4-1)v^2\Big]^2 - 16k^4(k^4-1)v^3 \\ &= \Big[(k^4+1)-4k^4v+(k^4-1)v^2\Big]^2 - 4(k^8-1)v^2. \end{aligned} \tag{26}$$

Completing the square with respect to $K=k^4$ *(for* $A(v)=(v-1)^2(v^2-6v+1)\neq 0$*) yields the generic monomial-discriminant identity*

$$4A(v)\,G(k,-1+v) = \big(2A(v)k^4+B(v)\big)^2 - 128v^3(v-1)^2, \tag{27}$$

where $B(v)=-2(v^2-1)(v^2-4v+1)$ *and the generic shifted discriminant* $\Delta_K(-1+v)=128v^3(v-1)^2$ *becomes a perfect square upon setting* $v=2u^2$ *(*$t=-1+2u^2$*), recovering (25).*

Proof. First, expanding $(b^2+c^2-d^2)^2-2(d^2-b^2)(d^2-c^2)$ gives

$$\big(b^4+c^4+d^4+2b^2c^2-2b^2d^2-2c^2d^2\big)-\big(2d^4-2b^2d^2-2c^2d^2+2b^2c^2\big)=b^4+c^4-d^4.$$

On S_{-s^2}, we have $b^4+c^4-d^4=s^2a^4$, so $(b^2+c^2-d^2)^2-s^2a^4=2(d^2-b^2)(d^2-c^2)$, which factors as (20). Introducing λ via (21), adding the two equations in (21) and multiplying by λ eliminates sa^2:

$$2\lambda(b^2+c^2-d^2)=2\lambda^2(d^2-b^2)+(d^2-c^2) \iff 2\lambda(\lambda+1)b^2+(2\lambda+1)c^2=(2\lambda^2+2\lambda+1)d^2,$$

proving (22).

(1) Substituting $\lambda=(k^2-1)/2$ into (22) gives $2\lambda(\lambda+1)=(k^4-1)/2$, $2\lambda+1=k^2$, and $2\lambda^2+2\lambda+1=(k^4+1)/2$; multiplying by 2 yields $(k^4-1)b^2+2k^2c^2=(k^4+1)d^2$, which is smooth for $k\notin\{0,\pm1\}$. At $(b,c,d)=(k,1,k)$, we have $(k^4-1)k^2+2k^2=(k^4+1)k^2$. Writing affine coordinates $B=b/d$ and $C=c/d$ with base point $(B_0,C_0)=(1,1/k)$, pass the line $C-\frac{1}{k}=-t(B-1)$ through $(1,1/k)$. Substituting $C=\frac{1}{k}-t(B-1)$ into $(k^4-1)B^2+2k^2C^2-(k^4+1)=0$ yields

$$(B-1)\Big[\big((k^4-1)+2k^2t^2\big)B+\big((k^4-1)-2k^2t^2-4kt\big)\Big]=0.$$

Rescaling the slope parameter $t\mapsto\frac{k^4-1}{2k}t$ and clearing the common factor $2k^2(k^4-1)$ gives $b(k,t),c(k,t),d(k,t)$ in (23).

(2) Subtracting the first equation of (21) from the second gives $2sa^2=\frac{1}{\lambda}(d^2-c^2)-2\lambda(d^2-b^2)$. Substituting $\lambda=(k^2-1)/2$ and (23), the common factor $2(k^2-1)$ cancels out and leaves $sa^2=G(k,t)$ with $G(k,t)$ given by (24). Collecting powers of k^4 in $G(k,t)$ directly yields the second line of (24).

(3) Viewing $G(k,t)=\alpha_2(t)K^2+\alpha_1(t)K+\alpha_0(t)$ as a quadratic in $K=k^4$, where $\alpha_2(t)=t^2(t^2-4t-4)$, $\alpha_1(t)=-2t(t+2)(t^2-2t-2)$, and $\alpha_0(t)=(t^2+2t+2)^2$, its discriminant is

$$\Delta_K(t)=\alpha_1(t)^2-4\alpha_2(t)\alpha_0(t)=4t^2\Big[(t+2)^2(t^2-2t-2)^2-(t^2-4t-4)(t^2+2t+2)^2\Big]=128t^2(t+1)^3.$$

Setting $t+1=2u^2$ (i.e., $t=2u^2-1$) turns $\Delta_K(2u^2-1)=128(2u^2-1)^2(2u^2)^3=\big(32u^3(2u^2-1)\big)^2$ into a square in $\mathbb{Q}[u]$. Applying the quadratic formula in $K=k^4$ splits $G(k,2u^2-1)=L_+(k,u)\,L_-(k,u)$ as in (25). Writing $L_\pm(k,u)=X(k,u)\pm Y(k,u)$ with $X=\frac{1}{2}(L_++L_-)$ and $Y=\frac{1}{2}(L_+-L_-)$ gives $G(k,2u^2-1)=X(k,u)^2-Y(k,u)^2$.

(4) Substituting $t=-1+v$ into (24) gives $G(k,-1+v)=A(v)k^8+B(v)k^4+C(v)$ with $A(v)=(v-1)^2(v^2-6v+1)$, $B(v)=-2(v^2-1)(v^2-4v+1)$, and $C(v)=(v^2+1)^2$. Expanding the two right-hand sides of (26) reproduces $A(v)k^8+B(v)k^4+C(v)$ identically, and multiplying by $4A(v)$ gives $4A(v)G(k,-1+v)=(2A(v)k^4+B(v))^2-(B(v)^2-4A(v)C(v))$, where $B(v)^2-4A(v)C(v)=\Delta_K(-1+v)=128(v-1)^2v^3$, proving (27). □

Remark 2.2 (New Primitive Integer Solution on S_{-225} ($s = 15$) via Theorem 2.1). Evaluating (23)–(24) at $k = 1/4$ and $t = -4/5$ yields $G(1/4, -4/5) = 15 \cdot (43/160)^2$, so $s = 15$ ($n = -225$). Clearing denominators and dividing out the common factor gives the primitive integer solution $(a, b, c, d) = (86, 45, 334, 109)$ to $-225a^4 + b^4 + c^4 = d^4$, verified by exact arithmetic:

$$-225 \cdot 86^4 + 45^4 + 334^4 = -12{,}307{,}683{,}600 + 4{,}100{,}625 + 12{,}444{,}741{,}136 = 141{,}158{,}161 = 109^4.$$

Applying $\mathcal{A}_9^{\pm}$ to $(86, 45, 334, 109)$ produces the two higher-height primitive solutions in the bottom rows of Table 1.

3 Discriminant-Zero Locus of the Biquadratic Conic-Square Pencil

We now analyze the quadratic pencil $(a(t), b(t), c(t), d(t))$ on S_{-s^2} and classify the discriminant-zero locus $\Delta_Q = A_2^2 - 4A_4A_0 = 0$ across all three algebraic regimes.

3.1 Joint Parity Analysis of the Inhomogeneous Quadratic Ansatz

Consider the inhomogeneous quadratic ansatz

$$b(t) = t^2 + \beta t - 1, \qquad d(t) = t^2 + \delta t + 1, \qquad c(t) = (t^2 + 1) + \gamma t, \qquad a(t) = (t^2 + 1) + \alpha t, \tag{28}$$

with $\gamma = -s\alpha$ and $\delta - \beta = s\alpha^2$. We first establish a parity and Euclidean-division proposition that governs the behavior of (28) under polynomial division by $(t^2 + 1)^2$.

Proposition 3.1 (Joint Parity Theorem and Exact Quotient/Remainder Decomposition)**.** *Let $F(t) = d(t)^4 - b(t)^4 - c(t)^4 + s^2 a(t)^4$ for the polynomials in (28).*

1. ***Joint $(t, \text{parameter})$-Parity Invariance:*** *Under the involution $(t, \alpha, \beta, \gamma, \delta) \mapsto (-t, -\alpha, -\beta, -\gamma, -\delta)$, all four polynomials $a(t), b(t), c(t), d(t)$ are strictly invariant (even). Consequently, in $F(t)$, every coefficient of an odd power of t (t^1, t^3, t^5, t^7) has odd total degree in $(\alpha, \beta, \gamma, \delta)$, while every coefficient of an even power of t (t^0, t^2, t^4, t^6, t^8) has even total degree in $(\alpha, \beta, \gamma, \delta)$. In particular, the inhomogeneous relation $\delta - \beta = s\alpha^2$ mixes degree 1 in (δ, β) with degree 2 in α.*

2. ***Euclidean Division by $(t^2 + 1)^2$:*** *Dividing $F(t)$ by $(t^2 + 1)^2$ in $\mathbb{Q}[s, \alpha, \beta, \gamma, \delta][t]$ yields $F(t) = (t^2 + 1)^2 Q_{\text{quo}}(t) + R_{\text{rem}}(t)$, where under $\gamma = -s\alpha$ and $\delta = \beta + s\alpha^2$, the quotient $Q_{\text{quo}}(t) = \sum_{j=0}^{4} A_j t^j$ has coefficients*

$$\begin{aligned} A_4 &= s^2 - 1, \qquad A_3 = 4s\alpha(\alpha + s + 1), \qquad A_2 = 2\big(3s^2\alpha^4 + 6s\alpha^2\beta + s^2 + 3\big), \\ A_1 &= 4\big(s^3\alpha^6 + 3s^2\alpha^4\beta + s^2(s+1)\alpha^3 + 3s\alpha^2\beta^2 + s\alpha^2 + s(s+1)\alpha + 6\beta\big), \\ A_0 &= s^4\alpha^8 + 4s^3\alpha^6\beta + 6s^2\alpha^4\beta^2 - s^2(s^2-1)\alpha^4 + 4s\alpha^2\beta^3 + 24\beta^2 + s^2 - 17, \end{aligned} \tag{29}$$

   and the exact remainder is

$$\begin{aligned} R_{\text{rem}}(t) = &-(2t^2+1)\big(A_0 - (s^2-1)\big) - 4t(t^2+1)\big(s^3\alpha^6 + 3s^2\alpha^4\beta + s^2(s+1)\alpha^3 + 3s\alpha^2\beta^2\big) \\ &+ 8\beta t^3(\beta^2 - 6) - 16\beta t. \end{aligned}$$

3. ***Discriminant-Zero Locus of (A_4, A_2, A_0) Across Regimes:*** *For the even-degree biquadratic coefficients (A_4, A_2, A_0) in (29), the degree-4 term in β cancels identically in $\Delta_{\text{quo}}(\alpha, \beta, s) = A_2^2 - 4A_4A_0$:*

$$\begin{aligned} -\frac{1}{4}\Delta_{\text{quo}}(\alpha, \beta, s) = &\, 4s(s^2-1)\alpha^2\beta^3 + 6\big(s^2(s^2-7)\alpha^4 + 4(s^2-1)\big)\beta^2 \\ &+ 4s\alpha^2\big(s^2(s^2-10)\alpha^4 - 3(s^2+3)\big)\beta \\ &+ s^4(s^2-10)\alpha^8 - s^2(s^4+4s^2+19)\alpha^4 - 24s^2 + 8 = 0. \end{aligned} \tag{30}$$

   This locus separates into two distinct algebraic regimes:

- Degenerate Quadratic Regime ($s=\pm1$): *When $s=\pm1$, we have $A_4=s^2-1=0$, so the β^3 coefficient vanishes and $\Delta_{\mathrm{quo}}(\alpha,\beta,\pm1)=A_2^2=4(3\alpha^4\pm6\alpha^2\beta+4)^2=0$ collapses to a doubled rational curve $\beta=\mp\frac{3\alpha^4+4}{6\alpha^2}$.*
- Generic Cubic Regime ($s\in\mathbb{Q}^\times\setminus\{\pm1\}$): *When $s\neq0,\pm1$ and $\alpha\neq0$, equation (30) has non-zero leading coefficient $4s(s^2-1)\alpha^2$ in β and defines an irreducible cubic polynomial in $\mathbb{Q}(s,\alpha)[\beta]$.*

Proof. (1) Under $\iota:(t,\alpha,\beta,\gamma,\delta)\mapsto(-t,-\alpha,-\beta,-\gamma,-\delta)$, each linear term $\alpha t,\beta t,\gamma t,\delta t$ is invariant while $t^2\pm1$ is even in t. Hence $\iota^*F(t)=F(t)$. Writing $F(t)=\sum_{r=0}^{8}f_r(\alpha,\beta,\gamma,\delta)t^r$, invariance under ι means $f_r(-\alpha,-\beta,-\gamma,-\delta)(-1)^r=f_r(\alpha,\beta,\gamma,\delta)$, so f_r has odd total degree in $(\alpha,\beta,\gamma,\delta)$ when r is odd and even total degree when r is even.

(2) Write $E=t^2+1$, so $t^2-1=E-2$. Then $a(t)=E+\alpha t$, $c(t)=E-s\alpha t$, $b(t)=E+\beta t-2$, and $d(t)=E+\delta t$. Expanding $F(t)$ in powers of $E=t^2+1$ and replacing $t^2=E-1$ in all terms of degree <2 in E yields the Euclidean division $F(t)=E^2Q_{\mathrm{quo}}(t)+R_{\mathrm{rem}}(t)$, where substituting $\delta=\beta+s\alpha^2$ produces (29) and the stated remainder $R_{\mathrm{rem}}(t)$.

(3) In $A_2=2(3s^2\alpha^4+6s\alpha^2\beta+s^2+3)$, the only term involving β is $12s\alpha^2\beta$, so A_2^2 has no β^4 or β^3 terms and its β^2 coefficient is $144s^2\alpha^4\beta^2$. Meanwhile, $A_4=s^2-1$ is independent of β and A_0 is cubic in β with leading term $4s\alpha^2\beta^3$. Therefore $\Delta_{\mathrm{quo}}=A_2^2-4A_4A_0$ has leading term $-16s(s^2-1)\alpha^2\beta^3$; dividing by -4 gives (30). When $s=\pm1$, $A_4=0$ and $\Delta_{\mathrm{quo}}=A_2^2=4(3\alpha^4\pm6\alpha^2\beta+4)^2$. When $s\neq0,\pm1$, to prove irreducibility of (30) in $\mathbb{Q}(s,\alpha)[\beta]$ by Gauss's Lemma over the UFD $R=\mathbb{Q}[s,\alpha]$, set $X=s\alpha^2\beta$. Multiplying (30) by $s^2\alpha^4$ yields the cubic polynomial in $R[X]$:

$$\begin{aligned}P_{\mathrm{cub}}(X;s,\alpha)=4(s^2-1)X^3+6\big(s^2(s^2-7)\alpha^4+4(s^2-1)\big)X^2+4s^2\alpha^4\big(s^2(s^2-10)\alpha^4-3(s^2+3)\big)X\\+s^6(s^2-10)\alpha^{12}-s^4(s^4+4s^2+19)\alpha^8-8s^2(3s^2-1)\alpha^4.\end{aligned}$$

If $P_{\mathrm{cub}}(X;s,\alpha)$ were reducible in $\mathbb{Q}(s,\alpha)[X]$, then by Gauss's Lemma it would factor in $\mathbb{Q}[s,\alpha][X]$ as $L(X;s,\alpha)Q(X;s,\alpha)$ with $\deg_X(L)=1$ and $\deg_X(Q)=2$, where the leading coefficients in X multiply to $4(s^2-1)$. Under the evaluation homomorphism $(s,\alpha)=(2,1)$, the leading coefficient $4(s^2-1)$ evaluates to $12\neq0$, so both $L(X;2,1)$ and $Q(X;2,1)$ retain their exact positive degrees 1 and 2 in $\mathbb{Q}[X]$. However, at $(s,\alpha)=(2,1)$ (where $X=2\beta$), $P_{\mathrm{cub}}(2\beta;2,1)=4(24\beta^3-108\beta^2-280\beta-360)=8(12\beta^3-54\beta^2-140\beta-180)$, which has no rational roots by the rational root test and is therefore irreducible in $\mathbb{Q}[\beta]$. Hence P_{cub} is irreducible over $\mathbb{Q}(s,\alpha)$, and for every fixed $s\in\mathbb{Q}^\times\setminus\{\pm1\}$ the polynomial (30) is irreducible in $\mathbb{Q}(\alpha)[\beta]$ (since $(s,\alpha)\mapsto(s,1)$ or $(2,1)$ preserves degree 3 in β). □

3.2 The Exact Reciprocal (t^2+1,t)-Pencil and Its Factored Discriminant Locus

Next, we establish the exact reciprocal quadratic pencil where $(t^2+1)^2$ divides $F(t)$ with **zero remainder** and the odd coefficients t^3 and t^1 of the quotient $Q(t)=F(t)/(t^2+1)^2$ are identical and vanish simultaneously.

Theorem 3.2 (Exact Reciprocal Biquadratic Pencil and Factored Discriminant Locus)**.** *Let $E=t^2+1$, and consider the 2-point reciprocal quadratic pencil on $S_n:na^4+b^4+c^4=d^4$:*

$$a(t)=u_a(t^2+1)+\alpha t,\quad b(t)=u_b(t^2+1)+\beta t,\quad c(t)=u_c(t^2+1)+\gamma t,\quad d(t)=u_d(t^2+1)+\delta t. \tag{31}$$

Then $F(t)=d(t)^4-b(t)^4-c(t)^4+na(t)^4=\sum_{k=0}^{4}C_k(t^2+1)^kt^{4-k}$, where

$$\begin{aligned}&C_4=u_d^4-u_b^4-u_c^4+nu_a^4, && C_3=4\big(u_d^3\delta-u_b^3\beta-u_c^3\gamma+nu_a^3\alpha\big),\\&C_2=6\big(u_d^2\delta^2-u_b^2\beta^2-u_c^2\gamma^2+nu_a^2\alpha^2\big), && C_1=4\big(u_d\delta^3-u_b\beta^3-u_c\gamma^3+nu_a\alpha^3\big),\\&C_0=\delta^4-\beta^4-\gamma^4+n\alpha^4.\end{aligned} \tag{32}$$

1. ***Exact Divisibility by*** $(t^2+1)^2$***:*** *The polynomial $F(t)$ is divisible by $(t^2+1)^2$ in $\mathbb{Q}[t]$ if and only if $C_0=0$ and $C_1=0$, i.e., $P_0=(\alpha:\beta:\gamma:\delta)\in S_n$ and $P_\infty=(u_a:u_b:u_c:u_d)\in T_{P_0}(S_n)$. When $C_0=C_1=0$, the exact quotient $Q(t)=F(t)/(t^2+1)^2$ is*

$$Q(t)=C_4(t^2+1)^2+C_3t(t^2+1)+C_2t^2=C_4t^4+C_3t^3+(2C_4+C_2)t^2+C_3t+C_4. \tag{33}$$

In particular, the t^3 and t^1 coefficients of $Q(t)$ are ***identical*** *(both equal to C_3), so the single linear condition $C_3=0$ (meaning $P_0\in T_{P_\infty}(S_n)$) annihilates both odd powers simultaneously and leaves the pure reciprocal biquadratic $Q(t)=A_4t^4+A_2t^2+A_0$ with $A_4=A_0=C_4$ and $A_2=2C_4+C_2$.*

2. ***Complete Factorization of the Discriminant-Zero Locus:*** *Because $A_4=A_0=C_4$ and $A_2=2C_4+C_2$, the discriminant $\Delta_Q=A_2^2-4A_4A_0$ factors identically into two components over $\mathbb{Q}$:*

$$\Delta_Q=(2C_4+C_2)^2-4C_4^2=C_2\big(C_2+4C_4\big)=0. \tag{34}$$

   - ***Component I*** *($C_2=0$)**:** Here P_∞ lies on the asymptotic inflection tangent to S_n at P_0 ($C_0=C_1=C_2=0$). Then $Q(t)=C_4(t^2+1)^2$ is a constant multiple of the square $(t^2+1)^2$, and setting $C_3=0$ recovers the principal asymptotic tangent locus of Section 1.*
   - ***Component II*** *($C_2+4C_4=0$)**:** Here $Q(t)=C_4(t^2-1)^2$ has double roots at $t=\pm1$, corresponding to a bitangent conic touching S_n at $t=\pm i$ and $t=\pm1$.*

Proof. Expanding $(u_xE+xt)^4=\sum_{k=0}^4\binom{4}{k}u_x^kx^{4-k}E^kt^{4-k}$ for each coordinate and summing with weights $(n,-1,-1,1)$ gives $F(t)=\sum_{k=0}^4C_kE^kt^{4-k}$ with C_k as in (32).

(1) Since $\gcd(E,t)=\gcd(t^2+1,t)=1$ in $\mathbb{Q}[t]$, E^2 divides $F(t)=E^2(C_4E^2+C_3Et+C_2t^2)+C_1Et^3+C_0t^4$ if and only if E^2 divides $C_1Et^3+C_0t^4$. Evaluating at $t=\pm i$ (where $E(i)=0$ and $t^4=1\neq0$) forces $C_0=0$; dividing the remaining term C_1Et^3 by E and evaluating again at $t=\pm i$ (where $t^3=\mp i\neq0$) forces $C_1=0$. When $C_0=C_1=0$, dividing $F(t)$ by $E^2=(t^2+1)^2$ leaves $Q(t)=C_4(t^2+1)^2+C_3t(t^2+1)+C_2t^2$, which expands to (33).

(2) Setting $C_3=0$ in (33) leaves $Q(t)=C_4t^4+(2C_4+C_2)t^2+C_4$, so $A_4=A_0=C_4$ and $A_2=2C_4+C_2$. Expanding $\Delta_Q=A_2^2-4A_4A_0=(2C_4+C_2)^2-4C_4^2=4C_4^2+4C_4C_2+C_2^2-4C_4^2=C_2(C_2+4C_4)$ gives (34). When $C_2=0$, $Q(t)=C_4(t^4+2t^2+1)=C_4(t^2+1)^2$; when $C_2=-4C_4$, $Q(t)=C_4(t^4-2t^2+1)=C_4(t^2-1)^2$. □

3.3 The Two-Factor Conic-Square Biquadratic Pencil and the Universal Conic $\beta^2-s\alpha^2=4$

We now show how the substitution $\gamma=-s\alpha$ generates a common biquadratic divisor $M_4(t)$ between $d(t)^2+b(t)^2$ and $c(t)^2+sa(t)^2$ for every rational $s\in\mathbb{Q}^\times$.

Theorem 3.3 (The Common Biquadratic Factor Pencil on S_{-s^2})**.** *On $S_{-s^2}:(d^2+b^2)(d^2-b^2)=(c^2+sa^2)(c^2-sa^2)$, consider the four quadratic polynomials in t:*

$$a(t)=(t^2+1)+\alpha t,\qquad c(t)=(t^2+1)-s\alpha t,\qquad b(t)=w(t^2-1)+\beta t,\qquad d(t)=(t^2-1)-w\beta t. \tag{35}$$

1. ***Simultaneous Cancellation of Odd Powers in*** c^2+sa^2 ***and*** d^2+b^2***:*** *The choice $\gamma=-s\alpha$ in $c(t)$ and $\delta=-w\beta$ in $d(t)$ cancels the t^3 and t^1 coefficients of both $c(t)^2+sa(t)^2$ and $d(t)^2+b(t)^2$ identically:*

$$\begin{aligned}c(t)^2+sa(t)^2&=(s+1)\Big[(t^2+1)^2+s\alpha^2t^2\Big],\\ d(t)^2+b(t)^2&=(w^2+1)\Big[(t^2-1)^2+\beta^2t^2\Big]=(w^2+1)\Big[(t^2+1)^2+(\beta^2-4)t^2\Big].\end{aligned} \tag{36}$$

2. ***Universal Rational Conic*** $\beta^2 - s\alpha^2 = 4$***:*** *The two quadratic combinations in (36) share the exact degree-4 biquadratic factor $M_4(t) = (t^2+1)^2 + s\alpha^2t^2 = (t^2-1)^2 + \beta^2t^2$ if and only if (α, β) lies on the conic $\beta^2 - s\alpha^2 = 4$. For every rational $s \in \mathbb{Q}^\times$ (including $s = 2, 3, 5, \dots$), this conic contains the rational base points $(\alpha, \beta) = (0, \pm 2)$ and is rationally parametrized over $\mathbb{Q}(u)$ by*

$$\alpha(u) = \frac{4u}{u^2 - s}, \qquad \beta(u) = \frac{2(u^2+s)}{u^2-s}. \tag{37}$$

3. ***Exact Quartic Quotient*** $Q_4(t)$***:*** *When $\beta^2 - s\alpha^2 = 4$, $M_4(t)$ divides $F(t) = d(t)^4 - b(t)^4 - c(t)^4 + s^2a(t)^4$ with zero remainder, leaving the degree-4 quotient $Q_4(t) = F(t)/M_4(t) = (w^2+1)(d^2-b^2) - (s+1)(c^2-sa^2)$:*

$$\begin{aligned} Q_4(t) = {} & (s^2 - w^4)t^4 + 4\big(s(s+1)\alpha - w(w^2+1)\beta\big)t^3 \\ & + \big((w^4-1)\beta^2 - s(s^2-1)\alpha^2 + 2s^2 + 2w^4 - 4\big)t^2 \\ & + 4\big(s(s+1)\alpha + w(w^2+1)\beta\big)t + (s^2 - w^4). \end{aligned} \tag{38}$$

In particular, when $s = w^2$ is a rational square (so $n = -w^4 \equiv -1 \pmod{\mathbb{Q}^{\times 4}}$), both the t^4 and t^0 coefficients $s^2 - w^4$ of $Q_4(t)$ vanish identically, reducing $Q_4(t) = 0$ to a quadratic equation in t.

Proof. (1) Expanding $c(t)^2 + sa(t)^2$ from (35) gives

$$\big((t^2+1)^2 - 2s\alpha t(t^2+1) + s^2\alpha^2t^2\big) + s\big((t^2+1)^2 + 2\alpha t(t^2+1) + \alpha^2t^2\big) = (s+1)\Big[(t^2+1)^2 + s\alpha^2t^2\Big].$$

Similarly, expanding $d(t)^2 + b(t)^2$ gives

$$\big((t^2-1)^2 - 2w\beta t(t^2-1) + w^2\beta^2t^2\big) + \big(w^2(t^2-1)^2 + 2w\beta t(t^2-1) + \beta^2t^2\big) = (w^2+1)\Big[(t^2-1)^2 + \beta^2t^2\Big],$$

and using $(t^2-1)^2 = (t^2+1)^2 - 4t^2$ yields (36).

(2) Comparing the brackets in (36), $(t^2+1)^2 + s\alpha^2t^2 = (t^2+1)^2 + (\beta^2-4)t^2$ holds identically in t if and only if $s\alpha^2 = \beta^2 - 4$, i.e., $\beta^2 - s\alpha^2 = 4$. Passing the line $\beta + 2 = u\alpha$ through the base point $(\alpha_0, \beta_0) = (0, -2)$ and substituting $\beta = u\alpha - 2$ into $\beta^2 - s\alpha^2 = 4$ gives $\alpha\big((u^2-s)\alpha - 4u\big) = 0$, whose non-zero root is $\alpha(u) = \frac{4u}{u^2-s}$ and $\beta(u) = u\alpha(u) - 2 = \frac{2(u^2+s)}{u^2-s}$, proving (37).

(3) Write $F(t) = (d^2+b^2)(d^2-b^2) - (c^2+sa^2)(c^2-sa^2)$. When $\beta^2 - s\alpha^2 = 4$, both $d^2+b^2 = (w^2+1)M_4(t)$ and $c^2+sa^2 = (s+1)M_4(t)$ are multiples of $M_4(t)$, so $F(t) = M_4(t)\,Q_4(t)$ where $Q_4(t) = (w^2+1)(d^2-b^2) - (s+1)(c^2-sa^2)$. Expanding $d^2-b^2 = (1-w^2)(t^2-1)^2 - 4w\beta t(t^2-1) + (w^2-1)\beta^2t^2$ and $c^2 - sa^2 = (1-s)(t^2+1)^2 - 2s(s+1)\alpha t(t^2+1) + s(s-1)\alpha^2t^2$ and collecting powers of t gives (38). When $s = w^2$, $s^2 - w^4 = 0$, so the t^4 and t^0 terms vanish and $Q_4(t) = t \cdot (\text{quadratic in } t)$. □

4 Higher-Degree Undetermined-Coefficient Elimination and Master T-Polynomial Classification

We now prove three obstruction and classification theorems governing polynomial parametrizations at higher degrees ($m \geq 4$), establishing the Fermat–Eckardt Divisibility Law, the UFD Binomial-Term Cancellation Obstruction, and the Vepir–Elkies Master T-Polynomial Classification Theorem.

4.1 The Fermat–Eckardt Divisibility Theorem and UFD Cancellation Obstruction

Theorem 4.1 (The Fermat–Eckardt Divisibility Theorem and UFD Binomial Obstruction)**.** *Let $b(p,q), c(p,q), d(p,q) \in \mathbb{Z}[p,q]$ be homogeneous polynomials of the same degree $m \geq 1$ with $\gcd(b,c,d) = 1$, and suppose $n(p,q)a(p,q)^4 + b(p,q)^4 + c(p,q)^4 = d(p,q)^4$ holds identically in $\mathbb{Q}[p,q]$ for non-zero homogeneous polynomials $n(p,q), a(p,q) \in \mathbb{Z}[p,q]$.*

1. ***Fermat–Eckardt Divisibility Law:*** *Every rational linear factor $L(p,q) = up - vq$ ($\gcd(u,v) = 1$) dividing $n(p,q)a(p,q)^4$ over $\mathbb{Q}$ must also divide the product $b(p,q)\,c(p,q)$. Specifically, at the root $(p_0 : q_0) = (v : u)$, we have $n(p_0,q_0)a(p_0,q_0)^4 = 0 \implies b(p_0,q_0)^4 + c(p_0,q_0)^4 = d(p_0,q_0)^4$. By Fermat's Last Theorem for exponent 4, the only rational points on $B^4 + C^4 = D^4$ in $\mathbb{P}^2(\mathbb{Q})$ are $(1:0:\pm1)$ and $(0:1:\pm1)$, so $(b(p_0,q_0) : c(p_0,q_0) : d(p_0,q_0))$ must lie at one of the four rational Eckardt planes, forcing $L(p,q) \mid b(p,q)$ (with $L \mid d \mp c$) or $L(p,q) \mid c(p,q)$ (with $L \mid d \mp b$).*

2. ***UFD Binomial-Term Cancellation Obstruction:*** *Write $d = X + Y$ and $b = X - Y$ with $X = \frac{d+b}{2}, Y = \frac{d-b}{2} \in \mathbb{Q}[p,q]$, so that*

$$d^4 - b^4 - c^4 = 8XY(X^2 + Y^2) - c^4 = 8X^3Y + 8XY^3 - c^4 = n(p,q)a(p,q)^4.$$

If $c(p,q)^4$ cancels either of the two additive terms $8X^3Y$ or $8XY^3$ identically in $\mathbb{Q}[p,q]$ (say $c^4 = 8XY^3$), then $\gcd(b,c,d) = 1$ in $\mathbb{Q}[p,q]$ implies $\gcd(X,Y) = 1$ in $\mathbb{Q}[p,q]$, and unique factorization in $\mathbb{Q}[p,q]$ forces $4 \mid m$, $X = 2\kappa_2 U(p,q)^4$, and $Y = \kappa_2 V(p,q)^4$ for coprime homogeneous polynomials $U, V \in \mathbb{Q}[p,q]$ of degree $m/4$ and $\kappa_2 \in \mathbb{Q}^\times$. This automatically forces the remaining term to be

$$d^4 - b^4 - c^4 = 8X^3Y = 4\big(2\kappa_2 U^3 V\big)^4,$$

so $n(p,q) = 4\big(\frac{2\kappa_2 U^3 V}{a(p,q)}\big)^4 \in 4\mathbb{Q}(p,q)^{\times 4}$. In particular, no constant $n \not\equiv 4 \pmod{\mathbb{Q}^{\times 4}}$ (when $\deg(a) = m$, recovering Carmichael's family [3] for $n = 4$) and no square-free non-constant polynomial $n(p,q)$ can arise from single-term cancellation $c^4 = 8XY^3$.

Proof. (1) Let $L(p,q) = up - vq$ be an irreducible linear polynomial in $\mathbb{Q}[p,q]$ dividing $n(p,q)a(p,q)^4$, with $\gcd(u,v) = 1$. Evaluating the homogeneous identity $n(p,q)a(p,q)^4 + b(p,q)^4 + c(p,q)^4 = d(p,q)^4$ at $(p_0,q_0) = (v,u) \in \mathbb{Z}^2 \setminus \{(0,0)\}$ gives $b(v,u)^4 + c(v,u)^4 = d(v,u)^4$ in $\mathbb{Z}$. Since $\gcd(b,c,d) = 1$ in $\mathbb{Q}[p,q]$, $L(p,q)$ cannot divide all three of $b(p,q), c(p,q), d(p,q)$, so $(b(v,u) : c(v,u) : d(v,u)) \in \mathbb{P}^2(\mathbb{Q})$ is a well-defined rational point on the Fermat quartic $B^4 + C^4 = D^4$. By Fermat's theorem on right triangles (Euler's proof for $N = 4$), $B^4 + C^4 = D^4$ has no solutions with $BC \neq 0$ in $\mathbb{Q}$. Therefore $b(v,u)\,c(v,u) = 0$. By the factor theorem for homogeneous polynomials in $\mathbb{Q}[p,q]$, $L(p,q) = up - vq$ divides $b(p,q)\,c(p,q)$. Since $L(p,q)$ is prime in the UFD $\mathbb{Q}[p,q]$, either $L \mid b$ (whereupon $c(v,u)^4 = d(v,u)^4 \implies L \mid (d-c)(d+c)$) or $L \mid c$ (whereupon $L \mid (d-b)(d+b)$).

(2) Suppose $d = X + Y$ and $b = X - Y$ with $c(p,q)^4 = 8XY^3$ in the UFD $\mathbb{Q}[p,q]$. Any irreducible factor dividing both X and Y divides $b = X - Y$, $d = X + Y$, and $c^4 = 8XY^3$, which is impossible since $\gcd(b,c,d) = 1$. Hence $\gcd(X,Y) = 1$ in $\mathbb{Q}[p,q]$. Since $\gcd(X,Y^3) = 1$ and $8XY^3 = c^4$, every irreducible factor $P(p,q) \in \mathbb{Q}[p,q]$ of X or Y must appear in X and in Y^3 with an exponent divisible by 4; since $\gcd(3,4) = 1$, the multiplicity of P in Y is also divisible by 4. Thus both X and Y are fourth powers up to scalars in $\mathbb{Q}^\times$: $X = \kappa_1 U(p,q)^4$ and $Y = \kappa_2 V(p,q)^4$ for coprime homogeneous polynomials $U, V \in \mathbb{Q}[p,q]$ of degree $m/4$ and $\kappa_1, \kappa_2 \in \mathbb{Q}^\times$. Then $8\kappa_1\kappa_2^3 U^4 V^{12} = c^4$ forces $8\kappa_1\kappa_2^3 \in \mathbb{Q}^{\times 4}$, so $\kappa_1\kappa_2^3 = 2r^4$ for some $r \in \mathbb{Q}^\times$, i.e., $\kappa_1 = 2(r/\kappa_2)^4\kappa_2$. Absorbing r/κ_2 into U gives $X = 2\kappa_2 U^4$ and $Y = \kappa_2 V^4$. Then the remaining term of $d^4 - b^4 - c^4$ is

$$8X^3Y = 8(2\kappa_2 U^4)^3(\kappa_2 V^4) = 64\kappa_2^4 U^{12} V^4 = 4\big(2\kappa_2 U^3 V\big)^4.$$

Hence $n(p,q)a(p,q)^4 = 4(2\kappa_2 U^3 V)^4$, which forces $n(p,q) = 4\big(\frac{2\kappa_2 U^3 V}{a(p,q)}\big)^4 \equiv 4 \pmod{\mathbb{Q}(p,q)^{\times 4}}$. In particular, if $\deg(a) = m$, then $\deg(2\kappa_2 U^3 V) = 4(m/4) = m = \deg(a)$ and $\gcd(a,b,c,d) = 1$ force $n(p,q) \in 4\mathbb{Q}^{\times 4}$ to be constant, while for any degree of $a(p,q)$, $n(p,q)$ cannot be a square-free non-constant polynomial. □

4.2 The Vepir–Elkies Master T-Polynomial Classification Theorem

Both Elkies' degree-8 family for $n = -12$ [6] and Vepir's degree-9 family for $n(p,q) = 20p^8 - q^8$ [16] arise from choosing X, Y, b so that all powers of p in $8XY(X^2 + Y^2) - b^4$ lie in a single arithmetic

progression of exponents. Specifically, setting $X(p,q) = p(up^{2k} + vq^{2k})$, $Y(p,q) = wp^kq^k \cdot p = wp^{k+1}q^k$, $d = X + Y$, $c = X - Y$, and $b(p,q) = q(mp^{2k} + Kq^{2k})$ (with $k = 4$ for degree 9, or $X = up^{2k} + vq^{2k}, Y = wp^kq^k$ with $k = 4$ for degree 8) gives the exact homogeneous identity

$$d(p,q)^4 - c(p,q)^4 - b(p,q)^4 = 8XY(X^2 + Y^2) - b^4 = q^{8k+4}\,P\big((p/q)^{2k}\big), \tag{39}$$

where $P(T)$ is the five-coefficient master polynomial in $T = (p/q)^{2k}$. Here we formulate and prove a complete classification theorem analyzing all coefficient-cancellation branches of $P(T)$ over $\mathbb{Q}$ and $\mathbb{C}$.

Theorem 4.2 (Classification of Maximal Coefficient Cancellation in the Vepir–Elkies Master Polynomial)**.** *Consider the master quartic polynomial in a single variable T (where $T = (p/q)^{2k}$):*

$$P(T) = 8wT(uT + v)\Big((uT + v)^2 + w^2T\Big) - (mT + K)^4 = \sum_{j=0}^{4} c_jT^j, \tag{40}$$

with $u, v, w, m \in \mathbb{Q}^\times$ and $K \in \mathbb{Q}$, where the five coefficients c_4, c_3, c_2, c_1, c_0 are

$$\begin{aligned} c_4 &= 8u^3w - m^4, & c_3 &= 4\big(6u^2vw + 2uw^3 - m^3K\big),\\ c_2 &= 2\big(12uv^2w + 4vw^3 - 3m^2K^2\big), & c_1 &= 4\big(2v^3w - mK^3\big), \qquad c_0 = -K^4. \end{aligned} \tag{41}$$

The polynomial $P(T)$ is invariant up to fourth-power scalars under the two-parameter scaling action $(u, v, w, m, K; T) \mapsto (\lambda\alpha^{-8}u,\ \lambda v,\ \lambda\alpha^{-4}w,\ \lambda\alpha^{-7}m,\ \lambda\alpha K;\ \alpha^8T)$, which multiplies $P(T)$ by $\lambda^4\alpha^4$ and transforms the ratio $K/w \mapsto \alpha^5(K/w)$. Dividing (u, v, w, m, K) by w normalizes $w = 1$ over $\mathbb{Q}$ without loss of generality. All solutions in $\mathbb{Q}^\times$ for which at least three of the four coefficients $\{c_4, c_3, c_2, c_1\}$ vanish simultaneously fall into the following exact classes:

1. ***Monomial-b Branch ($K = 0 \implies$ Elkies' Constant $n = -12$):*** *When $K = 0$, we have $c_0 = 0$, and c_3, c_2 share the common linear factor $(3uv + w^2)$:*

   $$c_3 = 8uw(3uv + w^2), \qquad c_2 = 8vw(3uv + w^2).$$

   Thus $c_3 = c_2 = 0 \iff w^2 = -3uv$ (the Difference-of-Two-Cubes Collapse $L^2 = 3XY$). Imposing $c_4 = 8u^3w - m^4 = 0$ then requires $(-3uv)m^4 = -24u^4vw$, which over $\mathbb{Q}$ has the unique normalized solution $(u, v, w, m, K) = (192, -1, 24, 192, 0)$, collapsing $P(T)$ to the single monomial $c_1T = -192T = -12 \cdot (2T^{1/4})^4$ and yielding $n \equiv -12 \pmod{\mathbb{Q}^{\times 4}}$ at degrees $8, 16, 24, \ldots$.

2. ***Binomial-b Branch ($K \neq 0$, normalized with $w = 1$):*** *Among the $\binom{4}{3} = 4$ three-element subsets of $\{c_4, c_3, c_2, c_1\}$ with $w = 1$ and $K \in \mathbb{Q}^\times$:*

   - ***Subcase A ($c_3 = c_2 = c_1 = 0 \implies$ Pure T^4-Binomial $P(T) = \frac{80}{27}T^4 - s^{20}$):*** *From $c_1 = 0$ we have $m = 2v^3/K^3$; substituting into $c_2 = 0$ gives $u = \frac{v^4}{K^4} - \frac{1}{3v}$, and substituting into $c_3 = 0$ yields $-\frac{8v^4(v^5+K^4)}{K^8} = 0$, i.e., $K^4 = -v^5$. Over $\mathbb{Q}^\times$, setting $s = -K/v \in \mathbb{Q}^\times$ parametrizes all rational solutions as $v = -s^4, K = s^5, m = -2s^{-3}, u = \frac{4}{3}s^{-4}$. Under the further normalization $K = 1$ ($s = 1$), the quintic $v^5 + 1 = (v+1)(v^4 - v^3 + v^2 - v + 1) = 0$ has the **unique** rational root $v = -1$, giving $(u, v, w, m, K) = (4/3, -1, 1, -2, 1)$ and the quartic binomial*

     $$P(T) = \frac{80}{27}T^4 - 1. \tag{42}$$

     Clearing denominators to $(u, v, w, m, K) = (4, -3, 3, -6, 3)$ in (39) with $T = p^8/q^8$ gives $d^4 - c^4 - b^4 = 3q^4(80p^{32} - 27q^{32})$ (or substituting $T = \frac{3p^2}{2q^2}$ into (42) and multiplying by the homogenizing factor $16q^8$ gives $16q^8P(\frac{3p^2}{2q^2}) = 16(15p^8 - q^8)$).

- ***Subcase B*** *($c_4 = c_3 = c_2 = 0 \implies$* ***Exclusion at Fixed*** *$K = 1$* ***and Vepir's Binomial*** *$20p^8 - q^8$* ***at*** *$K = -8/r^5$)**: With $w = 1$, the equation $c_4 = 8u^3 - m^4 = 0$ is rationally parametrized over $\mathbb{Q}^\times$ by*

$$u = \frac{r^4}{2}, \qquad m = r^3 \qquad (r \in \mathbb{Q}^\times), \tag{43}$$

since $8(r^4/2)^3 - (r^3)^4 = r^{12} - r^{12} = 0$. Substituting (43) into $c_3 = 0$ gives $v = \frac{2(Kr^5-1)}{3r^4}$, and substituting into $\frac{1}{2}c_2 = 0$ yields

$$\frac{1}{2}c_2 = -\frac{1}{3}Kr\big(Kr^5 + 8\big) = 0. \tag{44}$$

  (i) Fixed Normalization $K = 1$: *When $K = 1$ is fixed alongside $w = 1$, equation (44) becomes $-\frac{1}{3}r(r^5 + 8) = 0$, which has* ***no rational solution*** *$r \in \mathbb{Q}^\times$ since $r^5 = -8$ has no root in $\mathbb{Q}$. Moreover, over $\mathbb{C}$, any root of $r^5 = -8$ satisfies $r^{15} = -512$ and $v = -6/r^4$, giving $c_1 = 4(2v^3 - m) = -\frac{4(r^{15}+432)}{r^{12}} = \frac{320}{r^{12}} \neq 0$.*

  (ii) General Rational $K \in \mathbb{Q}^\times$ (Vepir's Degree-9 Binomial): *For general $K \in \mathbb{Q}^\times$, (44) has the unique rational solution $K = -8/r^5$ and $v = -6/r^4$, which reduces $P(T)$ to the linear binomial $P(T) = \frac{64(5r^8T-64)}{r^{20}}$. Choosing $r = 2$ (so $(u, v, w, m, K) = (8, -3/8, 1, 8, -1/4)$) and scaling by $\lambda = 8$ to clear denominators yields $(u, v, w, m, K) = (64, -3, 8, 64, -2)$ and $P(T) = 320T - 16 = 16(20T - 1)$. Substituting $T = p^8/q^8$ into (39) with $k = 4$ gives Vepir's exact 2-term binomial identity*

$$d(p,q)^4 - c(p,q)^4 - b(p,q)^4 = q^{36} \cdot 16\left(20\frac{p^8}{q^8} - 1\right) = \big(20p^8 - q^8\big)\big(2q^7\big)^4.$$

- ***Subcases C & D*** *($c_4 = c_3 = c_1 = 0$* ***and*** *$c_4 = c_2 = c_1 = 0$)**: Whenever both $c_1 = 0$ and $c_4 = 0$ hold simultaneously with $w = 1$ and $K \in \mathbb{Q}^\times$, $c_4 = 0 \implies u = r^4/2, m = r^3$ and $c_1 = 4(2v^3 - mK^3) = 0 \implies 2v^3 = r^3K^3 \implies (rK/v)^3 = 2$, which has* ***no rational solution*** *$r, v, K \in \mathbb{Q}^\times$ since $\sqrt[3]{2} \notin \mathbb{Q}$.*

Consequently, $c_4 = c_3 = c_2 = c_1 = 0$ has ***no solutions even over*** *$\mathbb{C}$ when $K \neq 0$: no constant-n family can arise from the binomial-b Vepir ansatz.*

Proof. Expanding $8wT(uT + v)^3 + 8w^3T^2(uT + v) - (mT + K)^4$ and collecting powers of T^4, T^3, T^2, T^1, T^0 yields (41).

(1) Suppose $K = 0$. Then $c_0 = 0$ and (41) simplifies to $c_4 = 8u^3w - m^4$, $c_3 = 8uw(3uv + w^2)$, $c_2 = 8vw(3uv + w^2)$, and $c_1 = 8v^3w$. Since $u, v, w \neq 0$, we have $c_1 \neq 0$, so vanishing of three of $\{c_4, c_3, c_2, c_1\}$ requires $c_4 = c_3 = c_2 = 0$. From $c_3 = c_2 = 0$ we obtain $w^2 = -3uv$, so $v = -w^2/(3u)$. Meanwhile, $c_4 = 0 \implies m^4 = 8u^3w = (2u)^3w$, which is a rational fourth power if and only if $w/(2u) = r^4$ for some $r \in \mathbb{Q}^\times$. Choosing the scale normalization $v = -1$ and $r = 1/2$ gives $w = u/8$ and $w^2 = 3u \implies u^2/64 = 3u \implies u = 192, w = 24$, and $m = 192$. Then $c_4 = c_3 = c_2 = c_0 = 0$ and $P(T) = c_1T = 8(-1)^3(24)T = -192T = -12(2pq^7)^4$ for $T = p^4/q^4$ after homogenizing by q^{32}.

(2) Suppose $K \neq 0$. Since every c_j in (41) is homogeneous of degree 4 in (u, v, w, m, K), dividing (u, v, w, m, K) by $w \in \mathbb{Q}^\times$ normalizes $w = 1$ over $\mathbb{Q}$ and divides $P(T)$ by w^4. With $w = 1$, (41) becomes

$$c_4 = 8u^3 - m^4, \qquad c_3 = 4(6u^2v + 2u - m^3K), \qquad c_2 = 2(12uv^2 + 4v - 3m^2K^2),$$
$$c_1 = 4(2v^3 - mK^3), \qquad c_0 = -K^4.$$

Subcase A ($c_3 = c_2 = c_1 = 0$): From $c_1 = 0$, we have $m = 2v^3/K^3$. Substituting $m = 2v^3/K^3$ into $c_2 = 0$ gives $12uv^2 + 4v - 12v^6/K^4 = 0$; since $v \neq 0$, dividing by $12v^2$ yields $u = \frac{v^4}{K^4} - \frac{1}{3v} = \frac{3v^5-K^4}{3K^4v}$.

Substituting $m=2v^3/K^3$ and $u=\frac{3v^5-K^4}{3K^4v}$ into $\frac{1}{4}c_3=6u^2v+2u-m^3K=0$ gives

$$6v\left(\frac{3v^5-K^4}{3K^4v}\right)^2+2\left(\frac{3v^5-K^4}{3K^4v}\right)-\frac{8v^9}{K^8}=-\frac{2v^4(v^5+K^4)}{K^8}=0.$$

Thus $v^5+K^4=0$. Setting $s=-K/v\in\mathbb{Q}^\times$ gives $v=-s^4$ and $K=s^5$. When normalized to $K=1$ ($s=1$), $v^5+1=(v+1)(v^4-v^3+v^2-v+1)=0$ (where the cyclotomic factor $\Phi_{10}(-v)=v^4-v^3+v^2-v+1$ is irreducible over $\mathbb{Q}$) has the unique rational root $v=-1$, yielding $m=2(-1)^3=-2$ and $u=1+\frac{1}{3}=\frac{4}{3}$. At $(u,v,w,m,K)=(4/3,-1,1,-2,1)$, we have $c_4=8(4/3)^3-(-2)^4=\frac{512}{27}-16=\frac{80}{27}$ and $c_3=c_2=c_1=0, c_0=-1$, so $P(T)=\frac{80}{27}T^4-1$, proving (42). Scaling (u,v,w,m,K) by 3 to $(4,-3,3,-6,3)$ gives $P(T)=240T^4-81$, and substituting $T=p^8/q^8$ into (39) with $k=4$ gives $d^4-c^4-b^4=3q^4(80p^{32}-27q^{32})$.

Subcase B ($c_4=c_3=c_2=0$): From $c_4=8u^3-m^4=0$ with $u,m\in\mathbb{Q}^\times$, we have $(2u)^3=m^4$. Setting $r=m/(2u)\in\mathbb{Q}^\times$ gives $m=2ur$, so $(2u)^3=(2u)^4r^4\implies 2u=r^{-4}$; replacing r by $1/r\in\mathbb{Q}^\times$ yields the exact rational parametrization $u=r^4/2$ and $m=r^3$ in (43), which satisfies $8u^3-m^4=8(r^{12}/8)-r^{12}=0$ identically. Substituting $u=r^4/2$ and $m=r^3$ into $\frac{1}{4}c_3=6u^2v+2u-m^3K=0$ gives

$$6\left(\frac{r^8}{4}\right)v+r^4-Kr^9=0\implies\frac{3}{2}r^8v=r^4(Kr^5-1)\implies v=\frac{2(Kr^5-1)}{3r^4}.$$

Substituting $u=r^4/2$, $m=r^3$, and $v=\frac{2(Kr^5-1)}{3r^4}$ into $\frac{1}{2}c_2=12uv^2+4v-3m^2K^2=0$ gives

$$6r^4\left(\frac{2(Kr^5-1)}{3r^4}\right)^2+4\left(\frac{2(Kr^5-1)}{3r^4}\right)-3K^2r^6=-\frac{Kr(Kr^5+8)}{3r^4}=0,$$

proving (44). If $K=1$ is fixed, (44) becomes $-\frac{1}{3}r(r^5+8)=0$, which has no rational solution $r\in\mathbb{Q}^\times$ since -8 is not a fifth power in $\mathbb{Q}$. Moreover, over $\mathbb{C}$, at any root of $Kr^5+8=0$ we have $Kr^5=-8$, so $v=\frac{2(-8-1)}{3r^4}=-\frac{6}{r^4}$ and $K=-\frac{8}{r^5}$, which gives

$$c_1=4(2v^3-mK^3)=4\left(2\left(-\frac{216}{r^{12}}\right)-r^3\left(-\frac{512}{r^{15}}\right)\right)=\frac{4(-432+512)}{r^{12}}=\frac{320}{r^{12}}\neq 0,$$

proving that $c_4=c_3=c_2=c_1=0$ has no solution in $\mathbb{C}$. For general $K\in\mathbb{Q}^\times$, $Kr^5+8=0$ gives the rational solution $K=-8/r^5$ and $v=-6/r^4$, where $P(T)=c_1T+c_0=\frac{320}{r^{12}}T-\frac{4096}{r^{20}}=\frac{64(5r^8T-64)}{r^{20}}$. At $r=2$, multiplying $(u,v,w,m,K)=(8,-3/8,1,8,-1/4)$ by $\lambda=8$ gives $(u,v,w,m,K)=(64,-3,8,64,-2)$ and $P(T)=320T-16=16(20T-1)$, which via (39) at $k=4$ ($T=p^8/q^8$) produces $(20p^8-q^8)(2q^7)^4$.

Subcases C & D ($c_4=c_3=c_1=0$ and $c_4=c_2=c_1=0$): In both remaining subsets, both $c_4=0$ and $c_1=0$ hold. From $c_4=8u^3-m^4=0$ we have $u=r^4/2$ and $m=r^3$ ($r\in\mathbb{Q}^\times$). Substituting $m=r^3$ into $c_1=4(2v^3-mK^3)=0$ gives $2v^3=r^3K^3$, so $(rK/v)^3=2$. Since 2 is not a cube in $\mathbb{Q}$ and $v\neq 0$, neither subset has any solution in $\mathbb{Q}^\times$. This exhausts all four 3-element subsets of $\{c_4,c_3,c_2,c_1\}$. □

4.3 Universal Exact Binomial and Trinomial $n(p,q)$ Engines

To construct exact, zero-remainder polynomial parametrizations for target binomials and trinomials across arbitrary degrees $m\geq 4$, we establish the following universal engines.

Theorem 4.3 (Universal Exact Binomial and Cyclotomic $n(p,q)$ Engines)**.**

1. ***Universal (U,V)-Binomial Engine for Any Polynomials $U(p,q),V(p,q)$:*** *For any coprime homogeneous polynomials $U(p,q),V(p,q)\in\mathbb{Z}[p,q]$ of the same degree $k\geq 1$, setting*

$$a(p,q)=2UV,\qquad b(p,q)=tU^3V,\qquad c(p,q)=hU^4-V^4,\qquad d(p,q)=hU^4+V^4 \tag{45}$$

(where $\deg(a) = 2k$ *and* $\deg(b) = \deg(c) = \deg(d) = 4k$*) satisfies* $n(p,q)a(p,q)^4 + b(p,q)^4 + c(p,q)^4 = d(p,q)^4$ *identically with zero remainder, where* $n(p,q)$ *is the exact 2-term binomial in* (U^8, V^8)*:*

$$n(p,q) = \left(\frac{8h^3 - t^4}{16}\right) U(p,q)^8 + \frac{h}{2}\, V(p,q)^8. \tag{46}$$

When $\gcd(U,V) = 1$*,* $\gcd(h,t) = 1$*, and* $2 \mid h$ *with* t *odd (so that* c, d *are odd whenever* V *is odd, and dividing* (a,b,c,d) *by* $\gcd(a,b,c,d) \mid 2$ *removes any common factor at* p, q*),* (45)–(46) *yields primitive polynomial solutions* $\gcd(a,b,c,d) = 1$*. (More generally, replacing* $a(p,q)$ *by* $2U^uV^v$ *with integers* $0 \le u, v \le 1$ *yields the polynomial binomial* $n_{u,v}(p,q) = \frac{8h^3-t^4}{16}U^{12-4u}V^{4-4v} + \frac{h}{2}U^{4-4u}V^{12-4v}$*, whereas for* $u > 1$ *or* $v > 1$ *the rational expression* $n_{u,v}(p,q)$ *acquires a non-constant denominator in* $\mathbb{Q}(p,q)$ *since* $\gcd(U,V) = 1$*.) In particular, choosing* $U = p + q$ *and* $V = p - q$ *(or* $U = p^2 + q^2, V = p^2 - q^2$*) in* (45)–(46) *yields primitive polynomial parametrizations at degrees* $4k \in \{4, 8, 12, 16, \dots\}$*.*

2. ***Universal Difference-of-Two-Cubes Hierarchy:*** *For any coprime homogeneous polynomials* $U(p,q), V(p,q)$ *of degree* $k \ge 1$*, setting*

$$d(p,q) = 3U^2 + 6UV - 4V^2, \qquad b(p,q) = 3U^2 - 6UV - 4V^2 \tag{47}$$

gives the exact polynomial identity $d(p,q)^4 - b(p,q)^4 = 1296U^7V - 3072UV^7$*. Consequently:*

- Three-Term Trinomial Regime: *For any choice of* $c(p,q) = 6U^rV^s$ *and* $a(p,q) = 2U^uV^v$ *with integers* $u, v, r, s \ge 0$ *satisfying* $4u \le 1, 4v \le 1, u \le r, v \le s$*, we have* $d^4 - b^4 - c^4 = n(p,q)a(p,q)^4$ *with*

$$n(p,q) = 81U^{7-4u}V^{1-4v} - 192U^{1-4u}V^{7-4v} - 81U^{4r-4u}V^{4s-4v}. \tag{48}$$

More generally, if $U = U_0^k$ *and* $V = V_0^k$ *are* k*-th powers of coprime degree-1 polynomials* $U_0(p,q), V_0(p,q)$ *(*$k \ge 4$*) and we set* $c(p,q) = 6U_0^rV_0^s$ *and* $a(p,q) = 2U_0^uV_0^v$ *with* $0 \le 4u \le k$*,* $0 \le 4v \le k$*,* $u \le r$*,* $v \le s$*, then* $d^4 - b^4 - c^4 = n(p,q)a(p,q)^4$ *with the polynomial trinomial*

$$n(p,q) = 81U_0^{7k-4u}V_0^{k-4v} - 192U_0^{k-4u}V_0^{7k-4v} - 81U_0^{4r-4u}V_0^{4s-4v}. \tag{49}$$

- Collapsed Monomial and Constant $n = -12$ Regime: *When* $4 \mid k$ *and* $(r,s) = (7k/4, k/4)$*, the first term* $1296U_0^{7k}V_0^k$ *of* $d^4 - b^4$ *and* $c(p,q)^4 = 1296U_0^{7k}V_0^k$ *cancel identically prior to division by* $a(p,q)^4$*, leaving* $d(p,q)^4 - b(p,q)^4 - c(p,q)^4 = -3072U_0^kV_0^{7k}$*. In this collapsed regime, for any integers* $u, v \ge 0$ *in the broader domain* $0 \le 4u \le k$ *and* $0 \le 4v \le 7k$*, setting* $a(p,q) = 2U_0^uV_0^v$ *yields* $d^4 - b^4 - c^4 = n(p,q)a(p,q)^4$ *with the exact single-monomial polynomial*

$$n(p,q) = -192U_0^{k-4u}V_0^{7k-4v}. \tag{50}$$

In particular, specializing $(4u, 4v) = (k, 7k)$ *(i.e.,* $u = k/4$ *and* $v = 7k/4$*, which satisfies* $0 \le 4u \le k$ *and* $0 \le 4v \le 7k$*) gives* $n(p,q) = -192$*, or* $n = -12$ *upon absorbing* $2^4 = 16$ *into* $a(p,q) = 4U_0^{k/4}V_0^{7k/4}$ *of homogeneous degree* $2k$*.*

3. ***Exact Arbitrary-Degree Trinomial & Binomial Constructions (***$m \ge 4$***):*** *For any degree* $m \ge 4$*, setting* $a(p,q) = 2p^iq^j$*,* $b(p,q) = tp^uq^v$*,* $c(p,q) = hp^m - q^m$*, and* $d(p,q) = hp^m + q^m$ *with* $4i \le m, 4j \le m$ *and* $u \ge i, v \ge j$ *yields the exact polynomial identity* $n(p,q)a(p,q)^4 + b(p,q)^4 + c(p,q)^4 = d(p,q)^4$ *(with* $\gcd(b,c,d) = 1$ *in* $\mathbb{Q}[p,q]$ *whenever* $h, t \ne 0$ *and* (u,v) *has* $u > 0, v > 0$*) with*

$$n(p,q) = \frac{8h^3p^{3m-4i}q^{m-4j} + 8hp^{m-4i}q^{3m-4j} - t^4p^{4u-4i}q^{4v-4j}}{16},$$

which collapses from a 3-term trinomial to a 2-term binomial whenever $4 \mid m$ *and* $(4u, 4v) = (3m, m)$ *or* $(m, 3m)$*.*

Proof. (1) Expanding $d(p,q)^4-c(p,q)^4$ for $d=hU^4+V^4$ and $c=hU^4-V^4$ via $(A+B)^4-(A-B)^4=8AB(A^2+B^2)$ with $A=hU^4$ and $B=V^4$ gives

$$d(p,q)^4-c(p,q)^4=8hU^4V^4\big(h^2U^8+V^8\big)=8h^3U^{12}V^4+8hU^4V^{12}.$$

Subtracting $b(p,q)^4=(tU^3V)^4=t^4U^{12}V^4$ combines the two $U^{12}V^4$ terms into $(8h^3-t^4)U^{12}V^4+8hU^4V^{12}=16U^4V^4\big(\frac{8h^3-t^4}{16}U^8+\frac{h}{2}V^8\big)=n(p,q)\,(2UV)^4$, proving (45)–(46). Since $d+c=2hU^4$ and $d-c=2V^4$ with $\gcd(U,V)=1$ in $\mathbb{Q}[p,q]$, any irreducible polynomial dividing b,c,d must divide both U and V, which is impossible; thus $\gcd(b,c,d)=1$ in $\mathbb{Q}[p,q]$, and over $\mathbb{Z}[p,q]$ the integer content $\gcd(a,b,c,d)$ is 1 whenever $\gcd(h,t)=1$ and $2\mid h$ with V primitive. Similarly, dividing by $(2U^uV^v)^4=16U^{4u}V^{4v}$ yields $n_{u,v}(p,q)$, which is a polynomial in $\mathbb{Q}[p,q]$ if and only if $4u\le 4$ and $4v\le 4$.

(2) Write $d=Q+L$ and $b=Q-L$ where $Q=3U^2-4V^2$ and $L=6UV$. Since $L^2=36U^2V^2=3(3U^2)(4V^2)$, we have

$$Q^2+L^2=\big(9U^4-24U^2V^2+16V^4\big)+36U^2V^2=9U^4+12U^2V^2+16V^4,$$

and therefore $Q(Q^2+L^2)=(3U^2-4V^2)\big((3U^2)^2+(3U^2)(4V^2)+(4V^2)^2\big)=(3U^2)^3-(4V^2)^3=27U^6-64V^6$. Multiplying by $8L=48UV$ gives

$$d^4-b^4=8QL(Q^2+L^2)=48UV(27U^6-64V^6)=1296U^7V-3072UV^7.$$

Subtracting $c^4=1296U^{4r}V^{4s}$ and dividing by $a^4=16U^{4u}V^{4v}$ yields (48), and substituting $U=U_0^k, V=V_0^k, c=6U_0^rV_0^s$, and $a=2U_0^uV_0^v$ yields $d^4-b^4-c^4=1296U_0^{7k}V_0^k-3072U_0^kV_0^{7k}-1296U_0^{4r}V_0^{4s}$, which divides by $16U_0^{4u}V_0^{4v}$ to give (49) when $4u\le k, 4v\le k, u\le r, v\le s$. When $4\mid k$ and $(r,s)=(7k/4,k/4)$, the first and third terms of $d^4-b^4-c^4$ cancel identically to leave $-3072U_0^kV_0^{7k}$, which is divisible in $\mathbb{Q}[p,q]$ by $a^4=16U_0^{4u}V_0^{4v}$ whenever $0\le 4u\le k$ and $0\le 4v\le 7k$, yielding (50).

(3) Expanding $(hp^m+q^m)^4-(hp^m-q^m)^4-(tp^uq^v)^4=8h^3p^{3m}q^m+8hp^mq^{3m}-t^4p^{4u}q^{4v}$ and dividing by $(2p^iq^j)^4=16p^{4i}q^{4j}$ produces the stated formula for $n(p,q)$, which is a polynomial in $\mathbb{Q}[p,q]$ since $m\ge 4i, m\ge 4j, u\ge i, v\ge j$. □

Table 2: Verified Zero-Remainder Binomial, Trinomial & Monomial Parametrizations $n(p,q)a^4+b^4+c^4=d^4$ Across Degrees 4 to 16.

Max Deg	**Exact** $n(p,q)$	$a(p,q)$	$b(p,q)$	$c(p,q)$	$d(p,q)$
4	$\frac{1}{2}(q^8-p^8)$	$2pq$	$2p^3q$	p^4-q^4	p^4+q^4
4	4 (constant)	$2pq^3$	$8p^3q$	$8p^4-q^4$	$8p^4+q^4$
5	$p(4p^{10}q-p^{11}+q^{11})$	$2pq$	$2p^4q$	$2p^5-q^5$	$2p^5+q^5$
6	$p^2(4p^{12}q^2-p^{14}+q^{14})$	$2pq$	$2p^5q$	$2p^6-q^6$	$2p^6+q^6$
7	$p^3(4p^{14}q^3-p^{17}+q^{17})$	$2pq$	$2p^6q$	$2p^7-q^7$	$2p^7+q^7$
8	-12 (constant)	$2pq^7$	$192p^7q$	$192p^8-24p^4q^4-q^8$	$192p^8+24p^4q^4-q^8$
8	4 (Family 4-I)	$2rs(r^6+5r^4s^2+4r^2s^4+2s^6)$	$r^8+6r^6s^2-12r^2s^6-4s^8$	$2rs(r^6+4r^4s^2+10r^2s^4+4s^6)$	$r^8+6r^6s^2+20r^4s^4+12r^2s^6+4s^8$
9	$20p^8-q^8$ (Vepir)	$2q^7$	$2q(32p^8-q^8)$	$p(64p^8-8p^4q^4-3q^8)$	$p(64p^8+8p^4q^4-3q^8)$
12	$3U^{24}+V^{24}$	$2U^3V^3$	$2U^9V^3$	$2U^{12}-V^{12}$	$2U^{12}+V^{12}$
15	$q^3(p^{33}+4p^3q^{30}-q^{33})$	$2p^3q^3$	$2p^3q^{12}$	$p^{15}-2q^{15}$	$p^{15}+2q^{15}$
16	4 (U,V deg 4)	$2UV^3$	$8U^3V$	$8U^4-V^4$	$8U^4+V^4$

5 Iterated Asymptotic Dynamics and the Infinite Master-Polynomial Tower on S_{-1}

Having reduced the principal asymptotic tangent self-map on S_{-s^2} from degree 17 to the canonical degree-9 self-map $\mathcal{A}_9^{\pm}$ in Section 1, we now establish the exact dynamical degree law of its second iterates $\mathcal{A}_9^{\mp}\circ\mathcal{A}_9^{\pm}$ and $\mathcal{A}_9^{\pm}\circ\mathcal{A}_9^{\pm}$ on S_{-s^2}, and we determine its explicit action on Euler's degree-7 rational curve $C_{\text{eul}}\subset S_{-1}$ alongside the complete infinite dihedral tower of single-polynomial families on S_{-1}.

5.1 Segre Factorization and Exact Degree-49 Reduction of the Alternating Iterate $\mathcal{A}_9^{\mp} \circ \mathcal{A}_9^{\pm}$

Recall from Theorem 1.1 that $\mathcal{A}_9^{\pm}(a_1, b_1, c_1, d_1) = \big(a_1 L_a^{\pm}, b_1 L_b^{\pm}, c_1 L_c^{\pm}, d_1 L_d^{\pm}\big)$, where $L_a^{\pm} = S^{\pm} - 4W$, $L_b^{\pm} = S^{\pm} - 4Z^{\pm}$, $L_c^{\pm} = S^{\pm} - 4U$, and $L_d^{\pm} = S^{\pm} - 4V^{\pm}$. Introducing the Segre coordinates

$$X_1 = sa_1^2 b_1^2, \qquad X_2 = c_1^2 d_1^2, \qquad Y_1 = sa_1^2 d_1^2, \qquad Y_2 = b_1^2 c_1^2, \tag{51}$$

the four degree-8 monomials factor on the Segre quadric $UW = V^{\pm} Z^{\pm} = X_1 X_2 Y_1 Y_2 = P$ as

$$U = X_1 Y_1, \quad V^{\pm} = \pm X_1 Y_2, \quad Z^{\pm} = \pm X_2 Y_1, \quad W = X_2 Y_2, \quad M_{\mp}(P_1) = S^{\pm} = (X_1 \pm X_2)(Y_1 \pm Y_2). \tag{52}$$

To prove multivariate irreducibility of the second-iterate multipliers rigorously from specialized univariate polynomials, we first establish a general degree-preserving specialization lemma for bihomogeneous polynomials.

Lemma 5.1 (Bihomogeneous Specialization Irreducibility Criterion)**.** *Let $F(X_1, X_2; Y_1, Y_2) \in \mathbb{Q}[X_1, X_2, Y_1, Y_2]$ be a non-zero bihomogeneous polynomial of bidegree (d, d) $(d \geq 1)$. Write $F = \sum_{j=0}^{d} C_j(Y_1, Y_2) X_1^j X_2^{d-j} = \sum_{k=0}^{d} \widetilde{C}_k(X_1, X_2) Y_1^k Y_2^{d-k}$, where each $C_j \in \mathbb{Q}[Y_1, Y_2]$ and $\widetilde{C}_k \in \mathbb{Q}[X_1, X_2]$ is homogeneous of degree d. Suppose that:*

- *(i) F is primitive with respect to both variable pairs, i.e., no non-constant irreducible factor of $\gcd(C_d(Y_1, Y_2), C_0(Y_1, Y_2))$ in $\mathbb{Q}[Y_1, Y_2]$ divides $F(X_1, X_2; Y_1, Y_2)$, and no non-constant irreducible factor of $\gcd(\widetilde{C}_d(X_1, X_2), \widetilde{C}_0(X_1, X_2))$ in $\mathbb{Q}[X_1, X_2]$ divides $F(X_1, X_2; Y_1, Y_2)$; and*
- *(ii) there exists $(y_1, y_2) \in \mathbb{Q}^2$ such that $C_d(y_1, y_2) \neq 0$ and the univariate specialization $f(x) = F(x, 1; y_1, y_2) \in \mathbb{Q}[x]$ of degree d is irreducible over $\mathbb{Q}$.*

Then $F(X_1, X_2; Y_1, Y_2)$ is irreducible in $\mathbb{Q}[X_1, X_2, Y_1, Y_2]$.

Proof. Because F is bihomogeneous of bidegree (d, d) in (X_1, X_2) and (Y_1, Y_2), any factorization $F = GH$ in $\mathbb{Q}[X_1, X_2, Y_1, Y_2]$ decomposes into bihomogeneous factors G of bidegree (d_1, e_1) and H of bidegree (d_2, e_2) with $d_1 + d_2 = d$ and $e_1 + e_2 = d$. Suppose first that one factor has $d_1 = 0$, so $G = G(Y_1, Y_2)$ is a homogeneous polynomial of degree $e_1 \geq 0$ in (Y_1, Y_2) alone. Then $G(Y_1, Y_2)$ divides every coefficient $C_j(Y_1, Y_2)$ $(0 \leq j \leq d)$; in particular, G divides $\gcd(C_d, C_0)$ as well as F. By hypothesis (i), no non-constant factor of $\gcd(C_d, C_0)$ divides F, so $e_1 = 0$ and $G \in \mathbb{Q}^{\times}$. By the same argument applied to $\widetilde{C}_d(X_1, X_2)$ and $\widetilde{C}_0(X_1, X_2)$, no factor can have $e_1 = 0$ and $d_1 > 0$.

Now suppose $F = GH$ with $d_1 \geq 1$ and $d_2 \geq 1$. Write $G = \alpha(Y_1, Y_2) X_1^{d_1} + \ldots$ and $H = \beta(Y_1, Y_2) X_1^{d_2} + \ldots$, so that $\alpha(Y_1, Y_2)\beta(Y_1, Y_2) = C_d(Y_1, Y_2)$. At $(Y_1, Y_2) = (y_1, y_2)$, hypothesis (ii) gives $\alpha(y_1, y_2)\beta(y_1, y_2) = C_d(y_1, y_2) \neq 0$, so both $\alpha(y_1, y_2) \neq 0$ and $\beta(y_1, y_2) \neq 0$. Consequently, the univariate specializations $g(x) = G(x, 1; y_1, y_2)$ and $h(x) = H(x, 1; y_1, y_2)$ preserve their exact positive degrees $\deg(g) = d_1 \geq 1$ and $\deg(h) = d_2 \geq 1$ and satisfy $f(x) = g(x)h(x)$ in $\mathbb{Q}[x]$, contradicting the irreducibility of $f(x)$. Thus F is irreducible in $\mathbb{Q}[X_1, X_2, Y_1, Y_2]$. □

Theorem 5.2 (Dynamical Degree Dichotomy of the Second Iterates on S_{-s^2})**.** *Let $P_1 = (a_1 : b_1 : c_1 : d_1) \in S_{-s^2}$, and let $P_2 = \mathcal{A}_9^{+}(P_1) = (a_1 L_a^{+} : b_1 L_b^{+} : c_1 L_c^{+} : d_1 L_d^{+})$ have degree-8 monomials*

$$\begin{aligned} U_2 &= U(L_a^{+})^4 (L_b^{+})^2 (L_d^{+})^2, \quad & V_2^{\epsilon} &= \epsilon V^{+} (L_a^{+})^2 (L_b^{+})^4 (L_c^{+})^2, \\ W_2 &= W(L_b^{+})^2 (L_c^{+})^4 (L_d^{+})^2, \quad & Z_2^{\epsilon} &= \epsilon Z^{+} (L_a^{+})^2 (L_c^{+})^2 (L_d^{+})^4, \end{aligned} \tag{53}$$

for $\epsilon \in \{+1, -1\}$. While the naive degree of the second iterate $\mathcal{A}_9^{\epsilon}(\mathcal{A}_9^{+}(P_1))$ is $9 \times 9 = 81$, the two compositions exhibit a sharp degree dichotomy:

1. ***Same-Sign Iterate $\mathcal{A}_9^+\circ\mathcal{A}_9^+$ (Irreducible Degree 81):*** *For $\epsilon=+1$, the sum $S_2^+ = U_2+V_2^+ + W_2 + Z_2^+ = M_-(P_2)$ factors in $\mathbb{Z}[X_1,X_2,Y_1,Y_2]$ as*

$$S_2^+ = M_-(P_1)\,H_{32}(P_1)^2, \tag{54}$$

where $H_{32}(P_1)=Q_{32}^+(P_1)=\Phi_8^{+,+}(P_1)\,\Phi_8^{+,-}(P_1)\,\Psi_{16}^+(P_1)$ is the degree-32 invariant polynomial from (12) in $M_-(P_1)=(X_1+X_2)(Y_1+Y_2)$, $M_+(P_1)=(X_1-X_2)(Y_1-Y_2)$, and $P=s^2a_1^4b_1^4c_1^4d_1^4$:

$$\begin{aligned} H_{32}(P_1) &= \big(M_-(P_1)^2-16P\big)^2 - 64P\,M_+(P_1)^2 \\ &= M_-(P_1)^4 - 32P\,M_-(P_1)^2 - 64P\,M_+(P_1)^2 + 256P^2. \end{aligned} \tag{55}$$

Because $H_{32}(P_1)$ is coprime to U_2,V_2^+,W_2,Z_2^+ and the four multipliers $S_2^+-4W_2, S_2^+-4Z_2^+, S_2^+-4U_2, S_2^+-4V_2^+$ are irreducible of bidegree $(9,9)$ in (X_1,X_2) and (Y_1,Y_2) by Lemma 5.1, $\mathcal{A}_9^+\circ\mathcal{A}_9^+$ has exact degree 81.

2. ***Alternating-Sign Iterate $\mathcal{A}_9^-\circ\mathcal{A}_9^+$ (Exact Reduction from Degree 81 to Degree 49):*** *For $\epsilon=-1$, all four coordinates of $\mathcal{A}_9^-(\mathcal{A}_9^+(P_1))$ are identically divisible in $\mathbb{Z}[s,a_1,b_1,c_1,d_1]$ by the degree-32 invariant $H_{32}(P_1)$ in (55). Dividing out $H_{32}(P_1)$ reduces $\mathcal{A}_9^-\circ\mathcal{A}_9^+$ to an exact degree-49 polynomial self-map:*

$$\big(\mathcal{A}_9^-\circ\mathcal{A}_9^+\big)_{\mathrm{red}}(P_1) = \Big(a_1L_a^+\,\Omega_a(P_1),\, b_1L_b^+\,\Omega_b(P_1),\, c_1L_c^+\,\Omega_c(P_1),\, d_1L_d^+\,\Omega_d(P_1)\Big), \tag{56}$$

where the irreducible degree-40 polynomials $\Omega_a,\Omega_b,\Omega_c,\Omega_d$ of bidegree $(5,5)$ in (X_1,X_2) and (Y_1,Y_2) are given in closed form by

$$\begin{aligned} \Omega_a(P_1) &= M_-(P_1)H_{32}(P_1) - 2(L_c^+)^2\Big[(X_1+X_2)Y_2(L_b^+)^2 + (Y_1+Y_2)X_2(L_d^+)^2\Big], \\ \Omega_b(P_1) &= -M_-(P_1)H_{32}(P_1) + 2(L_d^+)^2\Big[(X_1+X_2)Y_1(L_a^+)^2 + (Y_1+Y_2)X_2(L_c^+)^2\Big], \\ \Omega_c(P_1) &= M_-(P_1)H_{32}(P_1) - 2(L_a^+)^2\Big[(X_1+X_2)Y_1(L_d^+)^2 + (Y_1+Y_2)X_1(L_b^+)^2\Big], \\ \Omega_d(P_1) &= -M_-(P_1)H_{32}(P_1) + 2(L_b^+)^2\Big[(X_1+X_2)Y_2(L_c^+)^2 + (Y_1+Y_2)X_1(L_a^+)^2\Big]. \end{aligned} \tag{57}$$

Proof. Let $X_1(P_2)=sa_2^2b_2^2=X_1(L_a^+)^2(L_b^+)^2$, $X_2(P_2)=c_2^2d_2^2=X_2(L_c^+)^2(L_d^+)^2$, $Y_1(P_2)=sa_2^2d_2^2=Y_1(L_a^+)^2(L_d^+)^2$, and $Y_2(P_2)=b_2^2c_2^2=Y_2(L_b^+)^2(L_c^+)^2$ be the Segre coordinates of $P_2=\mathcal{A}_9^+(P_1)$, so that $U_2=X_1(P_2)Y_1(P_2)$, $V_2^\epsilon=\epsilon X_1(P_2)Y_2(P_2)$, $W_2=X_2(P_2)Y_2(P_2)$, $Z_2^\epsilon=\epsilon X_2(P_2)Y_1(P_2)$, and $S_2^\epsilon=\big(X_1(P_2)+\epsilon X_2(P_2)\big)\big(Y_1(P_2)+\epsilon Y_2(P_2)\big)$.

We first prove the fundamental Segre sum identities:

$$X_1(P_2)+X_2(P_2)=(X_1+X_2)\,H_{32}(P_1), \qquad Y_1(P_2)+Y_2(P_2)=(Y_1+Y_2)\,H_{32}(P_1). \tag{58}$$

To verify (58), write $M_-=(X_1+X_2)(Y_1+Y_2)$, so that $L_a^+L_b^+=(M_--4X_2Y_2)(M_--4X_2Y_1)=M_-^2-4X_2(Y_1+Y_2)M_-+16X_2^2Y_1Y_2$ and $L_c^+L_d^+=(M_--4X_1Y_1)(M_--4X_1Y_2)=M_-^2-4X_1(Y_1+Y_2)M_-+16X_1^2Y_1Y_2$. Using $(Y_1+Y_2)M_-=\frac{M_-^2}{X_1+X_2}$, we obtain

$$\begin{aligned} L_a^+L_b^+ &= \frac{(X_1-3X_2)M_-^2}{X_1+X_2}+16X_2^2Y_1Y_2, \\ L_c^+L_d^+ &= \frac{(X_2-3X_1)M_-^2}{X_1+X_2}+16X_1^2Y_1Y_2. \end{aligned}$$

Squaring both expressions and forming $X_1(L_a^+L_b^+)^2+X_2(L_c^+L_d^+)^2$, observe that

$$\begin{aligned} X_1(X_1-3X_2)^2+X_2(X_2-3X_1)^2 &= (X_1+X_2)^3, \\ 2X_1X_2^2(X_1-3X_2)+2X_2X_1^2(X_2-3X_1) &= -4X_1X_2(X_1+X_2)^2, \end{aligned}$$

$$X_1X_2^4+X_2X_1^4=X_1X_2(X_1+X_2)\big((X_1+X_2)^2-3X_1X_2\big).$$

Hence $(X_1+X_2)^2$ cancels from the denominator and leaves (X_1+X_2) times

$$M_-^4-64PM_-^2+256P^2\big((X_1+X_2)^2-3X_1X_2\big)=M_-^4-32PM_-^2-64PM_+^2+256P^2=H_{32}(P_1),$$

where we used $M_-^2-M_+^2=4(X_1X_2(Y_1+Y_2)^2+Y_1Y_2(X_1+X_2)^2-4P)$ to rewrite the middle term, establishing the first identity of (58); the second follows by the symmetry $(X_1,X_2)\leftrightarrow(Y_1,Y_2)$.

(1) Multiplying the two identities in (58) immediately yields

$$S_2^+=\big(X_1(P_2)+X_2(P_2)\big)\big(Y_1(P_2)+Y_2(P_2)\big)=(X_1+X_2)(Y_1+Y_2)\,H_{32}(P_1)^2=M_-(P_1)\,H_{32}(P_1)^2,$$

proving (54). Since $H_{32}(P_1)\big|_{P=0}=M_-(P_1)^4\neq 0$ while U_2,V_2^+,W_2,Z_2^+ vanish at $P=0$, we have $\gcd(H_{32}(P_1),U_2)=\gcd(H_{32}(P_1),W_2)=1$. In $F_{81}=S_2^+-4W_2$, which is symmetric under $(X_1,X_2)\leftrightarrow(Y_1,Y_2)$, the leading coefficient of X_1^9 is $C_9(Y_1,Y_2)=(Y_1+Y_2)^9$ and the coefficient of X_2^9 is $C_0(Y_1,Y_2)=(Y_1+Y_2)(Y_1^2-6Y_1Y_2+Y_2^2)^4$, so $\gcd(C_9,C_0)=Y_1+Y_2$. Neither Y_1+Y_2 nor X_1+X_2 divides F_{81} since $F_{81}\big|_{Y_2=-Y_1}=-4W_2\big|_{Y_2=-Y_1}\neq 0$ and $F_{81}\big|_{X_2=-X_1}=-4W_2\big|_{X_2=-X_1}\neq 0$. At $(Y_1,Y_2)=(2,1)$, we have $C_9(2,1)=3^9=19683\neq 0$ while the univariate specialization $p_9(x)=F_{81}(x,1;2,1)=19683x^9-187497x^8+810924x^7-2120596x^6+3691194x^5-4359102x^4+3409980x^3-1667940x^2+457083x-53217$ is irreducible in $\mathbb{Q}[x]$. By Lemma 5.1 (and Klein four-group symmetry on the remaining coordinates), $S_2^+-4W_2, S_2^+-4Z_2^+, S_2^+-4U_2, S_2^+-4V_2^+$ are irreducible of bidegree $(9,9)$, so $\mathcal{A}_9^+\circ\mathcal{A}_9^+$ has exact degree 81.

(2) For the alternating iterate $\mathcal{A}_9^-(\mathcal{A}_9^+(P_1))$ $(\epsilon=-1)$, the four multipliers are $S_2^--4W_2$, $S_2^-+4Z_2^+$, $S_2^--4U_2$, and $S_2^-+4V_2^+$, where $S_2^-=\big(X_1(P_2)-X_2(P_2)\big)\big(Y_1(P_2)-Y_2(P_2)\big)$. Set $A=X_1(P_2)+X_2(P_2)=(X_1+X_2)H_{32}(P_1)$ and $C=Y_1(P_2)+Y_2(P_2)=(Y_1+Y_2)H_{32}(P_1)$ by (58). Then $X_1(P_2)-X_2(P_2)=A-2X_2(P_2)=2X_1(P_2)-A$ and $Y_1(P_2)-Y_2(P_2)=C-2Y_2(P_2)=2Y_1(P_2)-C$. Using the elementary binomial identity $(A-2B)(C-2D)-4BD=AC-2(AD+BC)$, we obtain the exact algebraic factorizations:

$$\begin{aligned}
S_2^--4W_2&=AC-2\big(AY_2(P_2)+CX_2(P_2)\big)\\
&=H_{32}(P_1)\Big[M_-(P_1)H_{32}(P_1)-2(X_1+X_2)Y_2(P_2)-2(Y_1+Y_2)X_2(P_2)\Big]=H_{32}(P_1)\,\Omega_a(P_1),\\
S_2^-+4Z_2^+&=\big(A-2X_2(P_2)\big)\big(2Y_1(P_2)-C\big)+4X_2(P_2)Y_1(P_2)=-AC+2\big(AY_1(P_2)+CX_2(P_2)\big)\\
&=H_{32}(P_1)\,\Omega_b(P_1),\\
S_2^--4U_2&=\big(2X_1(P_2)-A\big)\big(2Y_1(P_2)-C\big)-4X_1(P_2)Y_1(P_2)=AC-2\big(AY_1(P_2)+CX_1(P_2)\big)\\
&=H_{32}(P_1)\,\Omega_c(P_1),\\
S_2^-+4V_2^+&=\big(2X_1(P_2)-A\big)\big(C-2Y_2(P_2)\big)+4X_1(P_2)Y_2(P_2)=-AC+2\big(AY_2(P_2)+CX_1(P_2)\big)\\
&=H_{32}(P_1)\,\Omega_d(P_1).
\end{aligned}$$

Dividing all four coordinates of $\mathcal{A}_9^-(\mathcal{A}_9^+(P_1))$ by $H_{32}(P_1)$ drops the degree from 81 to $81-32=49$, yielding (56)–(57). Expanding $\Omega_a(X_1,X_2;Y_1,Y_2)=\sum_{j=0}^5 C_j(Y_1,Y_2)X_1^jX_2^{5-j}$ (which is symmetric under $(X_1,X_2)\leftrightarrow(Y_1,Y_2)$) gives the explicit homogeneous degree-5 coefficients:

$$\begin{aligned}
C_5(Y_1,Y_2)&=(Y_1+Y_2)^2(Y_1-Y_2)(Y_1^2-14Y_1Y_2+Y_2^2),\\
C_4(Y_1,Y_2)&=-(Y_1+Y_2)(13Y_1^4-146Y_1^3Y_2+236Y_1^2Y_2^2-94Y_1Y_2^3+23Y_2^4),\\
C_3(Y_1,Y_2)&=-2(7Y_1^5+45Y_1^4Y_2-146Y_1^3Y_2^2+2Y_1^2Y_2^3-53Y_1Y_2^4+17Y_2^5),\\
C_2(Y_1,Y_2)&=2(7Y_1^5-71Y_1^4Y_2-2Y_1^3Y_2^2-150Y_1^2Y_2^3+27Y_1Y_2^4-3Y_2^5),\\
C_1(Y_1,Y_2)&=(Y_1+Y_2)^2(13Y_1^3+45Y_1^2Y_2+3Y_1Y_2^2+3Y_2^3),\\
C_0(Y_1,Y_2)&=-(Y_1+Y_2)^2(Y_1^3+21Y_1^2Y_2-9Y_1Y_2^2+3Y_2^3).
\end{aligned}$$

Here $\gcd(C_5,C_0)=(Y_1+Y_2)^2$, and Y_1+Y_2 does not divide Ω_a since $\Omega_a\big|_{Y_2=-Y_1}=512X_1^2X_2^2Y_1^5(X_1+X_2)\neq 0$ (and by symmetry $X_1+X_2\nmid\Omega_a$). At $(Y_1,Y_2)=(2,1)$, we have $C_5(2,1)=-207\neq 0$ and the specialization $p_5(x)=\Omega_a(x,1;2,1)=-207x^5+543x^4+610x^3-2954x^2+2637x-693$ is irreducible in $\mathbb{Q}[x]$. By Lemma 5.1 (and direct multivariate factorization in $\mathbb{Q}[X_1,X_2,Y_1,Y_2]$), $\Omega_a,\Omega_b,\Omega_c,\Omega_d$ are irreducible of bidegree $(5,5)$. □

5.2 Action of $\mathcal{A}_9^{\pm}$ on Euler's Curve and the Universal V_4-Symmetric Tower on S_{-1}

Applying the canonical degree-9 self-map $\mathcal{A}_9^{\pm}$ and the crossed-pairing skew-line involution Φ_{L_1,L_2} to Euler's degree-7 curve $C_{\mathrm{eul}}(p,q) \subset S_{-1}$ reveals that every resulting curve is generated by a single master polynomial $F_m(p,q)$ via the Klein four-group $V_4 = \{\mathrm{id}, \tau_q, \tau_{pq}, \tau_{pq} \circ \tau_q\}$ acting on (p,q) by $\tau_q(p,q) = (p,-q)$ and $\tau_{pq}(p,q) = (q,p)$.

Theorem 5.3 (Single-Polynomial V_4-Symmetric Hierarchy on S_{-1})**.** *On Euler's quartic surface $S_{-1} : -a^4 + b^4 + c^4 = d^4$, let $F_7(p,q) = q(p^6 - 3p^5q - 2p^4q^2 + p^2q^4 + q^6)$ denote Euler's degree-7 master polynomial, which generates $C_{\mathrm{eul}}(p,q)$ via*

$$\big(a_m(p,q),\, b_m(p,q),\, c_m(p,q),\, d_m(p,q)\big) = \Big(F_m(p,q),\, \epsilon_b F_m(p,-q),\, \epsilon_c F_m(q,p),\, \epsilon_d F_m(-q,p)\Big) \tag{59}$$

(or with the middle two entries or outer two entries transposed) for signs $\epsilon_b, \epsilon_c, \epsilon_d \in \{\pm 1\}$.

1. ***Degree-43 and Degree-45 Asymptotic Images*** *$\mathcal{A}_9^{\pm}(C_{\mathrm{eul}})$: Because $C_{\mathrm{eul}}(p,q) = \big(F_7(p,q), F_7(p,-q), F_7(q,p), F_7(-q,p)\big)$ meets the base locus $\mathrm{Bs}(\mathcal{A}_9^{+})$ of Corollary 1.2 with total multiplicity 20 along*

$$G_{20}^{+}(p,q) = 2p^2q^2(p^2-q^2)(p^2+q^2)(p^4+q^4)(p^2-pq-q^2)(p^2+pq-q^2)(p^4-p^2q^2+q^4)$$

and meets $\mathrm{Bs}(\mathcal{A}_9^{-})$ with total multiplicity 18 along

$$G_{18}^{-}(p,q) = 4p^3q^3(p^6+4p^4q^2-6p^3q^3+4p^2q^4+q^6)(p^6+4p^4q^2+6p^3q^3+4p^2q^4+q^6),$$

the naive degree $9 \times 7 = 63$ drops by 20 (resp. 18) to yield two primitive V_4-symmetric polynomial families on S_{-1} of ***degree 43*** *and* ***degree 45****, generated via (59) by $F_{43}^{\mathrm{asym}}(p,q) = -F_7(p,q)\,P_{36}(p,q)$ and $F_{45}^{\mathrm{asym}}(p,q) = -(F_7(p,q)/q)\,P_{39}(p,q)$, where*

$$\begin{aligned} P_{36}(p,q) = {} & p^{36} + 12p^{35}q + 15p^{34}q^2 + 96p^{33}q^3 + 93p^{32}q^4 + 216p^{31}q^5 + 299p^{30}q^6 - 84p^{29}q^7 \\ & + 489p^{28}q^8 - 660p^{27}q^9 + 243p^{26}q^{10} - 48p^{25}q^{11} - 1581p^{24}q^{12} - 72p^{23}q^{13} + 1518p^{22}q^{14} \\ & - 2604p^{21}q^{15} + 6090p^{20}q^{16} - 48p^{19}q^{17} - 5630p^{18}q^{18} + 3816p^{17}q^{19} + 6090p^{16}q^{20} - 876p^{15}q^{21} \\ & + 1518p^{14}q^{22} + 1680p^{13}q^{23} - 1581p^{12}q^{24} + 5532p^{11}q^{25} + 243p^{10}q^{26} + 972p^9q^{27} + 489p^8q^{28} \\ & - 264p^7q^{29} + 299p^6q^{30} + 12p^5q^{31} + 93p^4q^{32} + 15p^2q^{34} + q^{36}, \end{aligned} \tag{60}$$

$$\begin{aligned} P_{39}(p,q) = {} & p^{39} + 3p^{38}q + 3p^{37}q^2 - 12p^{36}q^3 - 9p^{35}q^4 - 19p^{33}q^6 + 87p^{32}q^7 + 60p^{31}q^8 \\ & - 177p^{30}q^9 + 18p^{29}q^{10} - 240p^{27}q^{12} + 252p^{26}q^{13} + 255p^{25}q^{14} - 261p^{24}q^{15} + 315p^{23}q^{16} \\ & - 1049p^{21}q^{18} + 264p^{20}q^{19} + 750p^{19}q^{20} - 264p^{18}q^{21} + 765p^{17}q^{22} - 2022p^{15}q^{24} + 261p^{14}q^{25} \\ & + 1800p^{13}q^{26} - 252p^{12}q^{27} - 486p^{11}q^{28} - 256p^9q^{30} + 177p^8q^{31} + 204p^7q^{32} - 87p^6q^{33} \\ & - 27p^5q^{34} + p^3q^{36} + 12p^2q^{37} - 3q^{39}. \end{aligned} \tag{61}$$

2. ***Explicit Master Polynomials*** *$F_{13}, F_{19}, F_{25}, F_{31}, F_{37}, F_{43}^{\mathrm{skew}}$* ***of the Dihedral Skew-Line Tower:*** *Composing Φ_{L_1,L_2} with sign/permutation symmetries on C_{eul} generates an infinite arithmetic-progression tower of V_4-symmetric primitive families at degrees $m = 6k+1 \in \{7, 13, 19, 25, 31, 37, 43, 49, \dots\}$ (where F_{43}^{skew} is distinct from the asymptotic image F_{43}^{asym} in Part 1), each satisfying $-F_m(p,q)^4 + F_m(p,-q)^4 + F_m(q,p)^4 - F_m(-q,p)^4 = 0$ identically in*

$\mathbb{Z}[p,q]$:

$$
\begin{aligned}
F_{13}(p,q) &= p^{13}+p^{12}q+2p^{11}q^2-4p^{10}q^3-3p^9q^4+3p^8q^5+7p^7q^6+4p^6q^7\\
&\quad -12p^5q^8-6p^4q^9+5p^3q^{10}-p^2q^{11}+pq^{12}+q^{13},\\
F_{19}(p,q) &= q\Big(p^{18}+6p^{17}q-6p^{16}q^2+12p^{15}q^3+6p^{14}q^4+36p^{13}q^5+19p^{12}q^6-12p^{11}q^7\\
&\quad -24p^{10}q^8+36p^9q^9-12p^8q^{10}+12p^7q^{11}-5p^6q^{12}+6p^5q^{13}-18p^4q^{14}+6p^2q^{16}+q^{18}\Big),\\
F_{25}(p,q) &= p\Big(p^{24}+7p^{22}q^2-2p^{20}q^4+3p^{19}q^5-25p^{18}q^6+63p^{17}q^7+43p^{16}q^8-36p^{15}q^9\\
&\quad -134p^{14}q^{10}-213p^{13}q^{11}+179p^{12}q^{12}+333p^{11}q^{13}-128p^{10}q^{14}-207p^9q^{15}+136p^8q^{16}+57p^7q^{17}\\
&\quad -97p^6q^{18}-9p^5q^{19}+4p^4q^{20}+9p^3q^{21}+10p^2q^{22}+3pq^{23}+q^{24}\Big),\\
F_{31}(p,q) &= -p^{31}+p^{30}q-8p^{29}q^2+8p^{28}q^3-3p^{27}q^4+15p^{26}q^5+76p^{25}q^6-91p^{24}q^7+142p^{23}q^8\\
&\quad +50p^{22}q^9-434p^{21}q^{10}+350p^{20}q^{11}+21p^{19}q^{12}-738p^{18}q^{13}+916p^{17}q^{14}+770p^{16}q^{15}\\
&\quad -830p^{15}q^{16}-478p^{14}q^{17}-303p^{13}q^{18}+126p^{12}q^{19}+718p^{11}q^{20}+266p^{10}q^{21}-254p^9q^{22}\\
&\quad -154p^8q^{23}-32p^7q^{24}-19p^6q^{25}+24p^5q^{26}+42p^4q^{27}+p^3q^{28}+11p^2q^{29}-pq^{30}+q^{31},\\
F_{37}(p,q) &= p\Big(p^{36}+3p^{34}q^2-12p^{32}q^4-6p^{31}q^5-25p^{30}q^6-129p^{29}q^7-3p^{28}q^8+36p^{27}q^9+597p^{26}q^{10}\\
&\quad +996p^{25}q^{11}-2091p^{24}q^{12}-1443p^{23}q^{13}+2391p^{22}q^{14}-876p^{21}q^{15}+810p^{20}q^{16}+5040p^{19}q^{17}\\
&\quad -3683p^{18}q^{18}-7572p^{17}q^{19}+1833p^{16}q^{20}+6063p^{15}q^{21}+774p^{14}q^{22}-2493p^{13}q^{23}-417p^{12}q^{24}\\
&\quad +231p^{11}q^{25}-195p^{10}q^{26}+204p^9q^{27}-96p^8q^{28}-57p^7q^{29}+140p^6q^{30}+36p^5q^{31}-27p^4q^{32}\\
&\quad -21p^3q^{33}-3p^2q^{34}-6pq^{35}+q^{36}\Big),\\
F_{43}^{\mathrm{skew}}(p,q) &= q\Big(p^{42}-9p^{41}q-14p^{40}q^2-36p^{39}q^3+37p^{38}q^4-252p^{37}q^5+183p^{36}q^6+486p^{35}q^7\\
&\quad -568p^{34}q^8-1260p^{33}q^9-886p^{32}q^{10}-180p^{31}q^{11}+893p^{30}q^{12}+441p^{29}q^{13}-2166p^{28}q^{14}\\
&\quad +1944p^{27}q^{15}-2061p^{26}q^{16}-216p^{25}q^{17}+3667p^{24}q^{18}-1836p^{23}q^{19}-2912p^{22}q^{20}-216p^{21}q^{21}\\
&\quad -1940p^{20}q^{22}+1944p^{19}q^{23}+1507p^{18}q^{24}+441p^{17}q^{25}-1818p^{16}q^{26}-180p^{15}q^{27}\\
&\quad -2949p^{14}q^{28}-1260p^{13}q^{29}+1973p^{12}q^{30}+486p^{11}q^{31}-616p^{10}q^{32}-252p^9q^{33}-406p^8q^{34}-36p^7q^{35}\\
&\quad -33p^6q^{36}-9p^5q^{37}-98p^4q^{38}+13p^2q^{40}+q^{42}\Big).
\end{aligned}
\tag{62}
$$

Proof. First, we prove the general V_4-equivariance preservation principle: suppose a rational curve $C_m(p,q)=(a_m,b_m,c_m,d_m)\subset S_{-1}$ satisfies the V_4-symmetry law (59), so that the involutions $\tau_q(p,q)=(p,-q)$ and $\tau_{pq}(p,q)=(q,p)$ act on (a_m,b_m,c_m,d_m) by signed coordinate permutations in the Klein four-group $V_4=\{\mathrm{id},(ab)(cd),(ac)(bd),(ad)(bc)\}\subset\mathrm{Aut}(S_{-1})$.

(1) Under $\mathcal{A}_9^{\pm}$ ($s=1$), the four degree-8 monomials $U=a_1^4b_1^2d_1^2$, $V^{\pm}=\pm a_1^2b_1^4c_1^2$, $W=b_1^2c_1^4d_1^2$, $Z^{\pm}=\pm a_1^2c_1^2d_1^4$ in (9) are permuted by $(ab)(cd)\in V_4$ via $(U,V^{\pm},W,Z^{\pm})\mapsto(V^{\pm},U,Z^{\pm},W)$ (for $\mathcal{A}_9^{+}$) and by $(ac)(bd)\in V_4$ via $(U,V^{\pm},W,Z^{\pm})\mapsto(W,Z^{\pm},U,V^{\pm})$. Consequently, $\mathcal{A}_9^{\pm}$ is V_4-equivariant up to signs, so $\mathcal{A}_9^{\pm}(C_m(p,q))$ again satisfies (59). Substituting $C_{\mathrm{eul}}(p,q)=\big(F_7(p,q),F_7(p,-q),F_7(q,p),F_7(-q,p)\big)$ into $\mathcal{A}_9^{+}$ and $\mathcal{A}_9^{-}$, we compute the base-locus intersections of C_{eul} with $\mathrm{Bs}(\mathcal{A}_9^{\pm})$ via Corollary 1.2:

- At $p=0$ and $q=0$, C_{eul} passes through the isolated Eckardt points $(0:0:1:1)$ and $(1:-1:0:0)$ of Corollary 1.2(2), contributing p^2q^2 to G_{20}^{+} and p^3q^3 to G_{18}^{-} (where the extra pq cancels q from $F_7(p,q)$ in F_{45}^{asym}).
- Along the four rational base lines $c_1^2-a_1^2=d_1^2-b_1^2=0$ of Corollary 1.2(1), the greatest common divisor of $F_7(q,p)^2-F_7(p,q)^2$ and $F_7(-q,p)^2-F_7(p,-q)^2$ factors as $(p^2-q^2)(p^2+q^2)(p^4+q^4)(p^2-pq-q^2)(p^2+pq-q^2)(p^4-p^2q^2+q^4)$ of degree 16, giving $G_{20}^{+}(p,q)$ of degree 20 for $\mathcal{A}_9^{+}$; similarly, along the base lines $c_1^2+a_1^2=d_1^2-b_1^2=0$ and $c_1^2-a_1^2=d_1^2+b_1^2=0$ of $\mathcal{A}_9^{-}$, the common factor is $(p^6+4p^4q^2-6p^3q^3+4p^2q^4+q^6)(p^6+4p^4q^2+6p^3q^3+4p^2q^4+q^6)$ of degree 12, giving $G_{18}^{-}(p,q)$ of degree 18 for $\mathcal{A}_9^{-}$.

Dividing the degree-63 polynomial $a_1(p,q)(S^{\pm}-4W)$ by $G^{+}_{20}(p,q)$ (resp. $G^{-}_{18}(p,q)$) yields $F^{\mathrm{asym}}_{43}(p,q)=-F_7(p,q)P_{36}(p,q)$ in (60) and $F^{\mathrm{asym}}_{45}(p,q)=-(F_7(p,q)/q)P_{39}(p,q)$ in (61).

(2) For the skew-line involution Φ_{L_1,L_2} in (84), the two skew lines $L_1=\{a+b=0,c+d=0\}$ and $L_2=\{a+c=0,b-d=0\}$ are stabilized (up to exchange $L_1\leftrightarrow L_2$) by the involution $(a,b,c,d)\mapsto(d,-c,-b,a)$ corresponding to $\tau_{pq}(p,q)=(q,p)$ on C_{eul}, while $\tau_q(p,q)=(p,-q)$ conjugates Φ_{L_1,L_2} to the $b\leftrightarrow c$ swapped involution. Hence every iterate in the dihedral orbit $(\Phi_{L_1,L_2}\circ\sigma)^k(C_{\mathrm{eul}})$ inherits the V_4-symmetry (59). On $\mathrm{Pic}(S_{-1})$, Φ_{L_1,L_2} acts as the reflection in $H-L_1-L_2$ (sending $H\mapsto 5H-4(L_1+L_2)$), while the signed coordinate permutations permute the line classes $\{L_i\}$. Starting from C_{eul} of degree $4\times 7=28$ in $\mathbb{P}^3$ (polynomial degree 7), applying Φ_{L_1,L_2} to the signed coordinate permutations of C_{eul} produces F_{13},F_{19},F_{31}, and subsequent steps from F_{13},F_{19},F_{25} produce F_{25} (from F_{13}), F_{37} (from F_{25}), F^{skew}_{43} (from F_{19}), and F_{49} (from F_{13}), generating the progression $m=6k+1\in\{7,13,19,25,31,37,43,49,\dots\}$ with the explicit master polynomials in (62), each of which satisfies $-F_m(p,q)^4+F_m(p,-q)^4+F_m(q,p)^4-F_m(-q,p)^4=0$ identically in $\mathbb{Z}[p,q]$. □

6 Dual Conic-Bundle Fibrations, 2-Adic Reciprocity Laws, and New Primitive Solutions on $S_{-9}, S_{-25}, S_{-100}$

We now unify the Four-Eckardt-Plane quadric pencil of Section 2 with the Gaussian sum-of-two-squares fibration on $S_{-s^2}: b^4+c^4=(d^2)^2+(sa^2)^2$, establish a sharp 2-adic and quadratic-reciprocity descent law across S_{-s^2}, and construct explicit primitive integer solutions and their degree-9 asymptotic images on S_{-9} ($s=3$), S_{-25} ($s=5$), and S_{-100} ($s=10$).

Theorem 6.1 (Duality of Eckardt and Gaussian Conic-Bundle Fibrations and 2-Adic Descent on S_{-s^2})**.**

1. ***Exact Bridge Between the Eckardt and Gaussian Pencils:*** *On $S_{-s^2}:(b^2)^2+(c^2)^2=(d^2)^2+(sa^2)^2$, the Gaussian sum-of-two-squares pencil with parameter $\tau\in\mathbb{P}^1(\mathbb{Q})$,*

$$(1-\tau^2)b^2+2\tau c^2=(1+\tau^2)d^2,\qquad 2\tau b^2-(1-\tau^2)c^2=(1+\tau^2)sa^2, \tag{63}$$

is identically equivalent to the Four-Eckardt-Plane pencil (21)–(22) under the Möbius transformation

$$2\lambda+1=\frac{1}{\tau}\iff\tau=\frac{1}{2\lambda+1}=\frac{c^2\pm sa^2}{b^2+d^2}. \tag{64}$$

In particular, the two rational Eckardt base-point sections of (22) correspond to:

- ***Section I*** *($b=d\implies 2\lambda+1=k^2,\tau=1/k^2$)**:** Gives Theorem 2.1 with $sa^2=G(k,t)$ in (24).*
- ***Section II*** *($c=d\implies\lambda=\frac{m^2}{1-m^2},\tau=\frac{1-m^2}{1+m^2}$)**:** Gives the dual conic $2m^2b^2-(m^4-1)c^2=(m^4+1)d^2$ ($m\in\mathbb{Q}\setminus\{0,\pm1\}$) with rational base point $(b:c:d)=(m:1:1)$, which is parametrized over $\mathbb{Q}(m,t)$ by*

$$\begin{aligned}b(m,t)&=-2m^3t^2+2(m^4-1)t-m(m^4-1),\\ c(m,t)&=2m^2t^2-4m^3t+m^4-1,\\ d(m,t)&=2m^2t^2-(m^4-1),\end{aligned} \tag{65}$$

reducing S_{-s^2} to the genus-1 double cover $sa^2=H(m,t)$ (satisfying $b(m,t)^4+c(m,t)^4-d(m,t)^4=H(m,t)^2$ identically), where

$$H(m,t)=4m^6t^4-8m^3(m^4+1)t^3+4(2m^8+3m^4-1)t^2-4m(m^8-1)t+m^2(m^4-1)^2. \tag{66}$$

2. ***2-Adic Valuation and Quadratic Reciprocity Law on S_{-s^2} and S_{-4} ($s=2$):*** *In any primitive integer solution $\gcd(a,b,c,d)=1$ to $-s^2a^4+b^4+c^4=d^4$ ($s\in\mathbb{Z}_{>0}$):*

   - ***Regime $s\equiv 2,3,5,6 \pmod 8$ (i.e., $4\nmid s$ and $s\not\equiv\pm1 \pmod 8$):*** *Both a and exactly one of $\{b,c\}$ (say b) must be even, while c,d are odd with $16\mid(d^4-c^4)$. (By contrast, when $4\mid s$, such as $s=4$ on S_{-16}, $16\mid s^2$ holds automatically, permitting a to be odd while one of $\{b,c\}$ is even and the other and d are odd, as in the primitive solution $(a,b,c,d)=(67,59,158,133)$ on S_{-16}.)*
   - ***Quadratic Reciprocity Descent for $s=2$ ($n=-4$):*** *For any non-zero primitive solution on $S_{-4}: -4a^4+b^4+c^4=d^4$, writing $a=2a_1$ and $b=2b_1$ yields*

$$(d-c)(d+c)(d^2+c^2)=16\big(b_1^4-4a_1^4\big)=16\big(b_1^2-2a_1^2\big)\big(b_1^2+2a_1^2\big)\neq 0.$$

   Under the coprimality condition $\gcd(a_1,b_1)=1$ (i.e., $\gcd(a,b)=2$), every solution on S_{-4} must satisfy $4\mid b$ ($b=4b_2$), $2\parallel a$ ($a=2a_1$ with a_1 odd), and $16\mid(d\mp c)$, reducing S_{-4} to $km(k^2+m^2)=8(4b_2^4-a_1^4)$ (with $k=(d+c)/2, m=(d-c)/2$), where every odd prime factor p of $(k^2+m^2)/\gcd(k,m)^2$ satisfies $p\equiv 1 \pmod 8$.

Proof. (1) Squaring and adding the two equations in (63) gives

$$\big((1-\tau^2)^2+4\tau^2\big)(b^4+c^4)=(1+\tau^2)^2(d^4+s^2a^4)\iff(1+\tau^2)^2(b^4+c^4-d^4-s^2a^4)=0,$$

so (63) lies on S_{-s^2} for every $\tau\in\mathbb{P}^1(\mathbb{Q})$. Substituting $\tau=\frac{1}{2\lambda+1}$ into the first equation of (63) and multiplying by $\frac{(2\lambda+1)^2}{2}$ gives

$$\frac{(2\lambda+1)^2-1}{2}b^2+(2\lambda+1)c^2=\frac{(2\lambda+1)^2+1}{2}d^2\iff 2\lambda(\lambda+1)b^2+(2\lambda+1)c^2=(2\lambda^2+2\lambda+1)d^2,$$

which coincides identically with (22). Similarly, substituting $\tau=\frac{1}{2\lambda+1}$ into the second equation of (63) reproduces the difference of (21), proving (64). At $c=d$, (22) becomes $2\lambda(\lambda+1)b^2=2\lambda^2d^2\iff(\frac{\lambda+1}{\lambda})b^2=d^2$. Setting $\frac{\lambda+1}{\lambda}=\frac{1}{m^2}$, i.e., $\lambda=\frac{m^2}{1-m^2}$ and $\tau=\frac{1-m^2}{1+m^2}$, transforms (22) into $2m^2b^2-(m^4-1)c^2=(m^4+1)d^2$ with base point $(b:c:d)=(m:1:1)$. Passing the line $b-md=t(c-d)$ through $(m:1:1)$ and substituting $b=md+t(c-d)$ into $2m^2b^2-(m^4-1)c^2-(m^4+1)d^2=0$ gives

$$(c-d)\Big[\big(2m^2t^2-(m^4-1)\big)c+\big(4m^3t-2m^2t^2-(m^4-1)\big)d\Big]=0,$$

whose non-trivial root gives (65); substituting (65) into the second equation of (63) (up to sign $sa^2\mapsto -sa^2$) yields $sa^2=H(m,t)$ as in (66).

(2) Recall that for any odd integer x, $x^2\equiv 1$ or $9 \pmod{16}$, so $x^4\equiv 1 \pmod{16}$, whereas for any even integer y, $y^4\equiv 0 \pmod{16}$. In $b^4+c^4-d^4=s^2a^4$ with $\gcd(a,b,c,d)=1$:

- Since each of b^4,c^4,d^4 is congruent to 0 or 1 $\pmod{16}$, the left-hand side satisfies $b^4+c^4-d^4\in\{-1,0,1,2\} \pmod{16}$.
- Suppose $4\nmid s$ and $s\not\equiv\pm1 \pmod 8$ (so $s\equiv 2,3,5,6 \pmod 8$). Then $s^2\equiv 4$ or $9 \pmod{16}$. If a were odd, then $s^2a^4\equiv s^2\equiv 4$ or $9 \pmod{16}$, which does not belong to $\{-1,0,1,2\} \pmod{16}$, a contradiction. Thus $a=2a_1$ must be even.
- With a even, $b^4+c^4-d^4=s^2a^4\equiv 0 \pmod{16}$. Since $\gcd(a,b,c,d)=1$, b,c,d cannot all be even, so $b^4+c^4\equiv d^4 \pmod{16}$ forces d and one of $\{b,c\}$ (say c) to be odd and the other ($b=2b_1$) to be even, with $16\mid(d^4-c^4)$. (When $4\mid s$, $s^2\equiv 0 \pmod{16}$ regardless of the parity of a, so a,b,d can be odd and c even, or a,c,d odd and b even.)

Specializing to $s=2$ ($n=-4$), substitute $a=2a_1$ and $b=2b_1$ into $-4a^4+b^4+c^4=d^4$ to obtain $(d-c)(d+c)(d^2+c^2)=16(b_1^4-4a_1^4)=16(b_1^2-2a_1^2)(b_1^2+2a_1^2)$. For any non-zero solution $(a_1,b_1)\neq(0,0)$, irrationality of $\sqrt{2}$ ensures $b_1^2-2a_1^2\neq 0$ and $b_1^2+2a_1^2>0$, so $d\neq\pm c$ and neither side vanishes. Since c,d are odd, let $g_{cd}=\gcd(c,d)$ (which is odd) and write $c=g_{cd}c_0, d=g_{cd}d_0$ with $\gcd(c_0,d_0)=1$ and c_0,d_0 odd. Because $g_{cd}\equiv\pm1,\pm3 \pmod 8$, we have $g_{cd}^2\equiv 1 \pmod 8$ and $g_{cd}^4\equiv 1 \pmod{16}$. Moreover, $d^2+c^2=g_{cd}^2(d_0^2+c_0^2)\equiv 2 \pmod 8$, so $v_2(d^2+c^2)=v_2(d_0^2+c_0^2)=1$ and the integer $M_{\text{odd}}=\frac{d_0^2+c_0^2}{2}$ is positive and odd. Suppose for contradiction that b_1 were odd. Then $b_1^4-4a_1^4\equiv 1-4a_1^4 \pmod 8$ is odd, so $v_2(d^4-c^4)=v_2(d_0^4-c_0^4)=4$. Since $v_2(d_0^2+c_0^2)=1$, we have $v_2(d_0^2-c_0^2)=3$, meaning $d_0\equiv\pm c_0+4 \pmod 8$. Squaring gives $d_0^2+c_0^2\equiv c_0^2+(c_0+4)^2=2c_0^2+8c_0+16\equiv 2+8=10 \pmod{16}$, so the odd positive integer $M_{\text{odd}}=\frac{d_0^2+c_0^2}{2}$ satisfies $M_{\text{odd}}\equiv 5 \pmod 8$. Every prime factor p of M_{odd} is odd and divides $d_0^2+c_0^2$ with $\gcd(c_0,d_0)=1$, so $\left(\frac{-1}{p}\right)=+1 \implies p\equiv 1$ or $5 \pmod 8$. Since a product of primes congruent to 1 (mod 8) is congruent to 1 (mod 8), $M_{\text{odd}}\equiv 5 \pmod 8$ must be divisible by at least one odd prime $p\equiv 5 \pmod 8$. Because $p\mid M_{\text{odd}}\mid\frac{d^4-c^4}{16}$, p divides $(b_1^2-2a_1^2)(b_1^2+2a_1^2)$. Under the coprimality condition $\gcd(a_1,b_1)=1$, p cannot divide both a_1 and b_1, so $b_1^2\equiv 2a_1^2 \pmod p$ or $b_1^2\equiv -2a_1^2 \pmod p$ with $p\nmid a_1$, which requires $\left(\frac{2}{p}\right)=+1$ or $\left(\frac{-2}{p}\right)=+1$. Both fail for $p\equiv 5 \pmod 8$, where $\left(\frac{2}{p}\right)=\left(\frac{-2}{p}\right)=-1$. This contradiction proves that $b_1=2b_2$ is even ($4\mid b$), which forces a_1 to be odd (since $\gcd(a_1,b_1)=1$) and $v_2(d^4-c^4)=4+v_2(16b_2^4-4a_1^4)=6$, so $v_2(d^2-c^2)=5$ and $16\mid(d\mp c)$. Writing $d=k+m$ and $c=k-m$ (so $k=(d+c)/2, m=(d-c)/2$ with $8\mid m$ or $8\mid k$) reduces the equation to $km(k^2+m^2)=8(4b_2^4-a_1^4)=8(2b_2^2-a_1^2)(2b_2^2+a_1^2)$. For any odd prime p dividing $(k^2+m^2)/\gcd(k,m)^2$, we have $\left(\frac{-1}{p}\right)=+1$, and since $\gcd(a_1,2b_2)=1$, $p\mid(2b_2^2-a_1^2)(2b_2^2+a_1^2)$ forces $\left(\frac{\pm 2}{p}\right)=+1$, so $\left(\frac{-1}{p}\right)=\left(\frac{2}{p}\right)=+1 \iff p\equiv 1 \pmod 8$. □

Table 3: New Primitive Integer Solutions and Degree-9 Asymptotic Images on S_{-9} ($s=3$), S_{-25} ($s=5$), and S_{-100} ($s=10$) (normalized projectively with $d_2>0$; every row satisfies $-s^2a_i^4+b_i^4+c_i^4-d_i^4=0$ identically).

Surface S_{-s^2}	**Primitive Seed** $P_1=(a_1,b_1,c_1,d_1)$	**Gaussian** $\tau_\pm=\frac{c_1^2\mp sa_1^2}{b_1^2+d_1^2}$	**Branch**	**Residual** gcd **of** (10) **& Primitive Image** $\mathcal{A}_9^\pm(P_1)=(a_2,b_2,c_2,d_2)$ $(d_2>0)$
$s=3$ ($n=-9$)	$(596,720,965,133)$	$-\frac{587}{2537},\frac{853}{229}$	$\mathcal{A}_9^+$	gcd = 1,649,000,400: $(-91641863106202348,-1240542874547388800,-142618053385951435,63022032130271221)$
$s=3$ ($n=-9$)	$(596,720,965,133)$	$-\frac{587}{2341},\frac{853}{229}$	$\mathcal{A}_9^-$	gcd = 234,404,400: $(-713045618311666196,-759880147925039760,-119504525728462872 5,464191709272981067)$
$s=3$ ($n=-9$)	$(860,1488,385,191)$	$-\frac{23}{25},\frac{9463}{90025}$	Seed	Verified primitive solution (gcd = 1): $-9\cdot 860^4+1488^4+385^4=191^4$
$s=5$ ($n=-25$)	$(38,70,73,23)$	$-\frac{31}{89},\frac{141}{61}$	$\mathcal{A}_9^+$	gcd = 110,360: $(-292282408274,-697859921570,-398738929249,531834678401)$
$s=5$ ($n=-25$)	$(38,70,73,23)$	$-\frac{31}{89},\frac{141}{61}$	$\mathcal{A}_9^-$	gcd = 688,080: $(-56345685068,-69458763640,-134346380813,99254444237)$
$s=5$ ($n=-25$)	$(226,30,713,663)$	$\frac{2321}{4041},\frac{189}{349}$	Seed	Verified primitive solution (gcd = 1): $-25\cdot 226^4+30^4+713^4=663^4$
$s=10$ ($n=-100$)	$(10,20,31,17)$	$-\frac{25}{53},\frac{97}{13}$	$\mathcal{A}_9^+$	gcd = 31,800: $(-9228490,-85111940,5865913,84816791)$
$s=10$ ($n=-100$)	$(10,20,31,17)$	$-\frac{3}{53},\frac{37}{13}$	$\mathcal{A}_9^-$	gcd = 96,200: $(-45602990,64322260,-152765117,106625581)$

7 Closed-Form Parametrizations of a Second Solution $P_2(P_0)$ from One Known Solution $P_0=(a_0,b_0,c_0,d_0)$

A central diophantine question for the diagonal quartic surface

$$S_n: na^4+b^4+c^4=d^4, \qquad \gcd(a,b,c,d)=1,$$

is the *Point-Duplication Problem*: given a single known non-trivial rational (or primitive integer) solution $P_0=(a_0,b_0,c_0,d_0)\in S_n(\mathbb{Q})$ with $a_0\neq 0$, so that $n=(d_0^4-b_0^4-c_0^4)/a_0^4$ is uniquely determined by P_0, construct an explicit closed-form polynomial or rational formula $P_2=\Psi(a_0,b_0,c_0,d_0)$ producing a second projectively distinct solution directly from the coordinates of P_0 (over $\mathbb{Q}(\sqrt{-n})$ for general $n\in\mathbb{Q}^\times$, and over $\mathbb{Q}$ whenever $-n=s^2\in\mathbb{Q}^{\times 2}$ or S_n carries a rational skew-line or quadric-pencil structure). In this section we establish:

1. the **Universal Tangent-Plane Taylor Expansion and Degree-9 Asymptotic Formula** over $\mathbb{Q}(\sqrt{-n})$ valid for every $n=(d_0^4-b_0^4-c_0^4)/a_0^4$, which is $\mathbb{Q}$-rational whenever $n=-s^2$ (Theorem 7.1);

2. the **Universal Diagonal-Quadric Tangent-Duplication Theorem** on any smooth genus-1 space quartic $E(P_0) = Q_1 \cap Q_2 \subset S_n$ passing through P_0, together with its exact reduction on the Four-Eckardt-Plane quadric pencil of S_{-s^2} to the (b_0, c_0)-symmetric degree-9 self-map $\mathcal{E}_9^{\pm}(P_0)$ and its exact coordinate-by-coordinate projective equivalence to $\mathcal{A}_9^{\mp}(P_0)$ (Theorem 7.2); and

3. the **Complete Classification of Second-Solution Generators Across All Geometric Regimes of** n ($n = -s^2$, $n = \pm 2s^2$, $n = 4$, $n = -1$, $n = -12$, and the coordinate-zero tacnodal locus $c_0 = 0$ or $d_0 = 0$, Theorem 7.3 and Table 4).

7.1 Universal Tangent-Plane Taylor Expansion and Degree-9 Asymptotic Formula for General n

Given a known point $P_0 = (a_0, b_0, c_0, d_0)$ with $a_0 d_0 \neq 0$, eliminating $n = (d_0^4 - b_0^4 - c_0^4)/a_0^4$ casts the surface equation of S_n purely in terms of the known coordinates (a_0, b_0, c_0, d_0):

$$\left(d_0^4 - b_0^4 - c_0^4\right) a^4 + a_0^4\left(b^4 + c^4 - d^4\right) = 0. \tag{67}$$

The projective tangent plane $T_{P_0}(S_n) \subset \mathbb{P}^3$ at P_0 is independent of n when written in homogeneous difference form:

$$T_{P_0}(S_n) : b_0^3(a_0 b - b_0 a) + c_0^3(a_0 c - c_0 a) - d_0^3(a_0 d - d_0 a) = 0. \tag{68}$$

Theorem 7.1 (Universal Tangent-Plane Taylor Expansion and Asymptotic Second-Solution Formula)**.** *Let $P_0 = (a_0, b_0, c_0, d_0) \in S_n(\mathbb{Q})$ be any known solution with $a_0 d_0 \neq 0$ and $n = (d_0^4 - b_0^4 - c_0^4)/a_0^4 \neq 0$.*

1. ***Exact Quartic Taylor Expansion on*** $T_{P_0}(S_n)$***:*** *Every line $L(u, v) \subset T_{P_0}(S_n)$ passing through P_0 (not contained in the plane $a = 0$) is parametrized in affine parameter t and direction $[u : v] \in \mathbb{P}^1$ by*

$$a(t) = a_0, \qquad b(t) = b_0 + u d_0^3 t, \qquad c(t) = c_0 + v d_0^3 t, \qquad d(t) = d_0 + (b_0^3 u + c_0^3 v)t. \tag{69}$$

Substituting (69) into $na(t)^4 + b(t)^4 + c(t)^4 - d(t)^4 = 0$ annihilates the t^0 and t^1 coefficients identically and yields the exact tangent-section equation

$$t^2\Big(c_2(u, v) + c_3(u, v)t + c_4(u, v)t^2\Big) = 0, \tag{70}$$

where the quadratic, cubic, and quartic Taylor forms at P_0 are given explicitly by

$$c_2(u, v) = 6d_0^2\Big[b_0^2(d_0^4 - b_0^4)\, u^2 - 2b_0^3 c_0^3\, uv + c_0^2(d_0^4 - c_0^4)\, v^2\Big], \tag{71}$$

$$c_3(u, v) = 4d_0\Big[b_0(d_0^8 - b_0^8)\, u^3 - 3b_0^6 c_0^3\, u^2 v - 3b_0^3 c_0^6\, uv^2 + c_0(d_0^8 - c_0^8)\, v^3\Big], \tag{72}$$

$$c_4(u, v) = (d_0^{12} - b_0^{12})\, u^4 - 4b_0^9 c_0^3\, u^3 v - 6b_0^6 c_0^6\, u^2 v^2 - 4b_0^3 c_0^9\, uv^3 + (d_0^{12} - c_0^{12})\, v^4. \tag{73}$$

2. ***Discriminant of the Second Fundamental Form and Universal Asymptotic Formula:*** *The quadratic tangent cone $c_2(u, v) = 0$ has reduced discriminant*

$$\Delta_{c_2} = b_0^6 c_0^6 - b_0^2 c_0^2 (d_0^4 - b_0^4)(d_0^4 - c_0^4) = -b_0^2 c_0^2 d_0^4 (d_0^4 - b_0^4 - c_0^4) = -a_0^4 b_0^2 c_0^2 d_0^4\, n. \tag{74}$$

Choosing the inflection directions $[u_\pm : v_\pm] = \left[c_0(b_0^2 c_0^2 \pm \sqrt{-n}\, a_0^2 d_0^2) : b_0(d_0^4 - b_0^4)\right]$ (for P_0 outside the base locus of Corollary 1.2 where $c_3(u_\pm, v_\pm)$ and $c_4(u_\pm, v_\pm)$ vanish simultaneously) so that $c_2(u_\pm, v_\pm) = 0$, the residual intersection of $L(u_\pm, v_\pm)$ with S_n is linear in t, namely $t_\pm = -c_3(u_\pm, v_\pm)/c_4(u_\pm, v_\pm)$. Clearing denominators and reducing modulo $na_0^4 = d_0^4 - b_0^4 - c_0^4$ yields the universal closed-form second-solution generator in $S_n\big(\mathbb{Q}(\sqrt{-n})\big)$ valid for any $n \in \mathbb{Q}^\times$:

$$P_2^{\pm}(P_0) = P_{\text{even}}(P_0) \pm \sqrt{-n}\, P_{\text{odd}}(P_0), \tag{75}$$

where $P_{\text{even}}(P_0)$ and $P_{\text{odd}}(P_0)$ are homogeneous degree-9 polynomials purely in (a_0,b_0,c_0,d_0):

$$P_{\text{even}}(P_0) = b_0^2d_0^2\Big(a_0(b_0^4-2c_0^4-d_0^4),\ b_0(b_0^4+2c_0^4-d_0^4),\\ c_0(-3b_0^4-2c_0^4+3d_0^4),\ d_0(b_0^4+2c_0^4-d_0^4)\Big), \tag{76}$$

$$P_{\text{odd}}(P_0) = a_0^2c_0^2\Big(a_0(b_0^4+d_0^4),\ b_0(b_0^4-3d_0^4),\ c_0(b_0^4+d_0^4),\ d_0(-3b_0^4+d_0^4)\Big). \tag{77}$$

In particular, $P_2^{\pm}(P_0)$ lies in $S_n(\mathbb{Q})$ for every generic point P_0 if and only if $n=-s^2$ ($s\in\mathbb{Q}^\times$) (or at special points where $P_{\text{odd}}(P_0)=0$), in which case (75) recovers the canonical degree-9 self-map $\mathcal{A}_9^{\pm}(P_0)$ of Theorem 1.1, while swapping $b_0\leftrightarrow c_0$ gives a second conjugate pair $\widetilde{\mathcal{A}}_9^{\pm}(P_0)=\sigma_{bc}\circ\mathcal{A}_9^{\pm}\circ\sigma_{bc}(P_0)$ yielding up to four distinct primitive rational second solutions from a single seed P_0.

Proof. (1) Along (69), $na(t)^4=na_0^4=d_0^4-b_0^4-c_0^4$ is constant in t. Expanding $b(t)^4+c(t)^4-d(t)^4$ by the binomial theorem gives:

- t^0 coefficient: $(d_0^4-b_0^4-c_0^4)+b_0^4+c_0^4-d_0^4=0$;
- t^1 coefficient: $4\big[b_0^3(ud_0^3)+c_0^3(vd_0^3)-d_0^3(b_0^3u+c_0^3v)\big]=0$;
- t^2 coefficient: $6\big[b_0^2d_0^6u^2+c_0^2d_0^6v^2-d_0^2(b_0^3u+c_0^3v)^2\big]=6d_0^2\big[b_0^2(d_0^4-b_0^4)u^2-2b_0^3c_0^3uv+c_0^2(d_0^4-c_0^4)v^2\big]=c_2(u,v)$;
- t^3 coefficient: $4\big[b_0d_0^9u^3+c_0d_0^9v^3-d_0(b_0^3u+c_0^3v)^3\big]=c_3(u,v)$;
- t^4 coefficient: $d_0^{12}(u^4+v^4)-(b_0^3u+c_0^3v)^4=c_4(u,v)$,

establishing (70)–(73).

(2) Viewing $\frac{1}{6d_0^2}c_2(u,v)=b_0^2(d_0^4-b_0^4)u^2-2b_0^3c_0^3uv+c_0^2(d_0^4-c_0^4)v^2$ as a binary quadratic form in (u,v), its quarter-discriminant is

$$\Delta_{c_2}=b_0^6c_0^6-b_0^2c_0^2(d_0^4-b_0^4)(d_0^4-c_0^4)=-b_0^2c_0^2d_0^4(d_0^4-b_0^4-c_0^4)=-a_0^4b_0^2c_0^2d_0^4\,n,$$

proving (74). By the quadratic formula, $c_2(u_\pm,v_\pm)=0$ for $u_\pm=b_0^3c_0^3\pm\sqrt{\Delta_{c_2}}=b_0c_0(b_0^2c_0^2\pm\sqrt{-n}\,a_0^2d_0^2)$ and $v_\pm=b_0^2(d_0^4-b_0^4)$ (or after dividing both by b_0). Along this direction, (70) reduces to $t^3\big(c_3(u_\pm,v_\pm)+c_4(u_\pm,v_\pm)t\big)=0$, giving $t_\pm=-c_3(u_\pm,v_\pm)/c_4(u_\pm,v_\pm)$. Setting $s=\sqrt{-n}$, this line $L(u_\pm,v_\pm)$ is the affine presentation of $P_0+tV_\pm(P_0)$ from Theorem 1.1; reducing modulo $s^2a_0^4=b_0^4+c_0^4-d_0^4$ in (10) replaces $U=s^2a_0^4b_0^2d_0^2$ by $(b_0^4+c_0^4-d_0^4)b_0^2d_0^2$, so that

$$\begin{aligned} U-3W&=b_0^2d_0^2(b_0^4-2c_0^4-d_0^4), & U+W&=b_0^2d_0^2(b_0^4+2c_0^4-d_0^4),\\ -3U+W&=b_0^2d_0^2(-3b_0^4-2c_0^4+3d_0^4), & V^\pm+Z^\pm&=\pm sa_0^2c_0^2(b_0^4+d_0^4),\\ V^\pm-3Z^\pm&=\pm sa_0^2c_0^2(b_0^4-3d_0^4), & -3V^\pm+Z^\pm&=\pm sa_0^2c_0^2(-3b_0^4+d_0^4),\end{aligned}$$

which yields (75)–(77) purely in terms of (a_0,b_0,c_0,d_0) and $\sqrt{-n}$. □

7.2 Universal Diagonal-Quadric Tangent-Duplication Theorem and the (b_0,c_0)-Symmetric Map $\mathcal{E}_9^{\pm}$

Whenever S_n carries a pencil of quadrics cutting out genus-1 space quartics $E=Q_1\cap Q_2\subset S_n$, a single known solution $P_0=(a_0,b_0,c_0,d_0)$ singles out the unique fiber $E(P_0)$ passing through P_0, and the Fermat tangent-chord duplication $P_0\mapsto -2P_0$ on $E(P_0)$ produces a second rational point $P_2(P_0)\in S_n(\mathbb{Q})$.

Theorem 7.2 (Universal Diagonal-Quadric Tangent-Duplication Formula and Projective Equivalence on S_{-s^2})**.** *Let $P_0 = (a_0, b_0, c_0, d_0) \in \mathbb{P}^3(\mathbb{Q})$ with $a_0b_0c_0d_0 \neq 0$, and write normalized coordinates $(X, Y, Z, W) = (a/a_0, b/b_0, c/c_0, d/d_0)$ so that P_0 corresponds to $(X, Y, Z, W) = (1, 1, 1, 1)$.*

1. ***Closed-Form Duplication on Any Diagonal Quadric Intersection:*** *Let $E_{u,v} \subset \mathbb{P}^3$ be the space quartic through $(1, 1, 1, 1)$ defined by two diagonal quadrics with parameters $(u, v) \in \mathbb{Q}^2$:*

$$Q_1 : Z^2 = (1-u)X^2 + uY^2, \qquad Q_2 : W^2 = (1-v)X^2 + vY^2, \tag{78}$$

which is a smooth curve of genus 1 whenever $u, v \in \mathbb{Q} \setminus \{0, 1\}$ and $u \neq v$. Then the Fermat tangent-chord duplication map $P_0 \mapsto P_2 = -2P_0$ on $E_{u,v}$ (which is well-defined up to coordinate signs under the 2-torsion group $(\mathbb{Z}/2\mathbb{Z})^3$ of sign changes $(\pm1, \pm1, \pm1, \pm1)$) is given in closed form for all (u, v) by the polynomial formulas

$$\begin{aligned}
a_2 &= a_0\Big[(u-v-1)^2 - 4v\Big],\\
b_2 &= b_0\Big[(u-v+1)^2 - 4(1-v)\Big],\\
c_2 &= c_0\Big[-3u^2 + 2uv + 2u + v^2 - 2v + 1\Big],\\
d_2 &= d_0\Big[(u+v-1)^2 - 4v(v-1)\Big],
\end{aligned} \tag{79}$$

which satisfy both quadrics Q_1 and Q_2 in (78) identically as polynomial identities in $\mathbb{Z}[u, v]$.

2. ***Reduction of the Four-Eckardt-Plane Quadric Duplication on S_{-s^2} to $\mathcal{E}_9^{\pm}(P_0)$:*** *On $S_{-s^2} : -s^2a^4 + b^4 + c^4 = d^4$, for any point $P_0 = (a_0, b_0, c_0, d_0)$ on the Zariski-open set where $a_0b_0c_0d_0(b_0^2 - d_0^2)(c_0^2 - d_0^2)(sa_0^2 + b_0^2)(sa_0^2 - b_0^2 - c_0^2 + d_0^2) \neq 0$, the unique fiber $E_{\pm}(P_0)$ of the Four-Eckardt-Plane quadric pencil (21) passing through P_0 is cut out by the proportions*

$$\frac{c^2 - sa^2}{d^2 - b^2} = \frac{c_0^2 - sa_0^2}{d_0^2 - b_0^2}, \qquad \frac{b^2 + sa^2}{d^2 - c^2} = \frac{b_0^2 + sa_0^2}{d_0^2 - c_0^2}, \tag{80}$$

which takes the normalized diagonal form (78) with parameters

$$u = \frac{b_0^2(sa_0^2 - c_0^2)(sa_0^2 + b_0^2 + c_0^2 - d_0^2)}{c_0^2(sa_0^2 + b_0^2)(sa_0^2 - b_0^2 - c_0^2 + d_0^2)}, \qquad v = \frac{b_0^2\big(s^2a_0^4 + sa_0^2(b_0^2 - c_0^2) - d_0^2(b_0^2 + c_0^2 - d_0^2)\big)}{d_0^2(sa_0^2 + b_0^2)(sa_0^2 - b_0^2 - c_0^2 + d_0^2)}. \tag{81}$$

Substituting (81) into (79) produces raw degree-17 polynomials which, upon reducing modulo $d_0^4 = b_0^4 + c_0^4 - s^2a_0^4$ and $(sa_0^2)^2 = b_0^4 + c_0^4 - d_0^4$, factor out the degree-8 parasite $2(sa_0^2 + b_0^2)(sa_0^2 - c_0^2)(b_0^2 - d_0^2)(c_0^2 - d_0^2)$ and extend by polynomial continuation to the globally defined (b_0, c_0)-symmetric degree-9 polynomial self-map $\mathcal{E}_9^{\pm}(P_0) = (a_2, b_2, c_2, d_2)$:

$$\begin{aligned}
a_2 &= a_0\Big[b_0^2c_0^2(b_0^4 + c_0^4 + 2d_0^4) \mp sa_0^2d_0^2(b_0^4 - c_0^4)\Big],\\
b_2 &= b_0\Big[b_0^2c_0^2(b_0^4 + c_0^4 - 2d_0^4) \mp sa_0^2d_0^2(b_0^4 + 3c_0^4)\Big],\\
c_2 &= c_0\Big[b_0^2c_0^2(b_0^4 + c_0^4 - 2d_0^4) \pm sa_0^2d_0^2(3b_0^4 + c_0^4)\Big],\\
d_2 &= -d_0\Big[b_0^2c_0^2(3b_0^4 + 3c_0^4 - 2d_0^4) \pm sa_0^2d_0^2(b_0^4 - c_0^4)\Big].
\end{aligned} \tag{82}$$

Moreover, in the polynomial ring $\mathbb{Q}[s, a_0, b_0, c_0, d_0]$, $\mathcal{E}_9^{\pm}(P_0)$ and the asymptotic tangent map $\mathcal{A}_9^{\mp}(P_0)$ satisfy the exact coordinate-by-coordinate polynomial transition identity

$$\big(b_0^4 - d_0^4\big)\,\mathcal{E}_9^{\pm}(P_0) \equiv \big(c_0^2d_0^2 \mp sa_0^2b_0^2\big)\,\mathcal{A}_9^{\mp}(P_0) \pmod{-s^2a_0^4 + b_0^4 + c_0^4 - d_0^4}, \tag{83}$$

establishing that $\mathcal{E}_9^{\pm}(P_0)$ and $\mathcal{A}_9^{\mp}(P_0)$ are projectively equal on S_{-s^2} everywhere outside the base locus of Corollary 1.2. (If one instead chooses the opposite 2-torsion sign on (c_2, d_2) in (79)–(82), the resulting point equals $\sigma_{ab}\big(\mathcal{A}_9^{\mp}(P_0)\big)$ where $\sigma_{ab}(a, b, c, d) = (-a, -b, c, d)$.)

Proof. (1) On $E_{u,v}=Q_1\cap Q_2$, the pencil $\lambda_1(Z^2-(1-u)X^2-uY^2)+\lambda_2(W^2-(1-v)X^2-vY^2)$ has four singular quadric cones at $\lambda_1/\lambda_2\in\{0,\infty,-(1-v)/(1-u),-v/u\}$, which are distinct if and only if $u,v\notin\{0,1\}$ and $u\neq v$; by the Jacobian criterion, $E_{u,v}$ is then a smooth curve of genus 1. The group $(\mathbb{Z}/2\mathbb{Z})^3$ of projective sign changes $(X:Y:Z:W)\mapsto(\pm X:\pm Y:\pm Z:\pm W)$ acts by translation by 2-torsion points in $\mathrm{Jac}(E_{u,v})[2]$, so the tangent-chord duplication map $P\mapsto -2P$ sends all eight points $(\pm1:\pm1:\pm1:\pm1)$ to the same point $(|X_2|:|Y_2|:|Z_2|:|W_2|)$ up to coordinate signs. Rationalize the first quadric $Z^2=(1-u)X^2+uY^2$ from $(X_0:Y_0:Z_0)=(1:1:-1)$ by setting $Z+X=t(X-Y)$, which yields

$$X(t)=-t^2+2t+u-1,\qquad Y(t)=-t^2-u+1,\qquad Z(t)=t^2+2(u-1)t-u+1,$$

where $(1:1:-1)$ lies at $t=\infty$ and $(1:1:1)$ lies at $t=1-u$. Substituting $(X(t),Y(t))$ into the second quadric $W^2=(1-v)X(t)^2+vY(t)^2$ gives the quartic

$$W(t)^2=t^4+4(v-1)t^3+2(2uv-u-4v+3)t^2+4(u-1)(v-1)t+(u-1)^2.$$

Matching the t^4,t^3,t^2 coefficients of $W(t)^2$ at $t=\infty$ with the quadratic $M_\infty(t)=t^2+2(v-1)t+c_0$ requires $2c_0+4(v-1)^2=2(2uv-u-4v+3)\implies c_0=2uv-u-2v^2+1$. With this choice of c_0, $W(t)^2-M_\infty(t)^2$ collapses to a linear polynomial in t that factors completely:

$$W(t)^2-M_\infty(t)^2=-4v(u-v)(v-1)\big(2t+u-v-1\big).$$

Evaluating $X(t),Y(t),Z(t),M_\infty(t)$ at the unique root $t_\infty=-(u-v-1)/2$ and multiplying (X,Y) by -4 and (Z,M_∞) by $+4$ (using the 2-torsion sign symmetry $(Z,W)\mapsto(-Z,-W)$ on $Q_1\cap Q_2$ to match the asymptotic branch orientation) gives

$$\begin{aligned}
-4X(t_\infty)&=(u-v-1)^2+8\left(\frac{u-v-1}{2}\right)-4(u-1)=(u-v-1)^2-4v,\\
-4Y(t_\infty)&=(u-v-1)^2+4(u-1)=(u-v+1)^2-4(1-v),\\
4Z(t_\infty)&=(u-v-1)^2-4(u-1)(u-v-1)-4(u-1)=-3u^2+2uv+2u+v^2-2v+1,\\
4M_\infty(t_\infty)&=(u-v-1)^2-4(v-1)(u-v-1)+4(2uv-u-2v^2+1)=(u+v-1)^2-4v(v-1),
\end{aligned}$$

proving (79).

(2) Rewriting the two equations of (21) as $\frac{b^2+c^2-d^2-sa^2}{d^2-b^2}=2\lambda=-1+\frac{c^2-sa^2}{d^2-b^2}$ and $\frac{b^2+c^2-d^2+sa^2}{d^2-c^2}=\frac{1}{\lambda}=-1+\frac{b^2+sa^2}{d^2-c^2}$ shows that the unique fiber of the Four-Eckardt-Plane pencil passing through P_0 is given by (80). Substituting $(a,b,c,d)=(a_0X,b_0Y,c_0Z,d_0W)$ into (80) gives the linear system in (Z^2,W^2):

$$\begin{aligned}
(d_0^2-b_0^2)c_0^2Z^2-(c_0^2-sa_0^2)d_0^2W^2&=(d_0^2-b_0^2)sa_0^2X^2-(c_0^2-sa_0^2)b_0^2Y^2,\\
(b_0^2+sa_0^2)c_0^2Z^2-(b_0^2+sa_0^2)d_0^2W^2&=-(d_0^2-c_0^2)sa_0^2X^2-(d_0^2-c_0^2)b_0^2Y^2.
\end{aligned}$$

Solving this 2×2 linear system by Cramer's rule gives $Z^2=(1-u)X^2+uY^2$ and $W^2=(1-v)X^2+vY^2$ with (u,v) as in (81). Substituting (81) into (79) and clearing the common denominator $c_0^4d_0^4(sa_0^2+b_0^2)^2(sa_0^2-b_0^2-c_0^2+d_0^2)^2$ produces degree-17 polynomials in (a_0,b_0,c_0,d_0). Reducing first modulo $d_0^4=b_0^4+c_0^4-s^2a_0^4$ factors out $2(sa_0^2+b_0^2)(sa_0^2-c_0^2)$, leaving degree-13 polynomials; then reducing $(sa_0^2)^2\equiv b_0^4+c_0^4-d_0^4$ factors out $(b_0^2-d_0^2)(c_0^2-d_0^2)$ from all four coordinates, leaving the degree-9 map $\mathcal{E}_9^+(P_0)$ (and $\mathcal{E}_9^-(P_0)$ under $s\mapsto-s$) in (82). Finally, to verify (83), multiply each coordinate of $\mathcal{A}_9^-(P_0)$ by $(c_0^2d_0^2-sa_0^2b_0^2)$ and substitute $(sa_0^2)^2=b_0^4+c_0^4-d_0^4$: for the a-coordinate,

$$\begin{aligned}
&\big(c_0^2d_0^2-sa_0^2b_0^2\big)\Big[b_0^2d_0^2(b_0^4-2c_0^4-d_0^4)-sa_0^2c_0^2(b_0^4+d_0^4)\Big]\\
&\qquad=b_0^2c_0^2d_0^4(b_0^4-2c_0^4-d_0^4)+b_0^2c_0^2(b_0^4+c_0^4-d_0^4)(b_0^4+d_0^4)\\
&\qquad\quad-sa_0^2d_0^2\Big[b_0^4(b_0^4-2c_0^4-d_0^4)+c_0^4(b_0^4+d_0^4)\Big]\\
&\qquad=(b_0^4-d_0^4)\Big[b_0^2c_0^2(b_0^4+c_0^4+2d_0^4)-sa_0^2d_0^2(b_0^4-c_0^4)\Big],
\end{aligned}$$

and the identical calculation on the b,c,d coordinates gives $(b_0^4-d_0^4)$ times the b,c,d coordinates of $\mathcal{E}_9^+(P_0)$ in (82), establishing (83) across all four coordinates. □

7.3 Complete Recipes for a Second Solution $P_2(P_0)$ Across All Regimes of n

Combining Theorems 7.1 and 7.2 with the skew-line involutions for $n=-1$ and $n=4$ and the Bremner–Choudhry–Ulas quadric fibrations [1] yields explicit closed-form second-solution generators across every known geometric regime of S_n.

Theorem 7.3 (Master Classification of Second-Solution Generators $P_0 \mapsto P_2(P_0)$)**.** *Let $P_0=(a_0,b_0,c_0,d_0)\in S_n(\mathbb{Q})$ be a single known non-trivial solution on $na^4+b^4+c^4=d^4$ with $a_0\neq 0$. Outside the explicit finite union of base, fixed, and 2-torsion subvarieties specified in each regime below, a second projectively distinct solution $P_2(P_0)$ is given in closed form (over $\mathbb{Q}(\sqrt{-n})$ for general $n\in\mathbb{Q}^\times$, and over $\mathbb{Q}$ in all rational regimes 1–5 below):*

1. ***Negative-Square Regime*** *$n=-s^2$ ($b_0^4+c_0^4-d_0^4=s^2a_0^4$): For any seed $P_0\notin \mathrm{Bs}(\mathcal{A}_9^{\pm})\cup\mathrm{Bs}(\widetilde{\mathcal{A}}_9^{\pm})$ (and outside the finite locus of fixed points of $\mathcal{A}_9^{\pm}$), P_0 generates **up to four projectively distinct primitive degree-9 $\mathbb{Q}$-rational solutions** via the asymptotic inflection maps $\mathcal{A}_9^{+}(P_0),\mathcal{A}_9^{-}(P_0)$ of (10) (equivalently $\mathcal{E}_9^{\mp}(P_0)$ of (82)) together with their $b_0\leftrightarrow c_0$ conjugates $\widetilde{\mathcal{A}}_9^{+}(P_0)$ and $\widetilde{\mathcal{A}}_9^{-}(P_0)$. On generic seeds (and function-field sections as in Theorem 5.2), iterating $\mathcal{A}_9^{\pm}$ and $\widetilde{\mathcal{A}}_9^{\pm}$ produces an infinite tree of primitive rational solutions of degrees 9^k (and 49 under alternating branches $\mathcal{A}_9^{\mp}\circ\mathcal{A}_9^{\pm}$).*

2. ***Single-Generator Skew-Line Regime*** *$n=-1$ ($b_0^4+c_0^4-d_0^4=a_0^4$): In addition to the four degree-9 rational solutions $\mathcal{A}_9^{\pm}(P_0),\widetilde{\mathcal{A}}_9^{\pm}(P_0)$ from (1), any point $P_0\notin L_1\cup L_2$ with $(a_0-b_0)(a_0-c_0)(b_0-d_0)(c_0+d_0)(a_0+b_0+c_0-d_0)\neq 0$ and $C_+(P_0)\neq 0$ immediately generates a **degree-5 $\mathbb{Q}$-rational second solution** via the crossed-pairing skew-line involution $\Phi_{L_1,L_2}(P_0)$:*

$$P_2(P_0)=(-C_-)\,(a_0,b_0,c_0,d_0)-C_+\,(-b_0+c_0+d_0,\ -a_0+c_0+d_0,\ a_0-b_0+d_0,\ -a_0+b_0+c_0), \tag{84}$$

where $-C_-=3(a_0-b_0)(a_0-c_0)(b_0-d_0)(c_0+d_0)$ and $C_+=a_0^3(b_0-c_0-d_0)-a_0(b_0^3-c_0^3-d_0^3)+b_0^3(c_0+d_0)-b_0(c_0^3+d_0^3)+c_0d_0(c_0^2-d_0^2)$. Composing with $b\leftrightarrow c$ generates the infinite dihedral orbit of Theorem 5.3(2) on non-torsion fibers.

3. ***Gaussian Conjugate Skew-Line Regime*** *$n=4$ ($4a_0^4+b_0^4+c_0^4=d_0^4$): Any seed P_0 with $(a_0,c_0)\neq(0,0)$ and $V_4(P_0)\neq 0$ generates a **degree-5 $\mathbb{Q}$-rational second solution** via the $\mathbb{Q}(i)$-conjugate skew-line involution $\Phi_{L,\bar{L}}(P_0)$:*

$$P_2(P_0)=\Big(U_4a_0-V_4(c_0-a_0),\ U_4b_0-V_4d_0,\ U_4c_0-V_4(c_0-2a_0),\ U_4d_0+V_4b_0\Big), \tag{85}$$

where $U_4=3(c_0^2-2a_0c_0+2a_0^2)^2$ and $V_4=-2\big(4a_0^4-4a_0^3c_0+2a_0c_0^3-c_0^4-b_0d_0(b_0^2+d_0^2)\big)$. Swapping $b_0\leftrightarrow c_0$ before applying $\Phi_{L,\bar{L}}$ yields a second independent degree-5 rational solution $\Phi_{L,\bar{L}}(a_0,c_0,b_0,d_0)$, and alternating $\Psi_4=\Phi_{L,\bar{L}}\circ\sigma_{bc}$ generates an infinite orbit on generic seeds.

4. ***Bremner–Choudhry–Ulas Regimes*** *$n=\pm 2s^2$ ($2s^2a_0^4+b_0^4+c_0^4=d_0^4$ **and** $-2s^2a_0^4+b_0^4+c_0^4=d_0^4$): By the algebraic factorizations*

$$\begin{aligned}(b^2+c^2-d^2)(b^2+c^2+d^2)&=2(bc-sa^2)(bc+sa^2) \qquad (n=+2s^2),\\ (d^2+b^2-c^2)(d^2-b^2+c^2)&=2(bc-sa^2)(bc+sa^2) \qquad (n=-2s^2),\end{aligned}$$

any known solution $P_0=(a_0,b_0,c_0,d_0)$ with $b_0c_0\neq\pm sa_0^2$ determines the unique quadric intersection $E(P_0)=Q_1\cap Q_2\subset S_{\pm 2s^2}$ passing through P_0:

$$\begin{aligned}Q_1(P_0):\ (b_0c_0-sa_0^2)(b^2+c^2-d^2)&=(b_0^2+c_0^2-d_0^2)(bc-sa^2),\\ Q_2(P_0):\ (b_0c_0+sa_0^2)(b^2+c^2+d^2)&=(b_0^2+c_0^2+d_0^2)(bc+sa^2)\end{aligned} \tag{86}$$

for $n=+2s^2$ (and with $b^2+c^2-d^2$ and $b^2+c^2+d^2$ replaced by $d^2+b^2-c^2$ and $d^2-b^2+c^2$, respectively, for $n=-2s^2$). Whenever $E(P_0)$ is smooth and P_0 is not a 2-torsion point of $E(P_0)$, parametrizing $Q_1(P_0)$ by rational projection from P_0 and taking the Fermat tangent at P_0 to the resulting binary quartic in $Q_2(P_0)$ produces an explicit $\mathbb{Q}$-rational second solution $P_2=-2P_0\in S_{\pm 2s^2}(\mathbb{Q})$.

5. ***Coordinate-Zero Tacnodal Regimes*** $c_0=0 \implies n=(d_0^4-b_0^4)/a_0^4$ ***(and*** $d_0=0 \implies n=-(b_0^4+c_0^4)/a_0^4$***):*** *When the known point has a vanishing coordinate $P_0=(a_0,b_0,0,d_0)$ with $a_0b_0d_0(d_0^4-b_0^4)\neq 0$, P_0 is a parabolic umbilical point where $c_2(u,v)=6b_0^2d_0^2(d_0^4-b_0^4)u^2$ is a degenerate rank-1 quadratic form. The tangent plane $T_{P_0}(S_n): b_0^3(a_0b-b_0a)=d_0^3(a_0d-d_0a)$ cuts out the genus-1 tacnodal curve parametrized by $a(T)=a_0$, $b(T)=b_0(1+d_0^3T)$, $d(T)=d_0(1+b_0^3T)$, and $c^4=T^2\big[6b_0^2d_0^2(b_0^4-d_0^4)+4b_0d_0(b_0^8-d_0^8)T+(b_0^{12}-d_0^{12})T^2\big]$. Note that over $\mathbb{R}$, $(b_0^4-d_0^4)>0$ is necessary for the bracket to take positive values; writing $c^2=TU$ requires a rational point (T_1,U_1) with $T_1U_1>0$ on the conic*

$$U^2=(b_0^4-d_0^4)\Big[(b_0^8+b_0^4d_0^4+d_0^8)\,T^2+4b_0d_0(b_0^4+d_0^4)\,T+6b_0^2d_0^2\Big] \tag{87}$$

such that T_1U_1 is a non-zero rational square $c_1^2\in\mathbb{Q}^{\times 2}$. Whenever (87) possesses a non-torsion rational point (T_1,U_1) with $T_1U_1=c_1^2\neq 0$, Fermat tangent duplication on $c^2=TU$ along the rationalized conic (87) generates a second non-zero rational solution $P_2\in S_n(\mathbb{Q})$.

Proof. (1) Part (1) is an immediate consequence of Theorems 1.1, 5.2, 7.1, and 7.2.

(2) On $S_{-1}: -a^4+b^4+c^4=d^4$, consider the two skew rational lines $L_1=\{a+b=0,c+d=0\}$ and $L_2=\{a+c=0,b-d=0\}$. For a point $P_0=(a_0:b_0:c_0:d_0)\in S_{-1}\setminus(L_1\cup L_2)$, the unique transversal line $M(P_0)$ meeting P_0,L_1, and L_2 connects $Q_1\in L_1$ and $Q_2\in L_2$, where taking the planes spanned by (P_0,L_1) and (P_0,L_2) gives $M(P_0)=\{P_0+tW(P_0)\}$ with direction vector

$$W(P_0)=(-b_0+c_0+d_0,\ -a_0+c_0+d_0,\ a_0-b_0+d_0,\ -a_0+b_0+c_0).$$

Notice that $P_0+W(P_0)=(a_0-b_0+c_0+d_0,\ -a_0+b_0+c_0+d_0,\ a_0-b_0+c_0+d_0,\ -a_0+b_0+c_0+d_0)\in L_2$ and $P_0-W(P_0)=(a_0+b_0-c_0-d_0,\ a_0+b_0-c_0-d_0,\ -a_0+b_0+c_0-d_0,\ a_0-b_0-c_0+d_0)\in L_1$. Thus the quartic polynomial $F_{-1}(P_0+tW(P_0))$ vanishes at $t=0$ (P_0), $t=1$ ($Q_2\in L_2$), and $t=-1$ ($Q_1\in L_1$), so $t(t^2-1)$ divides $F_{-1}(P_0+tW(P_0))$ modulo $F_{-1}(P_0)$. Specifically, in the polynomial ring $\mathbb{Z}[a_0,b_0,c_0,d_0][t]$, we have the exact identity

$$F_{-1}\big(P_0+tW(P_0)\big)-(1-2t^2-t^4)\,F_{-1}(P_0)=4(a_0+b_0+c_0-d_0)t(t^2-1)(C_-t+C_+),$$

where $C_-=-3(a_0-b_0)(a_0-c_0)(b_0-d_0)(c_0+d_0)$ and $C_+=a_0^3(b_0-c_0-d_0)-a_0(b_0^3-c_0^3-d_0^3)+b_0^3(c_0+d_0)-b_0(c_0^3+d_0^3)+c_0d_0(c_0^2-d_0^2)$. Consequently, on S_{-1} ($F_{-1}(P_0)=0$), the fourth intersection root is $t_2=-C_+/C_-$, and clearing the denominator C_- in $P_0+t_2W(P_0)$ yields (84).

(3) On $S_4: 4a^4+b^4+c^4=d^4$, consider the $\mathbb{Q}(i)$-conjugate skew lines $L=\{d-ib=0,c-(1+i)a=0\}$ and $\bar{L}=\{d+ib=0,c-(1-i)a=0\}$. For a point $P_0=(a_0:b_0:c_0:d_0)\in S_4(\mathbb{Q})$, the unique line meeting P_0,L, and $\bar{L}$ is defined over $\mathbb{Q}$ and parametrized by

$$P(t)=\big(a_0+t(c_0-a_0),\ b_0+td_0,\ c_0+t(c_0-2a_0),\ d_0-tb_0\big),$$

since at $t=\pm i$ we have $d(\pm i)\mp ib(\pm i)=(d_0\mp ib_0)\mp i(b_0\pm id_0)=0$ and $c(\pm i)-(1\pm i)a(\pm i)=c_0\pm i(c_0-2a_0)-(1\pm i)(a_0\pm i(c_0-a_0))=0$. Therefore $P(\pm i)\in L,\bar{L}\subset S_4$, so $t(t^2+1)$ divides $F_4(P(t))=4a(t)^4+b(t)^4+c(t)^4-d(t)^4$ modulo $F_4(P_0)=0$. Dividing $F_4(P(t))-(1-t^4)F_4(P_0)$ by $t(t^2+1)$ factors out $4(c_0^2-2a_0c_0+2a_0^2)$ and leaves the linear equation $U_4t+V_4=0$ with U_4 and V_4 as in (85); substituting $t=-V_4/U_4$ into $P(t)$ and clearing U_4 yields (85).

(4) For $n=+2s^2$, expanding $(b^2+c^2-d^2)(b^2+c^2+d^2)-2(bc-sa^2)(bc+sa^2)=(b^2+c^2)^2-d^4-2b^2c^2+2s^2a^4=2s^2a^4+b^4+c^4-d^4=0$ shows that every point on the quadric intersection (86) lies on S_{2s^2}, and similarly $(d^2+b^2-c^2)(d^2-b^2+c^2)-2(bc-sa^2)(bc+sa^2)=d^4-b^4-c^4+2s^2a^4=0$ for $n=-2s^2$. Since $P_0=(a_0,b_0,c_0,d_0)$ satisfies both equations of (86) by construction, P_0 is a rational point on $E(P_0)=Q_1(P_0)\cap Q_2(P_0)$, so its Fermat tangent-chord duplication $P_2=-2P_0$ produces a second rational solution on $S_{\pm 2s^2}$.

(5) When $c_0=0$, setting $u=1$ and $t=T$ in (69)–(73) gives $a(T)=a_0$, $b(T)=b_0(1+d_0^3T)$, $d(T)=d_0(1+b_0^3T)$, and $c_2(1,0)=6b_0^2d_0^2(d_0^4-b_0^4)$, $c_3(1,0)=4b_0d_0(d_0^8-b_0^8)$, $c_4(1,0)=d_0^{12}-b_0^{12}$. Since $na(T)^4+b(T)^4-d(T)^4+c^4=0$, solving for c^4 gives

$$c^4=-\Big(c_2(1,0)T^2+c_3(1,0)T^3+c_4(1,0)T^4\Big)=T^2(b_0^4-d_0^4)\Big[6b_0^2d_0^2+4b_0d_0(b_0^4+d_0^4)T+(b_0^8+b_0^4d_0^4+d_0^8)T^2\Big].$$

Writing $c^2 = \pm TU$ yields the conic (87), completing the proof. □

Table 4: Complete Classification of Closed-Form Second-Solution Parametrizations $P_0 = (a_0, b_0, c_0, d_0) \mapsto P_2(P_0)$ on $S_n : na^4 + b^4 + c^4 = d^4$.

Regime of n	**Defining Relation from P_0**	**Geometric Mechanism at P_0**	**Closed-Form Map $P_0 \mapsto P_2$**	**Field / Multiplicity from 1 Seed**
General $n \in \mathbb{Q}^\times$	$n = (d_0^4 - b_0^4 - c_0^4)/a_0^4$	Asymptotic tangent inflection $L_\pm \subset T_{P_0}(S_n)$	$P_{\text{even}}(P_0) \pm \sqrt{-n}\, P_{\text{odd}}(P_0)$ (75)	Deg 9 over $\mathbb{Q}(\sqrt{-n})$ (2 branches)
$n = -s^2$ ($s \in \mathbb{Q}^\times$)	$b_0^4 + c_0^4 - d_0^4 = s^2 a_0^4$	Rational asymptotic lines & Eckardt quadric pencil	$\mathcal{A}_9^{\pm}(P_0), \bar{\mathcal{A}}_9^{\pm}(P_0), \mathcal{E}_9^{\mp}(P_0)$ (10), (82)	Deg 9 over $\mathbb{Q}$ (4 distinct solutions)
$n = -1$ ($s = 1$)	$b_0^4 + c_0^4 - d_0^4 = a_0^4$	Crossed-pairing skew lines L_1, L_2 + asymptotic lines	$\Phi_{L_1,L_2}(P_0)$ (84) and $\mathcal{A}_9^{\pm}(P_0)$	Deg 5 (2 solutions) + Deg 9 (4 solutions) over $\mathbb{Q}$
$n = 4$	$4a_0^4 + b_0^4 + c_0^4 = d_0^4$	$\mathbb{Q}(i)$-conjugate skew-line transversal $L, \bar{L}$	$\Phi_{L,\bar{L}}(P_0)$ and $\Phi_{L,\bar{L}}(\sigma_{bc}P_0)$ (85)	Deg 5 over $\mathbb{Q}$ (2 distinct solutions)
$n = \pm 2s^2$	$\pm 2s^2 a_0^4 + b_0^4 + c_0^4 = d_0^4$	Bremner–Choudhry–Ulas quadric pencil through P_0	Tangent duplication $P_0 \mapsto -2P_0$ on (86)	Elliptic duplication on $E(P_0)$ over $\mathbb{Q}$
$c_0 = 0$ (Tacnodal)	$n = (d_0^4 - b_0^4)/a_0^4$	Parabolic tacnodal tangent section $T_{P_0}(S_n) \cap S_n$	Fermat duplication on $c^2 = TU$ (87)	Genus-1 double cover of conic over $\mathbb{Q}$

Remark 7.4 (Six Explicit Second Solutions Generated from Euler's Single Point $P_0 = (59, 133, 134, 158)$ on S_{-1}). Let $P_0 = (59, 133, 134, 158)$ be Euler's smallest positive primitive solution to $-a^4 + b^4 + c^4 = d^4$ ($n = -1, s = 1$). Applying the closed-form formulas of Theorem 7.3 directly to P_0 produces four distinct primitive second solutions (and six including $b \leftrightarrow c$ permutations), all verified by exact integer arithmetic:

1. **Degree-5 Skew-Line Involution $\Phi_{L_1,L_2}(P_0)$ (84):** Evaluating (84) at $P_0 = (59, 133, 134, 158)$ gives raw coordinates $(9173250000, 7785750000, -7652250000, 2177250000)$ with gcd = 750,000, yielding the primitive solution

$$\Phi_{L_1,L_2}(59, 133, 134, 158) = \big(12231,\ 10381,\ -10203,\ 2903\big) \qquad (\gcd = 750{,}000),$$

which satisfies $-12231^4 + 10381^4 + (-10203)^4 = 71{,}021{,}222{,}453{,}281 = 2903^4$. Applying Φ_{L_1,L_2} to the $b \leftrightarrow c$ swapped seed $(59, 134, 133, 158)$ gives raw coordinates $(8709962400, 7426922400, -7234077600, 2255882400)$ (gcd = 64,800), yielding a second primitive solution

$$\Phi_{L_1,L_2}(59, 134, 133, 158) = \big(134413,\ 114613,\ -111637,\ 34813\big) \qquad (\gcd = 64{,}800),$$

which satisfies $-134413^4 + 114613^4 + (-111637)^4 = 1{,}468{,}810{,}607{,}884{,}410{,}961 = 34813^4$.

2. **Degree-9 Asymptotic Tangent & Four-Eckardt-Plane Duplication Maps $\mathcal{A}_9^{\pm}(P_0)$ and $\mathcal{E}_9^{\mp}(P_0)$ (10), (82):** Evaluating (10) and (82) at $P_0 = (59, 133, 134, 158)$ gives the exact branch equalities

$$\mathcal{A}_9^{+}(P_0) = -\mathcal{E}_9^{-}(P_0) = \big(-161634227550911,\ 50577474852863,\ 186785699397086,\ 152526608743862\big),$$
$$\mathcal{A}_9^{-}(P_0) = -\mathcal{E}_9^{+}(P_0) = \big(-47301468159703,\ 54398746431911,\ 15168229732247,\ 44162725988761\big),$$

where the common projective scalar ratio between the raw polynomial evaluations of $\mathcal{E}_9^{+}(P_0)$ (gcd = 746,528) and $\mathcal{A}_9^{-}(P_0)$ (gcd = 599,072) across all four coordinates is $-\frac{746528}{599072} = -\frac{23329}{18721} = \frac{c_0^2 d_0^2 - a_0^2 b_0^2}{b_0^4 - d_0^4}$, confirming (83).

8 Literature Review and Diophantine Context of $na^4 + b^4 + c^4 = d^4$

The arithmetic geometry of diagonal quartic surfaces $S_n : na^4 + b^4 + c^4 = d^4$ in $\mathbb{P}^3$ sits at the confluence of classical diophantine analysis, the theory of K3 surfaces of high Picard rank, and modern obstructions to the Hasse principle. In this section we synthesize the historical evolution of the subject and place the theorems of Sections 1–7 in relation to prior work.

8.1 Euler's Quartic Conjecture, Classical Parametrizations, and Inflection Tangents

The systematic study of fourth-power diophantine equations began with Euler [7, 4], who in 1772 investigated equal sums of two fourth powers $A^4 + B^4 = C^4 + D^4$ (the surface S_{-1}) and constructed a

celebrated homogeneous polynomial parametrization of degree 7 by intersecting S_{-1} with transversal planes through pairs of skew rational lines, discovering the smallest positive primitive solution $59^4 + 158^4 = 133^4 + 134^4$. Simultaneously, Euler conjectured that the sum of three fourth powers can never equal a fourth power ($S_1(\mathbb{Q}) = \emptyset$ away from coordinate hyperplanes). In 1915, Carmichael [3] studied the surface $S_4 : 4a^4 + b^4 + c^4 = d^4$ via the identity $(hU^4 + V^4)^4 - (hU^4 - V^4)^4 = 8hU^4V^4(h^2U^8 + V^8)$, showing that choosing $h = 8$ makes the first term $4096U^{12}V^4 = (8U^3V)^4$ a fourth power and leaves $64U^4V^{12} = 4(2UV^3)^4$, which yields a degree-4 polynomial family on S_4. Theorem 4.1(2) above proves that Carmichael's $n \equiv 4 \pmod{\mathbb{Q}^{\times 4}}$ family is in fact the *unique* constant-n surface that can arise from single-term binomial cancellation $c^4 = 8XY^3$ in the UFD $\mathbb{Q}[p, q]$.

A decisive geometric advance for general diagonal quartics $AX_0^4 + BX_1^4 + CX_2^4 + DX_3^4 = 0$ occurred in 1944 when Richmond [12] proved that whenever the product of the four coefficients $ABCD$ is a rational square, the two principal asymptotic (inflection-tangent) directions at any rational point $P \in S(\mathbb{Q})$ are individually defined over $\mathbb{Q}$. Passing the inflection tangent line through P cuts the quartic surface with multiplicity at least 3 at P, so the residual fourth intersection point is rational and given by a rational self-map on S. In the standard literature (see Richmond [12] and Logan, McKinnon, and van Luijk [10]), this asymptotic tangent map is formulated in its unreduced degree-17 presentation $\mathcal{A}_\pm(P_1) = K_4(P_1)P_1 - K_3^\pm(P_1)V_\pm(P_1)$. Theorem 1.1 of the present paper shows that on S_{-s^2}, the degree-16 and degree-12 invariants K_4 and $K_3^\pm$ share the exact degree-8 polynomial factor $M_\pm(P_1) = (sa_1^2b_1^2 \mp c_1^2d_1^2)(sa_1^2d_1^2 \mp b_1^2c_1^2)$ identically on $\mathbb{P}^3$, reducing the map to the canonical degree-9 self-map $\mathcal{A}_9^\pm$ and exposing its 1-dimensional radical base scheme along eight projective lines (Corollary 1.2) as well as the degree-49 reduction of its alternating second iterate $\mathcal{A}_9^\mp \circ \mathcal{A}_9^\pm$ (Theorem 5.2).

8.2 Elliptic Fibrations, Disproof of Euler's Conjecture, and Zariski Density

Whereas computational searches by Ward [17] and Lander and Parkin [9] established only bounds on small solutions to $a^4 + b^4 + c^4 = d^4$, Elkies [5] disproved Euler's quartic conjecture in 1988 by exploiting a genus-1 conic-bundle fibration on $S_1 : a^4 + b^4 + c^4 = d^4$, discovering the primitive solution $2682440^4 + 15365639^4 + 18796760^4 = 20615673^4$ and proving that $S_1(\mathbb{Q})$ is dense in the real Euclidean topology. In a subsequent algorithmic study of rational points near curves via lattice reduction, Elkies [6] constructed a degree-8 polynomial family for $n = -12$ ($-12a^4 + b^4 + c^4 = d^4$). Recently, Vepir, Tomita, Berg, et al. [15, 16] discovered a degree-9 polynomial parametrization for the binomial $n(p, q) = 20p^8 - q^8$. Theorem 4.2 unifies both constructions into a single five-coefficient master polynomial $P(T)$ and classifies all rational three-coefficient cancellation branches (the $K = 0$ monomial-b branch for $n \equiv -12 \pmod{\mathbb{Q}^{\times 4}}$, the $K \neq 0$ Subcase A quartic $P(T) = \frac{80}{27}T^4 - s^{20}$, and the $K = -8/r^5$ Subcase B linear binomial yielding Vepir's $20p^8 - q^8$).

From the viewpoint of arithmetic geometry, diagonal quartic surfaces $S_n \subset \mathbb{P}^3$ are smooth K3 surfaces whose *geometric* Picard group $\mathrm{Pic}(S_{n,\overline{\mathbb{Q}}})$ over an algebraic closure $\overline{\mathbb{Q}}$ has maximal possible rank $\rho(S_{n,\overline{\mathbb{Q}}}) = 20$ and is generated over $\overline{\mathbb{Q}}$ by the classes of the 48 geometric straight lines lying on $S_{n,\overline{\mathbb{Q}}}$ (which are defined over the number field $\mathbb{Q}(\sqrt[4]{-n}, \zeta_8)$; see Pinch and Swinnerton-Dyer [11] and Bright [2]). Over $\mathbb{Q}$, the absolute Galois group $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ permutes these 48 line classes, and the *arithmetic* (Galois-invariant) Picard group $\mathrm{Pic}(S_n) \subseteq \mathrm{Pic}(S_{n,\overline{\mathbb{Q}}})^{\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})}$ has rank $1 \leq \rho(S_n) \leq 20$ depending on n (for example, $\rho(S_{-1}) = 8$ as computed by Pinch and Swinnerton-Dyer [11], whereas $\rho(S_n) = 1$ for generic n). The Galois action on $\mathrm{Pic}(S_{n,\overline{\mathbb{Q}}})$, the classification of elliptic pencils, and the arithmetic of rational curves on S_n were systematically developed by Swinnerton-Dyer [13, 14] and Pinch and Swinnerton-Dyer [11]. When the rational Picard rank $\rho(S_n)$ is at least 2 and S_n carries an elliptic fibration or a pair of skew lines defined over $\mathbb{Q}$ or a quadratic extension, rational curves and infinite dihedral orbits of sections can be generated iteratively. Theorem 5.3 makes this action completely explicit on S_{-1}, showing that both the skew-line dihedral tower at degrees $m = 6k + 1 \in \{7, 13, 19, 25, 31, 37, 43, 49, \dots\}$ and the asymptotic images at degrees 43 and 45 preserve a single-polynomial Klein four-group V_4-symmetry.

For surfaces where $ABCD$ is a rational square (such as S_{-s^2}), Logan, McKinnon, and van Luijk [10] proved that if S_{-s^2} possesses a single non-trivial rational point lying outside a finite union of special curves (where the asymptotic self-map is periodic or undefined), then the set of rational points $S_{-s^2}(\mathbb{Q})$ is Zariski dense. Complementing this result, Bremner, Choudhry, and Ulas [1] constructed explicit pencils of genus-1 space quartics $E = Q_1 \cap Q_2 \subset S_n$ for the regimes $n = \pm 2s^2$ (as well as parametric families of n with infinitely many rational points). In Section 2, Section 6, and Section 7.2, we establish the Four-Eckardt-Plane and Gaussian conic-bundle fibrations on S_{-s^2}, prove their equivalence under the Möbius transformation $2\lambda + 1 = 1/\tau$, and prove in Theorem 7.2 that Fermat tangent-chord duplication $P_0 \mapsto -2P_0$ along the Four-Eckardt-Plane quadric intersection $Q_1 \cap Q_2 \subset S_{-s^2}$ collapses from degree 17 to the degree-9 polynomial map $\mathcal{E}_9^{\pm}(P_0)$ that is projectively identical to the opposite asymptotic branch $\mathcal{A}_9^{\mp}(P_0)$.

8.3 Local-to-Global Principles and Brauer–Manin Obstructions

Finally, when n does not belong to one of the parametric or square-product regimes, the existence of rational points on S_n is governed by local solubility in $\mathbb{Q}_p$ (especially at the dyadic prime $p = 2$, as analyzed in Theorem 6.1(2)) and by the Brauer–Manin obstruction. Bright [2] classified the algebraic Brauer group $\mathrm{Br}_1(S_n)/\mathrm{Br}(\mathbb{Q})$ across all diagonal quartic surfaces over $\mathbb{Q}$, and Ieronymou and Skorobogatov [8] determined the odd-order transcendental Brauer–Manin obstructions on diagonal quartics. Table 5 summarizes the principal diophantine regimes of $S_n : na^4 + b^4 + c^4 = d^4$ across the literature and compares prior constructions with the exact reductions proved in this work.

Table 5: Synthesis of Classical and Modern Literature on $S_n : na^4 + b^4 + c^4 = d^4$ and Comparison with Results Established in This Paper.

Regime on S_n	**Key Literature**	**Prior Method**	**Contribution in This Paper**	**Results**
$n = -1$ ($A^4 + B^4 = C^4 + D^4$)	Euler [7], Dickson [4]	Skew-line planes; deg-7 curve C_{eul}	V_4-symmetric tower at deg $6k+1$ (7..49) & deg 43, 45 asymptotic curves	Thm. 5.3, 7.3(2)
$n = -s^2$ ($ABCD \in \mathbb{Q}^{\times 2}$)	Richmond [12], Logan et al. [10]	Unreduced deg-17 inflection map $\mathcal{A}_{\pm}$	Exact factor $M_{\pm}(P_1)$ drop to canonical deg-9 map $\mathcal{A}_9^{\pm}$; deg-49 iterate $\mathcal{A}_9^{\mp} \circ \mathcal{A}_9^{\pm}$	Thm. 1.1, Cor. 1.2, Thm. 5.2
$n = -s^2$ Conic Bundles	Swinnerton-Dyer [14], Tomita [15]	Special quadric sections	Universal (k,t)-Eckardt & Gaussian dual pencils; $t = 2u^2 - 1$ split	Thm. 2.1, Thm. 6.1(1)
$n = -s^2$ Quadric Duplication	Pinch–Swinnerton-Dyer [11]	Elliptic chord-tangent on $Q_1 \cap Q_2$	Closed-form deg-9 map $\mathcal{E}_9^{\pm}(P_0)$ & projective identity with $\mathcal{A}_9^{\mp}(P_0)$	Thm. 7.2
$n = 4$ ($4a^4 + b^4 + c^4 = d^4$)	Carmichael [3]	Binomial $c^4 = 8XY^3$; deg-4 family	UFD uniqueness theorem; deg-5 $\mathbb{Q}(i)$-conjugate skew-line map $\Phi_{L,\bar{L}}$	Thm. 4.1(2), Thm. 7.3(3)
$n = 1$ ($a^4 + b^4 + c^4 = d^4$)	Ward [17], Elkies [5]	Conic bundle & elliptic curve section	Degenerate $s = \pm 1$ vs. cubic $s \neq \pm 1$ biquadratic discriminant locus	Prop. 3.1, Thm. 3.3
$n = -12$ & $20p^8 - q^8$	Elkies [6], Vepir et al. [16]	Deg 8 ($n = -12$) & deg 9 ($20p^8 - q^8$)	Complete classification of maximal cancellation in master polynomial $P(T)$	Thm. 4.2, Thm. 4.3
$n = \pm 2s^2$	Bremner–Choudhry–Ulas [1]	Quadric pencils on $S_{\pm 2s^2}$	Unified point-duplication recipe $P_0 \mapsto -2P_0$ across all regimes of n	Thm. 7.3(4), Table 4
2-Adic & Brauer Obstructions	Bright [2], Ieronymou et al. [8]	Cohomological Brauer group $\mathrm{Br}(S_n)$	Explicit 2-adic valuation & quadratic-reciprocity descent on S_{-s^2}, S_{-4}	Thm. 6.1(2)

References

[1] A. Bremner, A. Choudhry, and M. Ulas, *Constructions of diagonal quartic and sextic surfaces with infinitely many rational points*, International Journal of Number Theory **10** (2014), no. 7, 1675–1698. Available at `arXiv:1402.4583`.

[2] M. J. Bright, *Brauer groups of diagonal quartic surfaces*, Journal of Symbolic Computation **41** (2006), no. 5, 544–558.

[3] R. D. Carmichael, *Diophantine Analysis*, Mathematical Monographs, No. 16, John Wiley & Sons, New York, 1915.

[4] L. E. Dickson, *History of the Theory of Numbers, Vol. II: Diophantine Analysis*, G. E. Stechert & Co., New York, 1934 (Euler's 1772 parametrization of $A^4 + B^4 = C^4 + D^4$, pp. 644–647).

[5] N. D. Elkies, *On $A^4 + B^4 + C^4 = D^4$*, Mathematics of Computation **51** (1988), no. 184, 825–835.

[6] N. D. Elkies, *Rational points near curves and small nonzero $|x^3 - y^2|$ via lattice reduction*, in *Algorithmic Number Theory (ANTS-IV)*, Lecture Notes in Computer Science **1838**, Springer, Berlin, 2000, pp. 33–63. Available at `arXiv:math/0005139`.

[7] L. Euler, *Resolutio formulae Diophanteae $ab(maa+nbb) = cd(mcc+ndd)$ per numeros rationales*, Novi Commentarii Academiae Scientiarum Petropolitanae **17** (1772), 64–83; reprinted in *Leonhardi Euleri Opera Omnia*, Ser. I, Vol. 3, Teubner, Leipzig, 1917, pp. 263–281.

[8] E. Ieronymou and A. N. Skorobogatov, *Odd order Brauer–Manin obstruction on diagonal quartic surfaces*, Advances in Mathematics **270** (2015), 181–205.

[9] L. J. Lander and T. R. Parkin, *Counterexamples to Euler's conjecture on sums of like powers*, Bulletin of the American Mathematical Society **72** (1966), 1079.

[10] A. Logan, D. McKinnon, and R. van Luijk, *Density of rational points on diagonal quartic surfaces*, Algebra & Number Theory **4** (2010), no. 1, 1–20. Available at `arXiv:0812.4779`.

[11] R. G. E. Pinch and H. P. F. Swinnerton-Dyer, *Arithmetic of diagonal quartic surfaces, I*, in *L-Functions and Arithmetic (Durham, 1989)*, London Mathematical Society Lecture Note Series **153**, Cambridge University Press, Cambridge, 1991, pp. 317–338.

[12] H. W. Richmond, *On the Diophantine equation $ax^4 + by^4 + cz^4 + dw^4 = 0$, the product abcd being a square number*, Journal of the London Mathematical Society **19** (1944), 193–194.

[13] H. P. F. Swinnerton-Dyer, *An enumeration of all varieties of degree 4*, American Journal of Mathematics **90** (1968), 1043–1063.

[14] H. P. F. Swinnerton-Dyer, *Arithmetic of diagonal quartic surfaces, II*, Proceedings of the London Mathematical Society (3) **80** (2000), no. 3, 513–544.

[15] S. Tomita, *Equal Sums of Quartics (In Context with the Richmond Equation)*, Computational Number Theory Notes, available at `maroon.dti.ne.jp/fermat/Sums_of_Quartics.pdf`.

[16] Vepir, S. Tomita, A. H. Berg, et al., *New infinite family of solutions to $n \cdot a^4 + b^4 + c^4 = d^4$?*, Mathematics Stack Exchange Question 5151446 (2026), available at `math.stackexchange.com/q/5151446`.

[17] M. Ward, *Euler's problem on sums of three fourth powers*, Duke Mathematical Journal **15** (1948), 827–837.